Matching and Incentive Effects of Test-Blind Admissions*

Phi Adajar
Massachusetts Institute of Technology
Job Market Paper

Nagisa Tadjfar
University of British Columbia

Kartik Vira
Institute for Fiscal Studies

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High-stakes final exams shape students' educational paths in two ways: universities use different sets of information to screen applicants, shaping the equilibrium match, and they may create incentives for students to exert effort. We study these forces using the staggered introduction of unconditional offers in the UK, which allowed students to matriculate regardless of results from their age-18 exams (A-levels). We first show there are no incentive effects: while students who accept an unconditional offer score 0.14*SD below those who decline one, conditional on observables, using a two-way fixed effects specification, we find that exposure to unconditional offers has a precisely estimated null effect on students' age-18 exam performance. Unconditional offers have positive match effects for recipients: exposure increases matriculation by 1.1pp, largely at middle-tercile universities. Using a structural model, we find that unconditional offers are anti-meritocratic: they benefit the lowest-achieving quartile of students at the expense of higher-achieving students and lower overall welfare by 0.5%. Post-qualification admissions, under which students apply after receiving their scores, raise welfare by 14.2% through better sorting. This gain is mostly attributable to increased information for both sides of the market. Our results inform current policy debates in the UK, highlighting how the distributional effects of admissions policies depend on student information.*

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1 Introduction

How do test-blind admissions shape higher education? Many countries, including China and India, admit millions of students to university each year using performance on high-stakes standardized exams. At the same time, many universities in the US, France, and Japan now admit students before or without standardized test scores. Policy-makers who decide whether university matching markets should use information on applicant test scores face tradeoffs in *match effects* and *incentive effects*. Changing the applicant screening process reallocates students across universities, generating *match effects*. On the other hand, *incentive effects* result from changes in student behavior in response to test-blind admissions. When university admissions no longer use test scores to select students, students may exert less effort studying, a form of moral hazard. Understanding the equilibrium effects of test-blind admissions requires empirical estimation of both match effects and incentive effects.

In this paper, we estimate the equilibrium effects of test-blind admissions using quasi-random variation in test-blind policies in the UK and a structural model of university applications. In 2013, nearly all UK students exclusively received university offers that were conditional on their age-18 exam scores, but by 2019, nearly 40% of university applicants received at least one *unconditional offer*, which guaranteed admission regardless of these exam scores. Students who accepted unconditional offers scored 0.14SD worse than students who declined them after controlling for observables. We disambiguate whether this gap is driven by incentive effects or selection into take-up on the basis of private information with reduced-form evidence, comparing students with the same application set but different *exposure* to unconditional offers. We find no incentive effects, indicating that the gap is driven by private information, not moral hazard. The same methodology shows that exposure to unconditional offers has positive match effects, and increases university attendance by 1.1pp and also the quality of university attended. However, a structural model to examine match effects on non-recipients shows that unconditional offers are anti-meritocratic, and benefit the lowest-achieving quartile of students at the expense of other quartiles, decreasing overall welfare by 0.5%. A counterfactual admissions system where students apply after receiving test scores — post-qualification admissions — raises student welfare by 14.2% by increasing the quality of match assignments.

We begin by describing the institutional context of UK university admissions and unconditional offers, documenting how they were used and their effects on universities. Universities may have used unconditional offers to attract higher-quality students who value the certainty of a university spot. Lower-ranked universities were more likely to use unconditional offers, and students who received an unconditional offer outperformed those who did not receive one by 0.4SD on their teacher-predicted age-18 grades; after a university began using these offers, the average teacher-predicted age-18 grades of enrolling students increased by 0.07SD. However, unconditional offers hurt universities on other metrics: after their introduction, the average final age-18 grades of enrolling students *decreased* by 0.02SD. Conditional on observables, students who took an unconditional offer scored 0.14SD *lower* on their age-18 exams than those who declined one. Students who accepted unconditional offers had *ex post* lower ability than their *ex ante* ability suggests, causing universities to experience a "winner's curse".

This "winner's curse" is consistent with two different explanations: moral hazard and private information. Under *moral hazard*, students may reduce test-taking effort after taking an unconditional offer, knowing that their end-of-year grades no longer matter. On the other hand, students may have *private information*: students who expect to do poorly on their end-of-year exams, and so to miss the grade requirements of their other offers, may self-select into taking these offers. Critics, including the Education Secretary, argued that unconditional offers reduced students' effort and distorted students' decisions, and Universities UK cited lower motivation among takers when it restricted the practice in 2021. Effectively, these narratives attribute the achievement gap between takers and decliners to moral hazard alone.

We distinguish between these two explanations with reduced-form evidence, and find no evidence of moral hazard. To do so, we make use of whether students were *exposed* to unconditional offers by applying to a program that was using them. We then use a two-way fixed effects design to compare the test scores of two students with the same application set, one exposed to unconditional offers and one not. To account for the fact that students may preferentially apply to programs that make unconditional offers, we provide evidence that students did not anticipate the first year in which a program began using unconditional offers, and use first-year exposure as our core identifying variation. We estimate a statistically precise null effect of unconditional offer exposure on later test scores. We conclude that the gap in achievement between unconditional offer takers and decliners is driven by private information, not moral hazard.

With the same design, we examine the match effects for recipients of unconditional offers, and find that exposure to unconditional offers increases university attendance and the quality of university. Students exposed to unconditional offers are 1.1pp more likely to matriculate into university than those who were not, driven by attendance at middle-tercile universities. Among the set of students that attend university, exposure to unconditional offers marginally raises the quality of university attended. This finding is consistent with the idea that in the absence of unconditional offers, takers would have missed grade conditions and enrolled at a less-selective university.

To study the equilibrium match effects, especially for students who declined or did not receive unconditional offers, we estimate a structural model of students and universities, focusing on students applying to mathematics programs. We specify a game between students and universities with four stages that mirror the application process in the UK: applications, offers, ranking, and acceptances. The model includes an unconditional-offer premium, which captures the value students place on an offer being guaranteed. Given the reduced-form results, we explicitly model private information observed by students as they rank offers and assume that there are no moral hazard effects. Students seek to optimize the utility of enrollment, while universities maximize the quality of their incoming cohort subject to capacity constraints and an exogenously determined level of unconditional offers. Identification comes from differences across students in three characteristics: school-program distance, admissions rates, and exposure to unconditional offers.

This structural model allows us to recover parameters about private information and the unconditional-offer premium which cannot be found through reduced-form evidence. We estimate that students can forecast two-thirds of the variance of their own age-18 test scores, suggesting that the extent of private information in this context is large. This information asymmetry between students and universities is consistent with the reduced-form results, and means that without age-18 test scores,

universities will face a "winner's curse". We additionally find that the unconditional-offer premium is small for the average student, from which we infer that the benefits of certainty are small in this setting and that unconditional offers have little ability to improve overall ex-ante welfare. However, both girls and lower-achieving students have a positive unconditional-offer premium, indicating that some students gain more from unconditional offers than others.

Finally, we use these estimates to assess student welfare and equilibrium university enrollment under different university admissions schemes, finding that unconditional offers are anti-meritocratic, but reducing information frictions improves the match. We study three admissions designs: a full ban on unconditional offers, unconditional offers used at their 2018 levels, and post-qualification admissions, under which students apply to university *after* receiving their age-18 grades. Post-qualification admissions were debated in the UK in 2022 and are again under discussion as generative AI changes the nature of assessment (Harvey-Brewin, 2026). The unconditional offer regime decreases overall student utility by 0.5%; the lowest-achieving quartile of students does benefit, but does so at the expense of higher-achieving students. In contrast, post-qualification admissions lead to an increase in total welfare of 14.2%, largely through improved allocation, with higher-achieving students benefiting more. Improved information for students and universities accounts for about two-thirds of the total gain in welfare. Unlike unconditional offers, post-qualification admissions improve outcomes for students of all abilities and are meritocratic, providing empirical support for a reform of the UK admissions system.

Our counterfactual simulations and reduced-form analysis contribute to the growing literature on the optimal design of college admissions systems. The shift to test-optional university admissions in the wake of COVID-19 — and the shift back to test-required admissions a few years later — has led to a rapid proliferation of papers analyzing the efficacy of such policies and impacts on students and cohort composition, mostly at selective institutions (Sacerdote, Staiger and Tine, 2025; Avery, Shi and Magouirk, 2025). Dessein, Frankel and Kartik (2025) focus on universities' choice to be test-optional or test-required; in their model, inter-university competition leads to an adversely selected applicant pool. Doğan and Erdil (2025) discuss the benefits of post-qualification admissions from a theoretical lens. Two recent papers examine the equilibrium effects of admissions systems: Borghesan (2025) uses a structural model to derive equilibria when universities stop requiring the SAT, while Barahona, Dobbin and Otero (2026) study the equilibrium effects of affirmative action in Brazil. We examine the equilibrium impacts of test-blind admissions, with evidence on both incentive and match effects for students across the ability distribution.

Our setting also allows us to study how admissions policies induce or deter student effort. A small literature shows that affirmative action can raise effort exerted by high-school students who benefit from such policies (Grau, 2018; Akhtari, Bau and Laliberté, 2024; Cotton, Hickman and Price, 2022). A recent paper from Krishna et al. (2026) shows that changing grade reporting on centralized tests can also induce effort. Existing evidence on how test-blind policies change student effort is correlational (Office for Students, 2019); we provide the first causal estimate, separating the effort response from selection.

The timing of admissions has been studied through a large theoretical literature; in matching markets where students are screened by multiple schools, enrollees are adversely selected. Most directly, Chade (2006) develops a model with imperfect signals in which being matched leads to an

"acceptance curse"; this is further developed by Lee (2009) in the university admissions setting.[1] In these papers, inter-university competition leads to adverse selection; in our setting, selection occurs primarily through students' private information about a bad future signal. In contrast to Lee (2009), who shows that the winner's curse leads to efficiency-improving early admissions, we demonstrate empirically that students' private information informs selection into take-up in a way that acts *against* early admissions. Finally, this literature is largely theoretical; we provide reduced-form evidence resulting from quasi-experimental variation in offer timing.

We also contribute to a literature on preference estimation in the context of application systems. We build upon the work of Agarwal and Somaini (2018), Larroucau and Rios (2020), Bleemer (2024), and Kapor (2025), which model students' application choice of portfolios or rank-order lists. While the literature abstracts away from the timing of offers, we incorporate offer types into the student-side choice problem directly. Further, we discuss how preferences under correlated admissions systems (such as those in Ali and Shorrer, 2025) can be estimated by adapting the methodology of Larroucau and Rios (2020), which explicitly requires independent admissions.

Our decomposition of the taker–decliner achievement gap relates to a literature in insurance and consumer credit that distinguishes adverse selection from moral hazard. The positive correlation test of Chiappori and Salanié (2000) describes how correlation between insurance take-up and outcomes is a sign of private information, but distinguishing the two forces requires additional exogenous variation, such as the field experiment studied by Karlan and Zinman (2009).[2] We bring this literature to the new domain of college admissions.

2 Empirical Setting, Data, and Descriptive Results

2.1 University applications in the UK

When students apply to university in the United Kingdom, students receive tentative offers that are based on their prior academic performance, but which are *conditional* on performance on an end-of-year exam. Initial applications include grades from exams taken at age 16, as well as teacher-predicted grades for age-18 exams that students have not completed when they apply. Students receive offers from universities and rank choices before taking their exams and receiving their final grades. As a result, programs have historically made "conditional offers," which are contingent on students achieving certain grades in their final exams; programs can withdraw offers from students who do not meet these conditions. This means that students face uncertainty about their university enrollment until they receive their final grades. Such a system stands in contrast to *post-qualification* admissions systems, such as those used in China, India, and Australia, where the match between students and universities only occurs once both parties are aware of all the relevant information, including test scores.

Approximately 250,000 18-year-old students from England, Wales, and Northern Ireland[3] ap-

[1] See also Chade, Lewis and Smith (2014) and Kloosterman and Troyan (2020); the latter focuses on student inferences after being accepted.

[2] See also the canonical models of Akerlof (1970) and Rothschild and Stiglitz (1976); further theoretical work in Chiappori et al. (2006); and Adams, Einav and Levin (2009), Einav, Finkelstein and Cullen (2010), and Einav and Finkelstein (2011) for additional empirical studies.

[3] We exclude Scottish students; their exam and tuition fee systems differ from those in other UK countries.

ply every year to university through the centralized Universities and Colleges Admissions Service (UCAS) platform. Because tuition fees for domestic students are uniform across universities,[4] the relevant tradeoffs for UK students are over academic fit and preferences for geography, amenities, and other factors. Table 1 summarizes the process as of 2012.[5]

Table 1: Timeline of UK university applications process as of 2012

January	Students **apply** to up to 5 programs, specific majors at specific universities.[a] Applications include age-16 and age-17 standardized test grades and teacher predictions of age-18 grades.
February–May	Programs make **offers** of places. Offers are contingent on the student meeting conditions on their age-18 grades.
June	Students **rank** up to two programs from which they receive offers; all others are forfeited.[b] Students then take age-18 exams.
August	Age-18 exam results are released, and students **match** to programs. Programs must accept all students who ranked their offer and met the offer conditions, and have discretion to accept students who failed to meet conditions. Students who are not accepted at either ranked program may go unmatched or join an after-market clearing process.[c]

[a] There are approximately 35,000 such programs across 160 different universities.
[b] These are termed the *firm* and *insurance* choice; we call them the first and second choice throughout.
[c] 71% of students who ranked programs are admitted to their first choice, 10% to their second choice, 10% go unmatched, and 9% are admitted to a different program through the clearing process.

2.2 Unconditional offers

In the 2010s, universities increasingly began using *unconditional offers*, a form of test-blind admissions. In contrast to conditional offers, an unconditional offer admits students to a program regardless of their age-18 exam results. Historically, these offers were primarily reserved for students applying to university after completing secondary schooling (such as students re-applying after a gap year), as these students have already completed their exams and their final grades are included in the initial application. In 2014, programs increasingly began to make unconditional offers to applicants who had *not* completed their exams.[6] Programs also began to make *binding unconditional offers*,[7] where the offer had no conditions only if the student ranked it as their first

[4] Tuition fees for domestic students are capped in England, Wales, and Northern Ireland, and essentially all universities charge fees at the cap. In England and Wales, the cap was set at £9,000 per year in 2012, and following irregular below-inflation increases, it currently stands at £9,790. Students have access to standardized government loans with income-contingent repayment plans for tuition and living costs.

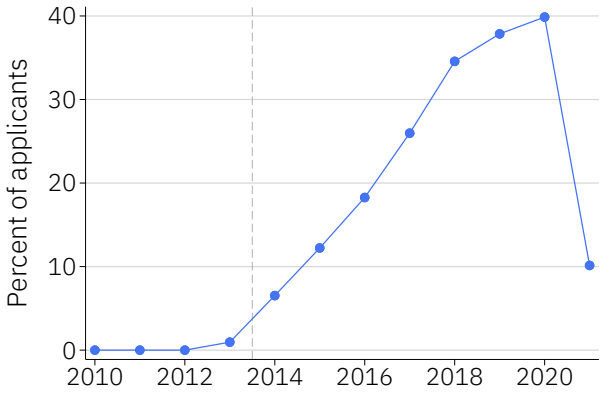
[5] There have been several changes in the university applications process during our analysis window. In 2007, the first year of our data, students could apply to up to 6 programs. This was reduced to 5 in 2008. In the late 2010s, the UK government phased out an intermediate exam taken at age 17 prior to university applications; refer to Tadjfar and Vira (2025) for more details.

[6] One reason behind this change is the lifting of strict enrollment caps. In 2014, enrollment caps were lifted at all universities (except alternative providers: private "for-profit" entities or other independent institutions that receive no UK government funding; they enroll less than 2% of UK undergraduates and are not considered in our results). In 2015, enrollment caps were lifted at all universities including alternative providers.

[7] In the UK, these were referred to as *conditional unconditional offers*, as they were conditional on the student ranking the offer first; we use the term "binding unconditional" for clarity.

choice. We pool together these two types of unconditional offers and refer to them as unconditional offers throughout. Figure 1 shows the rapid growth in unconditional offers from 2010 through 2019 for our analysis sample: in that final year, 38% of applicants received at least one unconditional offer.

Figure 1: Percent of applicants receiving an unconditional offer, 2010–2019.



Reactions to the growth of unconditional offers varied greatly. Critics, including the Education Secretary and the Association of School and College Leaders,[8] argued that offers were primarily a tool for universities to increase enrollment rather than a genuine benefit to students, and that the pressure to commit to a university before taking age-18 exams discouraged academic effort.[9] A particular concern was that binding unconditional offers pressured students to forfeit their higher-quality options, effectively limiting their ability to benefit from strong age-18 performance. Universities defending the practice argued that unconditional offers helped them attract high-achieving students and that anxious students valued a guaranteed admission offer; a 2019 UCAS survey found that 70% of applicants supported unconditional offers, largely because of how they reduced stress.

The widespread usage of unconditional offers ended in 2021 amid various bans. In 2020, amid concerns about pressure on students and disruption caused by the COVID-19 pandemic (which led to the cancellation of age-18 examinations that year), the UK government temporarily banned binding unconditional offers. Following this, in 2021, Universities UK, a group representing 142 universities, introduced a "Fair Admissions Code of Practice." Under this code, universities could no longer make unconditional offers to students who had not yet completed their age-18 examinations, effectively ending the practice for mainstream applicants.[10] Universities UK cited several reasons for this self-imposed ban, most notably the "unfair pressure" created by binding unconditional offers and the finding that unconditional offer takers were "less likely to report feeling motivated to perform well at school" (Universities UK, 2020).

[8] In the UK, "college" refers to secondary education institutions such as sixth form colleges and vocational colleges, equivalent to grades 11 and 12 in the US; it does *not* refer to universities or other tertiary education institutions.

[9] See, for example, coverage in *The Guardian* in 2018 and 2019.

[10] The code of practice also introduced requirements around transparency in admissions criteria and restrictions on the use of other incentive-based admissions practices.

2.3 Data sources

Our analysis draws on the Longitudinal Education Outcomes (LEO) database, produced by the UK Department for Education (DfE) and accessed via the Secure Research Service (SRS) of the Office for National Statistics, which allows us to link three administrative datasets from different data providers using anonymized individual identifiers (Office for National Statistics, 2025, 2026*a*,*b*).[11]

We observe the universe of applications to UK universities between 2007 and 2021 in the Universities and Colleges Admissions Service (UCAS) dataset. The application data includes the program identifiers (including both university and major), whether each program made an offer and the type of offer (conditional or unconditional), the student's ranking of their offers (first and second choices), and whether the student was ultimately admitted. The data also includes teacher-predicted and realized age-18 grades (A-levels, BTEC examinations), and the subjects students take in Years 12 and 13. Programs rely heavily on teacher-predicted age-18 grades to make initial offers, but recognize that these predictions are upward-biased and less predictive of university performance than realized scores (Murphy and Wyness, 2020).

The National Pupil Database (NPD) provides student-level data on educational attainment for students in England completing secondary school between 2009 and 2021. The NPD data provides additional demographic characteristics, age-16 standardized test results (GCSEs), age-17 results (AS-levels), and realized age-18 grades (A-levels, BTEC examinations).

Our research also uses data from the Higher Education Statistics Agency (HESA), which provides student-level data on university enrollment, degree program, degree type, program start and completion dates, and degree classification. The typical undergraduate degree lasts three years, after which students receive a degree classification — a coarse grade analogous to GPA, ranging from first-class honors through 2:1, 2:2, and third-class honors — which plays a significant role in graduate labor market outcomes (Britton et al., 2022). Our HESA sample spans academic years 2006–07 through 2020–21, and we observe outcomes for students matriculating from schools in England, Wales, Northern Ireland, and Scotland.

We restrict our main analysis sample to first-time applicants in the standard UCAS main scheme aged 18 as of August 31 of the application year. We also exclude students who apply only through alternative UCAS routes, or who cannot be linked to HESA data.[12] We limit our analysis to students applying in the 2010–2019 application years. For the reduced-form analysis in Section 3, we further restrict our sample to students with application sets in which no constituent program had used unconditional offers in or before 2013 and every constituent program averaged at least 20 offers per year between 2010 and 2012. Additional details on the construction of variables are provided in Appendix B.

2.4 Targeting and take-up

The rise in unconditional offers is driven by smaller and academically less-selective programs. Table A.1 reports characteristics of programs where more than 5% of offers made were unconditional

[11] Due to the sensitive nature of this data, all analysis was conducted in person in the UK at the Institute for Fiscal Studies or at SRS-designated Saferooms or Safepods.

[12] UCAS programs that do not link to HESA data are generally specialist programs at arts academies or music conservatoires rather than traditional university programs.

in some year between 2014 and 2019. Programs making unconditional offers were 39 percentage points *less likely* to be in the top tercile of university rankings, 11 percentage points less likely to be STEM programs, admitted students in 2010–2012 who scored 0.5SD lower on their age-18 exams, and had lower historical rates of being ranked as a student's first choice.

Unconditional offers were targeted toward high-achieving students but taken by weaker students. Columns (1)–(3) of Table 2 show the mean age-16 and age-18 grades (converted to z-scores) of students who received unconditional offers, compared to those who did not. Students who received unconditional offers were stronger than non-recipients on all academic characteristics; the largest difference occurs in teacher-predicted age-18 scores, where recipients scored 0.40SD higher than non-recipients. However, the difference between takers and decliners among recipients indicates that on the basis of observables, weaker students self-selected into taking unconditional offers.[13] This gap is especially large in the realized age-18 exam scores. As reported in column (6), unconditional offer takers scored 0.28SD lower on their age-18 exams than recipients who declined them; we call this the taker–decliner achievement gap.[14]

Table 2: Mean standardized test scores of unconditional offer recipients and takers

	Receiving an unconditional offer			Take-up, among offered		
	(1) No offer	(2) Received offer	(3) Difference	(4) Decliner	(5) Taker	(6) Difference
Age-16 English (z)	–0.007	0.086	0.093***	0.129	0.076	–0.053***
	(0.003)	(0.006)	(0.007)	(0.013)	(0.006)	(0.015)
Age-16 math (z)	0.022	0.062	0.040***	0.114	0.050	–0.064***
	(0.003)	(0.007)	(0.008)	(0.015)	(0.007)	(0.017)
Teacher-predicted age-18 (z)	–0.299	0.101	0.400***	0.251	0.068	–0.183***
	(0.003)	(0.006)	(0.007)	(0.014)	(0.007)	(0.015)
Achieved age-18 (z)	–0.282	–0.037	0.245***	0.191	–0.089	–0.280***
	(0.003)	(0.007)	(0.008)	(0.014)	(0.007)	(0.016)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Means, with standard errors in parentheses, among students in the first year in which any program in their application set makes unconditional offers. Columns (1)–(3) compare students who receive an unconditional offer with those who do not; columns (4)–(6) compare, among recipients, those who rank an unconditional offer against those who do not. A student ranks an unconditional offer if they rank an unconditional offer first or second, or a binding unconditional first. N is the full regression sample, covering all cohorts and exposure groups, and ranges from 1.33m to 1.63m across rows with the availability of each score.

2.5 Enrolling student composition

Programs targeted unconditional offers toward high-achieving students, but academically weaker students accepted them; we examine how these two forces affected programs' actual enrolling class.

Our analysis focuses on how enrolling class composition changed between the first year of usage and the years before unconditional offer use. Define $Z_{st} \in \{0, 1\}$ as an indicator for whether program s made more than 5% of its offers unconditional in year t, and let t_s^{first} be the first year in

[13] We define take-up as ranking a binding unconditional offer first, or ranking a standard unconditional offer first or second.

[14] The same qualitative pattern holds, with a smaller magnitude, among students exposed in a later year; see Table A.2 in the appendix.

which $Z_{st} = 1$. Table 3 reports coefficients from estimating

$$Y_{st} = \beta_1 Z_{st}^{\text{first}} + \beta_2 Z_{st}^{\text{later}} + \alpha_s + \lambda_t + \varepsilon_{st}, \tag{1}$$

where Y_{st} is an outcome of program s in year t, and α_s and λ_t are program and year fixed effects. The exposure indicator is split in two: $Z_{st}^{\text{first}} = \mathbf{1}\{t = t_s^{\text{first}}\}$, the first year a program makes unconditional offers, and $Z_{st}^{\text{later}} = Z_{st}\,\mathbf{1}\{t > t_s^{\text{first}}\}$, every subsequent year in which it makes them. The sample is program-years between 2010 and 2019, excluding programs that made unconditional offers before 2014 or averaged fewer than 20 offers per year in 2010–2012. The coefficient β_1 effectively compares enrolling class composition in the first year of unconditional offer usage to the pre-unconditional-offer period.

Table 3: Change in program characteristics after adopting unconditional offers

	(1) First year of unconditional offers	(2) Later years of unconditional offers
Panel A. Applications, offers, and enrollment		
Applications	−21.40***	−19.12***
	(2.049)	(1.906)
Offers made	−19.87***	−15.94***
	(1.736)	(1.614)
Enrollment	1.863***	3.856***
	(0.355)	(0.330)
Rate ranked first	0.035***	0.039***
	(0.002)	(0.002)
Offer rate	0.003	0.001
	(0.002)	(0.002)
Panel B. Test scores of the enrolling class (z)		
Age-16 English	0.012**	0.008
	(0.005)	(0.005)
Age-16 math	0.012**	0.010*
	(0.006)	(0.005)
Teacher-predicted age-18	0.070***	0.070***
	(0.006)	(0.006)
Expected age-18	0.055***	0.056***
	(0.005)	(0.004)
Achieved age-18	−0.024***	−0.042***
	(0.006)	(0.006)
Panel C. Degree outcomes of the enrolling class, four years after entry		
Holds a degree	−0.004	0.013
	(0.006)	(0.008)
Left without a degree	0.005	0.001
	(0.003)	(0.004)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each row is a separate regression of the named program-year outcome on indicators for the first year and for later years in which the program makes unconditional offers, with program and year fixed effects; columns (1) and (2) report the two coefficients. Expected age-18 attainment is a forecast of the enrolling class's achieved score from the characteristics observed at the time of application, contrasting with achieved age-18 attainment. N ranges from 45,183 to 46,826 across rows in Panels A and B. Panel C covers application cohorts through 2016 ($N = 30{,}300$, rounded to the nearest 10).

Panel A of Table 3 suggests that programs started making unconditional offers in a "bad year" to offset fewer applications. In the year unconditional offers were first used, programs had 21.4 fewer applications compared to years without unconditional offer usage. Despite receiving fewer applications, these programs saw a statistically significant *increase* in enrollment of 1.9 students while

holding the aggregate offer rate (including both conditional and unconditional offers) constant.

Panel B shows that programs enrolled students with better academic attributes observable at the time of application. The first three rows of Panel B show that in their first year of making unconditional offers, programs enrolled students with 0.01SD higher age-16 English and math scores and 0.07SD higher teacher-predicted grades. Absent any student private information or moral hazard effects, projecting these observables onto realized age-18 test scores leads to expected age-18 scores that are 0.06SD higher, as shown in the fourth row of Panel B.[15]

However, the actual academic performance of these students did not match the expected improvement. The last row of Panel B shows that the realized age-18 test scores of enrolling students were in fact 0.02SD *lower* than in cohorts where programs did not use unconditional offers. This large discrepancy between *expected* age-18 grades and *realized* age-18 grades suggests that students either had substantial private information about how they would fare in the actual exams relative to their teacher-predicted grades and sorted into accepting the unconditional offers accordingly, exerted *less* effort preparing for the exams than they otherwise would have in the absence of an unconditional offer, or both.

Panel C shows that students admitted through unconditional offers were not stronger on long-term metrics of student success. One possible explanation for the results in Panels A and B is that universities selectively chose to ignore results on students' age-18 exams, believing that performance on earlier tests was more valuable for predicting student success. However, this story is not supported by the data: in the first year of making unconditional offers, enrolling students were no more likely to graduate, and no less likely to drop out, than students who were enrolled without unconditional offers. As a result, we conclude that unconditional offers did not allow universities to enroll students with better long-term outcomes.

3 Match and Incentive Effects for Recipients

In this section, we examine how unconditional offers affected recipients of these offers. Section 3.1 discusses incentive effects, while Section 3.2 discusses match effects.

3.1 Incentive Effects

Table 3 poses a puzzle: in the year a program began making unconditional offers, its enrolling class was *stronger* on test scores observable at the time of offer-making than in other years, yet its realized age-18 scores were *lower*. The taker–decliner achievement gap in Table 2 shows this more directly: takers scored 0.28SD worse than decliners on their age-18 exams.

However, the taker–decliner achievement gap alone does not provide proof of incentive effects in our setting. Three components make up the taker–decliner gap: the causal change in achievement when a student takes an unconditional offer, holding observables and unobservables constant (moral hazard, equal to the average treatment effect on treated students); the difference in

[15] We estimate this by regressing actual age-18 grades on teacher-predicted age-18 grades, age-16 grades, and demographic variables using results from in-sample students between 2010 and 2013. The fitted values from this regression are used as the expected age-18 achievement.

students' unobservable characteristics (private information); and the difference in students' observable characteristics (the composition effect). Formally, let T_i indicate take-up by student i, $Y_i(1)$ and $Y_i(0)$ be i's age-18 score with and without take-up, X_i be the characteristics programs observe when making offers, and $\mu_t(x) \equiv \mathbb{E}[Y(0) \mid T = t, X = x]$ be the mean score of students with observed take-up status t and characteristics x if they were to decline an unconditional offer. The taker–decliner achievement gap δ decomposes as[16]

$$\delta \equiv \mathbb{E}[Y \mid T = 1] - \mathbb{E}[Y \mid T = 0] \tag{2}$$

$$= \text{Moral Hazard} + \text{Private Information} + \text{Composition Effect}, \tag{3}$$

where

$$\begin{aligned} \text{Moral Hazard} &\equiv \mathbb{E}[Y(1) - Y(0) \mid T = 1], \\ \text{Private Information} &\equiv \mathbb{E}[\mu_1(X) - \mu_0(X) \mid T = 1], \\ \text{Composition Effect} &\equiv \mathbb{E}[\mu_0(X) \mid T = 1] - \mathbb{E}[\mu_0(X) \mid T = 0]. \end{aligned}$$

Of key interest is the *conditional* taker–decliner achievement gap,

$$\delta_X \equiv \delta - \text{Composition Effect} = \text{Moral Hazard} + \text{Private Information}, \tag{4}$$

which represents the portion of δ that cannot be predicted solely on the basis of observable characteristics; in effect, δ_X is the taker–decliner achievement gap that came as a surprise to programs. We estimate δ_X by estimating the gap in age-18 test scores between takers and decliners with increasingly strict controls for observable characteristics, and report results in Table 4. Conditioning on all characteristics observed at the time of application, the gap between takers and decliners shrinks to -0.139SD. If we "over-condition" by including controls for application sets (unobserved by programs), the gap shrinks to -0.127SD, indistinguishable from the -0.139SD gap with only observed characteristics. This correlation between take-up and outcomes conditional on observables indicates the presence of either private information or moral hazard; in Appendix C, we develop a theoretical model to show this in our setting.[17] As described in Section 2.2, programs and policy-makers attributed the taker–decliner achievement gap to moral hazard.

To estimate the moral hazard effect of unconditional offers, we use quasi-random variation in *exposure* to unconditional offers resulting from the idiosyncratic timing of unconditional offer adoption at the program level. If exposure to unconditional offers were random, then comparing students who are exposed with those who are not could recover the treatment effect of unconditional offers. To account for possible confounds, including differences across years and in students' unobservable characteristics, we estimate a two-way fixed effects specification at the student level which controls for year and a student's application set.[18] Application sets can reflect preferences

[16] By iterated expectations, $\mathbb{E}[\mu_t(X) \mid T = t] = \mathbb{E}[Y(0) \mid T = t]$, so the right-hand sum simplifies to $\mathbb{E}[Y(1) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0]$, which equals δ since $Y = TY(1) + (1 - T)Y(0)$.

[17] See also Chiappori and Salanié (2000), who make this point in the context of insurance markets.

[18] We define application sets as unique combinations of {major × university} pairs applied to, aggregating across different programs within the same major at the same institution.

Table 4: The taker–decliner achievement gap

	$\hat{\delta}$	$\hat{\delta}_X$		
	(1) No controls	(2) + Academic attainment	(3) + Demographics, school	(4) + Application-set FE
Uncond. take-up	−0.280***	−0.133***	−0.139***	−0.127***
	(0.016)	(0.011)	(0.011)	(0.038)
Composition effect		−0.147	−0.141	−0.153
N	1,630,305	1,522,513	1,519,370	308,671

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column regresses the achieved age-18 score on an indicator for take-up — an unconditional offer ranked first or second, or a binding unconditional ranked first — adding, cumulatively, blocks of the characteristics a program observes when it makes the offer. Column (1) estimates the raw taker–decliner achievement gap $\hat{\delta}$; columns (2)–(4) estimate $\hat{\delta}_X$, the gap net of covariates. The composition effect row reports $\hat{\delta} - \hat{\delta}_X$: the part of the gap a program could have predicted at the point of offer. Column (4) additionally absorbs the application set, which programs do not observe. Standard errors are clustered at the application set.

and unobserved private information, and by conditioning on them (and other student characteristics), we can account for differences across students. The resulting treatment effect of unconditional offers corresponds to moral hazard, allowing us to distinguish it from private information.

We now formally describe our two-way fixed effects specification. Let $a(i)$ and $t(i)$ denote the application set and year of student i; for convenience, we write these as a and t. Define $Z_{at} \in \{0, 1\}$ as an indicator for whether any program in application set a made more than 5% of its offers unconditional in year t, and let t_a^{first} be the first year in which set a is exposed. We then estimate the two-way fixed effects specification

$$Y_i = \alpha_{a,P_{at}} + \gamma_1 Z_{at}^{\text{first}} + \gamma_2 Z_{at}^{\text{later}} + \lambda_t + X_i'\beta + \varepsilon_i, \tag{5}$$

where Y_i is the age-18 test score of student i, X_i a vector of academic and demographic controls (age-16 grades, teacher-predicted age-18 grades, socioeconomic quintile, and gender), and λ_t year fixed effects. The term $\alpha_{a,P_{at}}$ contains application set × post-first-exposure fixed effects, where $P_{at} = \mathbf{1}\{t > t_a^{\text{first}}\}$ indicates years after a set's first exposure. The exposure indicator is split into two parts: $Z_{at}^{\text{first}} = \mathbf{1}\{t = t_a^{\text{first}}\}$, the first year an application set is exposed to unconditional offers, and $Z_{at}^{\text{later}} = Z_{at}\,\mathbf{1}\{t > t_a^{\text{first}}\}$, every subsequent year in which the set is exposed. Standard errors are clustered at the application set level.

Our primary coefficient of interest is γ_1, the causal effect of first-year exposure to unconditional offers in one's application set, rather than the later-year coefficient γ_2. The coefficient γ_1 is causal under the assumption of strict exogeneity of treatment for first-year exposure. The core intuition for why this is likely to hold in this setting is that the first year of exposure comes as a "surprise" to students. Programs did not announce that they were using unconditional offers in advance, so students had little ability to predict which programs would introduce unconditional offers in that year. This assumption is consistent with the balance tests in Table A.3; in the year that unconditional offers began, application sets had no changes in observable characteristics in five of the six tests. In contrast, if students preferentially applied to programs that had given unconditional offers in the past, then γ_2 would not capture only the treatment effect of exposure, but also changes in applicant composition.

Figure A.1 shows the growth of both first-year and later exposure. Exposure spreads quickly once it begins: by 2018, 66.8% of students are exposed to unconditional offers through a program in their application set. From 2016 onward, most of those students are in sets exposed in an earlier year rather than newly exposed, and first-year exposure is the weaker treatment: 17.1% of first-year-exposed students receive an unconditional offer, against 31.0% of those exposed in later years.

Estimates from Equation (5) are reported in Table 5, and provide strong evidence against the moral hazard hypothesis and incentive effects. Exposure to unconditional offers has a precisely estimated null effect on students' age-18 test scores. The precision of our estimates allows us to rule out declines of more than 0.010SD in final exam scores.

Table 5: Effect of exposure to unconditional offers

	(1) Receives an offer	(2) Uncond. take-up	(3) Achieved age-18 (z)
First exposure	0.166***	0.143***	0.005
	(0.004)	(0.004)	(0.007)
Constant	0.014***	0.013***	0.259***
	(0.002)	(0.002)	(0.002)
N	368,874	366,288	366,288

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column is a separate regression on an indicator for the first year in which any program in the student's application set makes unconditional offers. Column (1) is *receipt*: whether the student is made an unconditional offer at all. Column (2) is *take-up*: an unconditional offer ranked first or second, or a binding unconditional ranked first — a choice the student makes over the offers they hold. All columns include academic and demographic controls, year fixed effects, and application set × post-first-exposure fixed effects. The sample is years 2010–2019, main scheme applicants, programs averaging at least 20 offers per year over 2010–2012, with no program in the application set using unconditional offers in 2013 or earlier. Standard errors are clustered at the application set, of which there are 97,730. Columns (2) and (3) are $\hat{\pi}$ and $\hat{\rho}$ in Table 6.

A more direct estimate of moral hazard can be derived through a local average treatment effect interpretation of our results. Estimates in Table 5 represent an intent-to-treat effect of exposure on test scores; however, as described in Equation (2), moral hazard is the average treatment effect on the treated. We can derive this quantity by interpreting Table 5 as components of an instrumental variable specification, instrumenting take-up with exposure. Doing so requires the standard instrumental variable assumptions. Relevance holds from column (2) of Table 5, with $F = 1{,}555$. The exclusion restriction requires that exposure move achieved scores only through take-up. This is plausible: decliners would have no reason to change their behavior in response to having had such an offer available given that they declined. Independence follows from the strict exogeneity argument described above. As noted by Bloom (1984), given that unexposed students have close to zero take-up,[19] the IV estimate corresponds to the average treatment effect on the treated — exactly the moral hazard term of interest. Under these assumptions, we find that

$$\text{Moral Hazard} = \mathbb{E}[Y(1) - Y(0) \mid T = 1] = \frac{\rho}{\pi} = 0.033\text{SD}. \tag{6}$$

[19] In Table A.4, we relax this assumption as a robustness check.

We conclude that take-up of unconditional offers does not cause students to achieve lower grades.

We use these estimates to test for and bound the extent of private information held by students. Starting from Equation (4), express private information as the wedge

$$\text{Private Information} \;=\; \delta_X - \frac{\rho}{\pi}, \qquad\qquad w \;\equiv\; \rho - \pi\delta_X \;=\; -\,\pi\,\text{Private Information}. \tag{7}$$

Testing for private information is then equivalent to testing $w > 0$. We report the results of this exercise in Table 6. The wedge is positive and precisely estimated, $\hat{w} = 0.024$ with a bootstrapped standard error of 0.007, rejecting $w \leq 0$: the causal effect of take-up — moral hazard — cannot explain why takers underperform decliners. To bound the share, let $\theta = (\rho/\pi)/\delta_X$ be the share of the conditional taker–decliner achievement gap composed of moral hazard.[20] We can bound θ by finding the smallest θ_0 at which $w(\theta_0) = \rho - \theta_0\pi\delta_X$ is significantly positive using a one-sided t-test. We find that $\theta \leq 0.346$: in other words, at least 65.4% of the taker–decliner gap is due to students' private information.

Table 6: Decomposing the taker–decliner achievement gap

	Estimate	SE	95% CI
Panel A. Estimated inputs			
$\hat{\pi}$, effect of exposure on take-up	0.143	(0.003)	[0.138, 0.151]
$\hat{\rho}$, effect of exposure on the achieved score (z)	0.005	(0.007)	[−0.010, 0.019]
$\hat{\delta}_X$, taker–decliner achievement gap net of covariates (z)	−0.134	(0.011)	[−0.156, −0.114]
Panel B. Test for private information			
$\hat{w} = \hat{\rho} - \hat{\pi}\hat{\delta}_X$	0.024	(0.007)	[0.009, 0.039]
Panel C. Implied split of the gap			
Moral hazard (ATT) $= \hat{\rho}/\hat{\pi}$	0.033	(0.048)	[−0.068, 0.128]
Private information $= \hat{\delta}_X - \hat{\rho}/\hat{\pi}$	−0.167	(0.049)	[−0.260, −0.064]
Moral-hazard share $\theta = (\hat{\rho}/\hat{\pi})/\hat{\delta}_X$	−0.244	(0.365)	≤ 0.346

Table reports decomposition of the taker–decliner achievement gap using the test described in Section 3.1, yielding point estimates and bounds on θ, the share of the gap composed of moral hazard. The gap is estimated on 1,700,457 students, while exposure effects are estimated on 366,288 students in non-singleton application sets. Standard errors are from a cluster bootstrap over application sets with 200 replications. The last column reports two-sided 95% percentile intervals, except in the θ row, where it reports the one-sided 95% upper bound obtained by inverting $w(\theta_0)$: the largest share not rejected by the data. Robustness checks reported in Table A.5.

These results are robust to alternative sample definitions, alternative thresholds for what counts as using unconditional offers, dropping the post-first-exposure block, and clustering at the least-selective university in the application set rather than at the application set, as reported in Table A.5. As shown in Table A.4, the decomposition is also robust to relaxing the assumption of no always-takers. Additionally, we estimate treatment effects of exposure on age-18 test scores using a quantile regression, and report results in Table A.6; because results are insignificant across the ability distribution, we conclude that the null effect in Table 5 is not driven by an increase in age-18 scores for high achievers and a decrease for low achievers, or vice versa.

[20] The estimate of δ_X differs slightly between Table 4 and Table 6; the former is estimated on all unconditional offer recipients, while the latter is estimated only on students in their year of first exposure.

3.2 Match effects

The previous subsection shows that unconditional offers do not impact students' academic achievement, and so there are no incentive effects in our setting. We now turn to match effects, and examine how unconditional offers affected students' matriculation and outcomes. To do so, we use Equation (5) to estimate the causal effect of unconditional offer exposure on students' trajectories through higher education.

We first examine results on university enrollment. Results are shown in panel (a) of Figure 2 and column (1) of Table 7. On net, unconditional offer exposure leads to an increase of 1.1pp in university matriculation, with the increase concentrated at middle-tercile universities. Recall that the effect of exposure on unconditional offer receipt is 16.6pp; again treating these results as an IV, we infer that students receiving an unconditional offer are 6.4pp more likely to matriculate at university.

Exposure also increases the quality of the program students attend. We define a program's average achievement as the average age-18 test scores of students in that program during 2010–2013, before unconditional offers began. We then estimate Equation (5) with the average achievement of the student's program as the dependent variable, restricted to students who enroll in university. Among university enrollees, exposure to unconditional offers leads students to matriculate at programs whose average achievement is 0.008SD higher. One story consistent with this finding is that in the absence of an unconditional offer, a taker would have underachieved on their age-18 test scores, and enrolled at a non-selective program. Relative to that counterfactual, an unconditional offer, even from a middle-tier program, can raise average university quality.

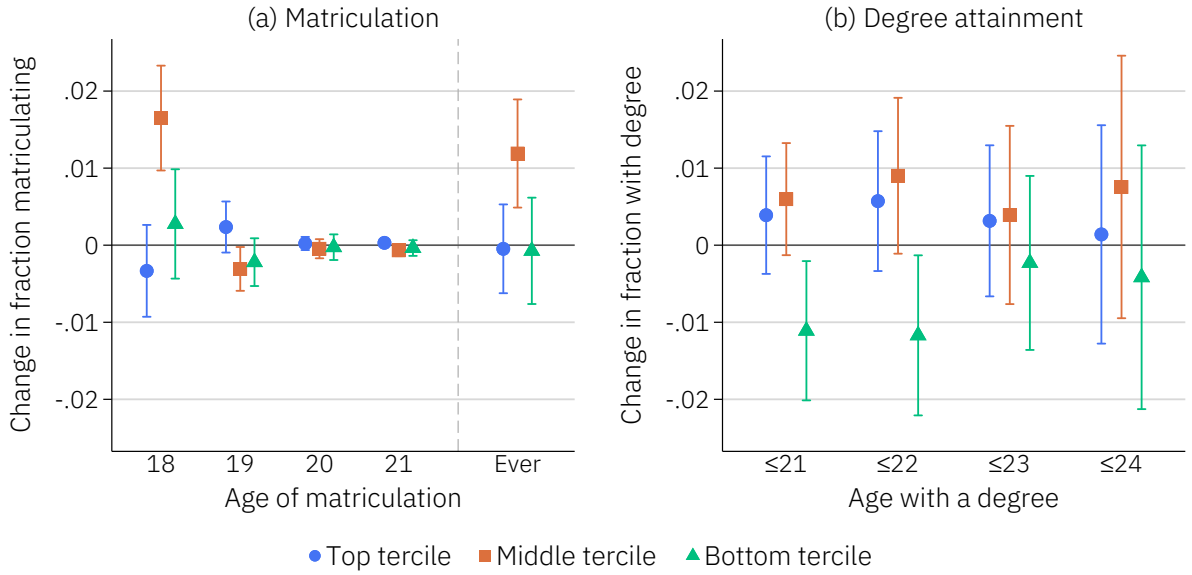
We find smaller effects on students' long-term educational trajectories. We estimate Equation (5) on years of university attended and degree completion by age 24, restricted to students for whom we observe data at age 24. Results are reported in panel (b) of Figure 2 and columns (3) and (4) of Table 7. Students exposed to unconditional offers spend an additional 0.052 years enrolled in university; however, this does not translate into a significant increase in university degree completion. Panel (b) of Figure 2 shows that there is a significant decrease in degree completion at bottom-tercile universities; there is suggestive evidence that this is offset by an increase in degree completion at middle- and top-tercile universities, though estimates are not significant at the 5% level.

Table 7: Effect of unconditional offer exposure on matriculation, program quality, years enrolled, and degree attainment

	(1) Ever matriculates	(2) Program quality (z)	(3) Years enrolled by 24	(4) Degree by 24
First exposure	0.011***	0.008**	0.052*	0.005
	(0.003)	(0.004)	(0.028)	(0.009)
Mean of dep. var.	0.926	0.301	3.044	0.798
N	368,870	296,990	126,810	126,810

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column is a separate estimation of Equation (5) on the outcome in its header, with the controls and fixed effects of Table 5, reporting the coefficient on first-year exposure; standard errors are clustered at the application set. Columns (1) and (4) are probabilities; column (2) is the average achieved z-score of the program a student matriculates at, as measured by average student achievement in 2010–2013, on a sample restricted to university matriculants; column (3) is years enrolled by age 24. N is rounded to the nearest 10.

Figure 2: Effect of unconditional offer exposure on matriculation and degree attainment, by university tercile. (a) Matriculation at ages 18–21, and ever; (b) holding an undergraduate degree by ages 21–24.



4 A Structural Model of Application and Enrollment

Section 3 shows that unconditional offers have small positive effects on students' university matriculation and no effect on their academic achievement. Three questions remain that the reduced form cannot answer: how much students value an offer being unconditional (the unconditional-offer premium), how one student's unconditional offer affects other students, and how the match would change under other admissions systems. We answer them with a structural model of the UK university admissions market, a four-stage game that mirrors the institutional timeline in Table 1. We estimate students' preferences using techniques from Larroucau and Rios (2020) and programs' offer and admission rules with probits, and we model students' private signals as in Kapor (2025).

The reduced-form results shape private information and moral hazard within the model. Given the implied extent of private information, in the model, students receive a private signal of their academic ability when ranking programs. Further, because we find no incentive effects in our setting, our model does not allow students to respond to an unconditional offer by reducing their effort.

4.1 Structural setup

The market has four stages. In Stage 1, each student applies to up to five programs, which form her portfolio A_i. In Stage 2, programs make conditional and unconditional offers, and student i holds the offer set $B_i \subseteq A_i$. In Stage 3, she observes a private signal of her age-18 exam performance and ranks up to two options, $C_i \subseteq B_i \cup \{s_0\}$, where s_0 is the outside option of not enrolling. In Stage 4, she takes her age-18 examinations and enrolls at her highest-ranked program that admits her, or takes the outside option, whose utility is normalized to zero.

We describe the stages in reverse order, as chronologically earlier stages rely on students' ex-

pectations about later stages. Let $\mathcal{I}$ index students and $\mathcal{S}$ index programs.

4.1.1 Stage 4: test results and match

In the final stage, students take their age-18 examinations, which reveal their performance $r_i \sim N(0, \sigma_r^2)$. A conditional offer that a student ranked converts to a place if and only if her augmented score clears the program's final bar,

$$z_i^{\text{offer}\prime}\gamma_s^{\text{admit}} \;+\; \alpha_s r_i \;+\; \mu_i \;\geq\; \bar{\pi}_s^R, \tag{8}$$

where z_i^{offer} collects the characteristics programs see at application (age-16 grades, teacher-predicted age-18 grades, age-18 test composition, Low-SES,[21] and indicators for missing data), γ_s^{admit} are program s's final-stage weights on them, and $\bar{\pi}_s^R$ is its conversion cutoff. The term $\mu_i \sim N(0,1)$ is the part of an application that every program sees and we do not, such as the teacher recommendation letter and the personal statement. An unconditional offer converts with certainty. A binding unconditional offer is unconditional only if ranked first; ranked second, it carries the program's usual condition.

4.1.2 Stage 3: ranking

In Stage 3, students observe a private signal of their age-18 exam performance and rank up to two options, (c_1, c_2), from their offer set B_i and the outside option s_0; offers they do not rank are forfeited. Student i's utility for enrolling at s is

$$U'_{is} \;=\; U_{is} \;+\; z_i'\lambda_i \cdot \mathbf{1}[s \text{ is an unconditional offer}] \;+\; \varepsilon_{is}, \tag{9}$$

where U_{is} is her Stage-1 utility (Section 4.1.4), $\varepsilon_{is} \sim N(0, \sigma_\varepsilon^2)$ is an i.i.d. taste shock, and $z_i'\lambda_i$ is the unconditional-offer premium. The vector z_i contains a constant, the student's standardized teacher-predicted age-18 grades, and an indicator for being a girl. The shock ε_{is} is realized after the offer set is observed and captures what the student learns between applying and ranking, such as impressions from campus visits.

During this stage, students receive a private signal about their academic ability. The signal is $\psi_i \sim N(0, \sigma_\psi^2)$, where $r_i = \psi_i + e_i$ and $e_i \sim N(0, \sigma_r^2 - \sigma_\psi^2)$ is independent of ψ_i. The student observes ψ_i; programs do not. The share σ_ψ^2/σ_r^2 measures how well a student can forecast her own age-18 exam performance.

Students face uncertainty about their actual admission, so when ranking programs they must solve a portfolio problem. Write $a_{is} \equiv \Pr(\text{admit } s \mid \psi_i, B_i)$ for the probability that Equation (8) holds for program s, and $a_{i,c_2\setminus c_1} \equiv \Pr(\text{admit } c_2,\ \text{reject } c_1 \mid \psi_i, B_i)$.[22] The student chooses (c_1, c_2) to maximize the expected value of eventual enrollment,

$$\max_{c_1, c_2 \in B_i \cup \{s_0\}} W_i(c_1, c_2), \qquad W_i(c_1, c_2) = a_{ic_1}\, U'_{ic_1} \;+\; a_{i,c_2\setminus c_1}\, U'_{ic_2}. \tag{10}$$

[21] Low-SES students are those from the two most deprived socioeconomic quintiles.

[22] These probabilities condition on offer sets B_i because the pattern of offers and rejections allows a student to update her beliefs about μ_i given the setup of Stage 2.

We assume that students know the offer process, including every program's weights and cutoffs, and so hold true beliefs about their chances of admission.[23]

4.1.3 Stage 2: offers

In Stage 2, programs decide which applicants receive offers. Program s scores applicant i by

$$\pi_{is} \;=\; z_i^{\text{offer}\prime}\gamma_s \;+\; \mu_i, \tag{11}$$

where z_i^{offer} and μ_i are as in Section 4.1.1, and the offer-stage weights γ_s can differ from γ_s^{admit} because r_i is not yet known. Program s makes an offer if and only if $\pi_{is} \geq \bar{\pi}_s$, a threshold set to fill expected capacity, and the offer is unconditional if and only if $\pi_{is} \geq \bar{\pi}_s^U$, where $\bar{\pi}_s^U > \bar{\pi}_s$, so the highest-scoring offer holders receive unconditional offers. Note that offers are "nested" with respect to μ_i, in the sense that a student with covariates z_i^{offer} faces a cutoff of $\bar{\mu}_{is}$ at each program, and whenever she receives an offer from a program that sets a high $\bar{\mu}_{is}$, she also receives an offer from a program with a lower $\bar{\mu}_{is}$ cutoff.

4.1.4 Stage 1: initial applications, portfolio choice

In Stage 1, each student applies to up to five programs. Student i's indirect utility from enrolling at program s is

$$U_{is} \;=\; \delta_s \;-\; d_{is} \;+\; \beta_i^c\,\text{close}_{is} \;+\; \beta^m m_{is} \;+\; v_{is}, \qquad v_{is} \sim N(0,\sigma_v^2), \tag{12}$$

with v_{is} independent across student–program pairs and $U_{i0} = 0$. The program fixed effects δ_s absorb everything about a program that all students value alike; Table 8 reports their mean as the constant. Distance d_{is} is measured in kilometers from the student's home postcode to the institution, and its coefficient is normalized to -1, so utility is measured in kilometers of avoided travel. The indicator $\text{close}_{is} = \mathbb{1}[d_{is} < 40]$ marks programs the student can attend while living at home, and m_{is} marks programs in her home UK region.[24]

The value of living at home varies across students:

$$\beta_i^c \;=\; \bar{\beta}^c \;+\; \Gamma^c\,\text{Low-SES}_i \;+\; \eta_i^c, \qquad \eta_i^c \sim N\!\left(0,\sigma_\eta^2\right), \tag{13}$$

where Low-SES_i is the Low-SES indicator; Low-SES students are likely to value living at home more.

When applying, a student knows z_i^{offer} but not μ_i, so her chance of an offer from s is

$$p_{is} \;=\; \Pr\!\left[\text{offer from } s \mid z_i^{\text{offer}}\right] \;=\; \Phi\!\left(z_i^{\text{offer}\prime}\gamma_s - \bar{\pi}_s\right). \tag{14}$$

She chooses the portfolio

$$A_i \;=\; \arg\max_{A\subseteq\mathcal{S},\,|A|\leq 5} V(A), \qquad V(A) \;=\; \mathbb{E}[W_i(C_i(B))], \tag{15}$$

where $C_i(B)$ is her optimal ranking given offer set B (Section 4.1.2) and the expectation is over offer

[23] As discussed in Agarwal and Somaini (2018), utility and beliefs about admission probability cannot be jointly estimated, so we pin the latter at its true value.

[24] The 12 International Territorial Level 1 regions, as defined by the Office for National Statistics.

sets.

4.2 Model assumptions

Solving Equation (15) exactly requires evaluating millions of portfolios, each of which involves finding the optimal ranking for every possible offer set. Two assumptions make the portfolio value a simple function of admission probabilities and utilities.

> *Assumption 1 (Portfolio evaluation).* When choosing a portfolio, students value it as though all held offers can be ranked. Further, they do not anticipate the arrival of their private signal ψ_i or the Stage-3 taste shock.

A portfolio is then worth the expected utility of the best program that admits the student, so she need not solve the ranking problem for every offer set, while she still faces the risk that a conditional offer does not convert. A fully rational student would instead reason through every possible offer set, grade outcome, and taste shock.

> *Assumption 2 (Nested admission).* When choosing a portfolio, students treat admission events as nested across programs: a student admitted to a program is also admitted to every program with a higher probability of admitting her.

Offers are nested in the model, because the common holistic component μ_i enters every program's offer score (Equation (11)). Nesting is natural in the UK context, where admission depends heavily on academic grades; in the data, 92.5% of students who receive an offer from a given program receive offers from all less-selective programs they apply to.

> *Implication 1 (Portfolio valuation).* Under Assumption 1 and Assumption 2, index the $K = |A_i| \leq 5$ programs in A_i so that $q_{is_{(1)}} \leq q_{is_{(2)}} \leq \ldots \leq q_{is_{(K)}}$, where q_{is} is the probability that s makes student i an offer and admits her, and set $q_{is_{(0)}} \equiv 0$. Then the expected utility of A_i is
>
> $$V(A_i) \;=\; \sum_{B \subseteq A_i} P_i(B) \max_{b \in B} U_{ib} \;=\; \sum_{k=1}^{K} \left(q_{is_{(k)}} - q_{is_{(k-1)}}\right) \max_{j \geq k} U_{is_{(j)}},$$
>
> where $P_i(B)$ is the probability that the programs admitting student i are exactly those in B.

The first equality follows from Assumption 1. The second follows from Assumption 2: a student admitted to $s_{(k)}$ is also admitted to every $s_{(j)}$ with $j > k$, so with probability $q_{is_{(k)}} - q_{is_{(k-1)}}$ her admitted set is $\{s_{(k)}, \ldots, s_{(K)}\}$. Together they make $V(A_i)$ linear in admission probabilities, and under Implication 2 below the inner maximum is $U_{is_{(k)}}$.

The probability used in Implication 1 is the *total* chance of admission q_{is}. We do this to account for programs with many offers but demanding grade conditions, or vice versa. Because μ_i enters both Equation (11) and Equation (8), receiving an offer and meeting its condition are positively dependent, so q_{is} is not just the product of the probability of an offer and the probability of admission,

given an offer. Instead, it is the joint probability

$$q_{is} \;=\; p_{is}\left[g_{is} + (1 - g_{is})\Pr(\text{admit } s \mid \text{offer from } s)\right],$$

where g_{is} is the probability that an offer from s is unconditional, and the conditional admission probability integrates over μ_i truncated at the offer threshold and over r_i given z_i^{offer}.

Our second implication allows us to infer how a student would rank the programs in her portfolio, even though no ranking is observed. It follows from Implication 1 and portfolio optimality.

> *Implication 2 (No dominated programs).* Under Assumption 1 and Assumption 2, let A_i be an optimal portfolio in which every program strictly raises $V(A_i)$, ordered as in Implication 1 with distinct admission probabilities $q_{is_{(1)}} < \cdots < q_{is_{(K)}}$. Then
>
> $$U_{is_{(1)}} > U_{is_{(2)}} > \cdots > U_{is_{(K)}}.$$
>
> Moreover, if $q_{is} < q_{is'}$ and $U_{is} < U_{is'}$ for some program s', then $s \notin A_i$.

Our final assumption imposes restrictions on student behavior after receiving offers.

> *Assumption 3 (No moral hazard).* Realized performance r_i is independent of B_i, conditional on A_i and z_i^{offer}.

In line with our results from Section 3, we assume that students do not change their effort in response to the offers they hold.[25]

4.3 Estimation procedure

We estimate the model in three blocks, in order: the supply side in Stages 2 and 4 (offer and conversion probabilities), Stage 1 (student utilities, from applications), and Stage 3 (the coefficient λ_i on the unconditional-offer premium and the signal variance σ_ψ^2, from rankings). Each block conditions on the blocks before it; Appendix F.6 discusses the validity of this sequential estimator, together with convergence diagnostics, and Appendix F gives further details on the procedure.

4.3.1 Supply side

We estimate two probabilities separately: receiving an offer, which we observe for every applicant, and converting an offer into a place, which we observe only for students who rank the program, the "at-risk" sample. Both are Firth-penalized probits, which handle the perfect separation at the most- and least-selective programs. The offer probit is fit separately for each program and cohort; in cells where every applicant or no applicant received an offer, we set the offer probability to 0.999 or 0.001. Although μ_i is a common shock, each probit is correctly specified, because μ_i is integrated out of every marginal probability. Let these probabilities be $\hat{p}_{is}$ and $\hat{a}_{is}$, respectively:

$$\hat{a}_{is} \;=\; \Phi\big(z_i^{\text{offer}\prime}\hat{\gamma}_s^{\text{admit}} + \hat{a}_s\hat{r}_i + \hat{c}_s^R\big), \qquad \hat{p}_{is} \;=\; \Phi\big(z_i^{\text{offer}\prime}\hat{\gamma}_s + \hat{c}_s\big),$$

[25] Moral hazard, if present, would make a stronger case for post-qualification admissions in Section 5.

where the fitted intercepts $\hat{c}_s = -\hat{\bar{\pi}}_s$ and $\hat{c}_s^R = -\hat{\bar{\pi}}_s^R$ recover the offer and admission cutoffs of Equation (14) and Equation (8).

Students' Stage-1 admission probability combines the two as in Implication 1:

$$\hat{q}_{is} = \hat{p}_{is}\left[\hat{g}_{is} + (1-\hat{g}_{is})\,\hat{a}_{is}^{o}\right],$$

where $\hat{g}_{is}$ is the probability that an offer from s is unconditional, implied by s's unconditional-offer cutoff, and $\hat{a}_{is}^{o}$ is the conversion probability conditional on holding the offer. Because the offer and conversion indices share μ_i, we compute $\hat{a}_{is}^{o}$ from their bivariate normal distribution, with correlation $1/\sqrt{1+\hat{\alpha}_s^2\sigma_r^2}$, rather than setting it to $\hat{a}_{is}$.

4.3.2 Demand side, Stage 1

Inequalities which are linear in utilities and probabilities allow us to estimate the demand side. Following Larroucau and Rios (2020), an optimal portfolio survives every "one-shot swap" (OSS): replacing one chosen program with an unchosen one does not raise its value,

$$V(A_i; q_i, U_i) \geq V((A_i \setminus \{s\}) \cup \{\ell\};\, q_i, U_i) \quad \text{for all } s \in A_i,\ \ell \notin A_i. \tag{16}$$

For a student who applies to fewer than five programs, adding a program must not raise the value of her portfolio either: $V(A_i; q_i, U_i) \geq V(A_i \cup \{\ell\}; q_i, U_i)$ for all $\ell \notin A_i$. With independent admission chances, these inequalities characterize the optimal portfolio.[26] With correlated admissions, they are necessary but not sufficient conditions for optimality, because rejection by one program is informative about the others; the exact conditions are the dynamic-programming inequalities of Ali and Shorrer (2025). Unfortunately, the Ali–Shorrer inequalities grow combinatorially with the number of programs; as such, we impose the smaller set of OSS inequalities to identify our parameters of interest. Appendix E characterizes the gap between the OSS inequalities and the Ali–Shorrer inequalities, and its implications for our estimates.[27]

The observed portfolio has no closed-form choice probability, so we sample Stage-1 utilities for every student–program combination by Markov chain Monte Carlo. Student utilities must satisfy inequalities $\mathcal{L}_i U_i > 0$, where $\mathcal{L}_i$ consists of three blocks of inequalities: a non-negativity constraint, $U_{is} > 0$ for all $s \in A_i$; the ranking constraint implied by Implication 2; and the one-shot swap inequalities described in Equation (16) (addition inequalities for students who apply to fewer than five programs). We draw from the joint posterior of the Stage-1 parameters and the utilities $\{U_i\}$ by Gibbs sampling; Appendix F.2 gives the blocking, and Appendix F.7 gives the priors and run lengths.

We additionally allow students to have imprecise estimates of admission probabilities. We treat q_{is} as latent and resample it with the utilities, proposing the offer-probit coefficient vector from its asymptotic distribution and accepting the proposal if the observed portfolio still satisfies the OSS (or addition) inequalities. The conversion-probit component of q_{is} is held fixed at its point estimate.

[26] Larroucau and Rios (2020) show that, with independent admission chances, a portfolio that no one-shot swap improves is optimal, so it matches the outcome of the Marginal Improvement Algorithm of Chade and Smith (2006).

[27] A separate interpretation of our methodology is that by using the OSS inequalities, rather than the full dynamic-programming inequalities, our parameters are set-identified, rather than point-identified. In practice, given the standard errors of our estimates, the identified set is a good approximation for point-identification.

To account for the fact that believed admission odds may be incorrectly ordered, we use only the OSS (or addition) constraints and non-negativity constraints to define $\mathcal{L}_i$ when redrawing q.[28]

4.3.3 Demand side, Stage 3

The ranking stage conditions on the Stage-1 estimates of Section 4.3.2, drawing the live-at-home coefficients β_i^c and the program fixed effects δ from their Stage-1 posterior so that Stage-1 uncertainty carries into the Stage-3 estimates. We estimate the Stage-3 parameters $(\sigma_\varepsilon^2, \lambda_i, \sigma_\psi^2)$ by Gibbs sampling subject to the student's revealed ranking: that student i ranked (c_1, c_2) rather than any other pair from B_i implies

$$W_i(c_1, c_2) \;\geq\; W_i(c_1', c_2') \quad \text{for every other pair } (c_1', c_2'), \tag{17}$$

with U_{is}' defined as in Equation (9). There are at most 26 such inequalities. As in Stage 1, we draw U_{is}' for every student–program combination subject to these constraints.

The signal ψ_i is latent. Each sweep draws ψ_i by a Metropolis step given the observed first and second choices and the take-up decision, recomputing the conversion probabilities in Equation (17) under the student's posterior, and then draws σ_ψ by slice sampling under a half-t prior, given $\{\psi_i\}$ and the realized $\{r_i\}$, keeping every observed ranking feasible. The remaining parameter draws are conjugate.

4.3.4 Sample

We use the same sample restrictions as in Section 3, with the additional restriction that students must have applied only to mathematics programs. We impose this additional restriction due to the combinatorial nature of the program choice problem: with over 35,000 programs, there are over 4×10^{20} different portfolios. This restriction makes our problem tractable, as we can constrain our choice set to 59 distinct mathematics programs. The supply-side probits are fit on the full applicant population, for additional precision. The Stage-1 sampler is run for 50,000 iterations (5,000 discarded as burn-in) and the Stage-3 sampler for 250,000 iterations (25,000 discarded); Appendix F.7 gives priors and further implementation detail.

4.4 Identification

Supply-side identification is standard: each per-program probit is identified by the excluded instrument of distance, following Kapor (2025). We focus on the demand side, where the argument follows Larroucau and Rios (2020), Agarwal and Somaini (2020), and Kapor (2025).

We pin the scale of students' preferences by fixing the distance coefficient to -1, so that every parameter is measured in kilometers of avoided travel. Utility's location is pinned by the outside option, normalized to zero.

[28] If the test for accepting a redrawn q also required the ranking constraints, every accepted draw would preserve the estimated order of programs by selectivity, and by Implication 2 each student's utilities over her chosen programs would have to follow that order. We therefore leave the ranking constraints out of the test, so a student's believed order of programs can differ from the estimated one; for accepted draws, all constraints, including the ranking constraints, are rebuilt under the new q.

Two features identify student preferences. First, distance is excluded from programs' utility over students, so it acts as a special regressor (Agarwal and Somaini, 2020), anchoring its units free of supply-side contamination. Second, the remaining utility parameters are identified from the OSS inequalities (Larroucau and Rios, 2020); here, identification comes from year-to-year variation in cutoffs and grade requirements, which "trace out" regions of utility.

Private information is identified from differences in Stage-3 behavior among students holding the same menu of offers with the same observable characteristics. Three margins in the data identify σ_ψ^2: the take-up rate of unconditional offers, the taker–decliner achievement gap, and the ranking choices observed by all students. In essence, the covariance between students' realized grades and their choice of ranking offers informs us about private information. As an example, if students who rank more-selective programs also outperform their expected age-18 grades, we infer that they possess private information about their academic ability.

The unconditional-offer premium is identified by variation in the availability of unconditional offers that is unrelated to student preferences. As in Section 3, the supply-side rollout of unconditional offers is plausibly exogenous, so comparing the rankings of exposed and unexposed cohorts identifies the coefficient λ_i.

4.5 Results

Table 8: Structural parameter estimates

Preferences over programs	Estimate	SD	*Unconditional-offer premium* λ_i	Estimate	SD
Distance (per km)	−1	(anchor)	At mean predicted ability	11.9	20.4
Can live at home	−40.9	3.0	Gradient, per SD of score	−118.1	18.0
× Low-SES Γ^c	173.2	5.4	Female	106.2	30.9
SD across students σ_η	259.1	2.3			
Same region	71.2	1.9	*Ranking taste shock*		
Constant	−130.3	2.2	SD σ_ε	482.0	4.1
Idiosyncratic shock σ_ν	291.7	1.1	*Information*		
			Private-signal variance σ_ψ^2	0.221	0.006
			Variance of age-18 scores σ_r^2	0.331	—
			Share privately observed	0.67	—

Posterior means and posterior standard deviations from the Stage-1 and Stage-3 samplers of Section 4.3.2 and Section 4.3.3, for students applying only to mathematics programs. The distance coefficient is normalized to -1, so utility parameters are in kilometers. Only live-at-home carries a random coefficient and a Low-SES interaction. σ_η, σ_ν, σ_ε are square roots of sampled variances, with delta-method SDs; the posterior SD is recovered as $\mathrm{MCSE} \times \sqrt{\mathrm{ESS}}$. Rows without an SD are fixed or derived quantities for which the sampler reports no posterior standard deviation. Convergence statistics are reported in Table A.7; notably, the convergence diagnostics for σ_ψ^2 indicate that its SD is likely understated.

Table 8 reports estimates from our structural model. Because the distance coefficient is normalized to -1, every parameter is a willingness to travel, in kilometers.

High-SES students prefer to live away from home: their mean value of living at home is -41km. This mean masks heterogeneity, both idiosyncratic and related to SES: the standard deviation of η_i^c is 259km, and Low-SES students value living at home at 132km on average. Students also prefer a university in their home region of the UK, valuing region match at 71km.

The unconditional-offer premium falls steeply with predicted ability and is larger for girls. At mean predicted ability, a boy values an unconditional offer at 12km (posterior SD 20.4km), and a

girl values it at 118km. The premium falls by 118km per standard deviation of predicted ability, so it is negative for boys more than 0.10SD above mean predicted ability and for girls more than 1.0SD above it.

Although the unconditional-offer premium is small for the average student, private information still shapes who takes one: students with low signals preferentially take unconditional offers. We estimate that age-18 achievement has a variance of $\sigma_r^2 = 0.331$ and that students privately observe two-thirds of it ($\sigma_\psi^2/\sigma_r^2 = 0.67$). This aligns with Section 3: because students hold much private information about their own ability, they can self-select into unconditional offers on the basis of their private signals.

5 General Equilibrium Effects and Policy Implications

Results from our structural model allow us to examine the general equilibrium effects of unconditional offers and evaluate alternative policy designs. We focus on three key questions. First, how do unconditional offers affect the welfare of students who declined or never received an unconditional offer? Second, do unconditional offers differentially affect students with different characteristics? Third, how is the student-program match affected by more information about students' ability and preferences, when given to students, programs, or both?

To answer these questions, we focus on three designs of the UK admissions system. In the first (a), there is a full ban on unconditional offers, as in 2010–2013. In the second (b), unconditional-offer shares are held at each program's 2018 level, the year with the most unconditional offers in this sample. Under this design, students placed on an unconditional offer receive the unconditional-offer premium. Finally, under post-qualification admissions (c), students apply in a single-stage process *after* age-18 results are released, so that r_i is common knowledge to both students and programs. We simulate each design in 200 markets, drawing from the posterior of the structural parameters. Appendix F.8 provides additional implementation details.

Post-qualification admissions are of particular interest for two reasons. First, they are the only design in which students' private signals play no role in the admissions process. Second, post-qualification admissions were proposed as a policy change in 2022, but were not implemented due to concerns about a compressed application timeline. Our simulations allow us to evaluate the potential benefits of this policy, and to identify the mechanisms through which it affects students' welfare.

We report welfare in kilometers of travel, assigning the outside option (utility zero) to unplaced students; the tables report levels under each design, and the text reports differences from the ban, with percentages relative to each group's welfare under the ban.

Table 9 reports welfare (Panel A, in kilometers) and placement rates (Panel B, in percent) under each of the three designs split along three dimensions. First, we report the overall welfare for all students. Next, we split estimates by students' unconditional-offer status in simulation (b): whether they take an unconditional offer, decline one, or do not receive one. Finally, we split estimates by students' quartile of academic attainment on their age-18 exams. In Panel A, we also report student utility under design (b) but excluding the unconditional-offer premium as (b′) to isolate changes to match quality from the unconditional-offer premium.

Table 9: Student welfare under counterfactual application systems

		By uncond. offer under 2018 shares			By achieved attainment			
Design	All	Taker	Decliner	No uncond.	Q1	Q2	Q3	Q4
Panel A. Welfare (km)								
(a) Ban on unconditional offers	1,319	1,059	1,676	1,268	645	1,156	1,612	1,865
(b) Unconditional shares of 2018	1,313	1,260	1,666	1,252	657	1,144	1,598	1,853
(b′) Excluding the premium	1,309	1,149	1,666	1,252	640	1,142	1,599	1,854
(c) Post-qualification admissions	1,506	1,241	1,845	1,458	654	1,362	1,866	2,143
Panel B. Placement rate (%)								
(a) Ban on unconditional offers	82.0	83.0	93.3	80.0	58.0	79.8	92.7	97.7
(b) Unconditional shares of 2018	82.1	100.0	92.8	79.3	58.6	79.5	92.5	97.6
(c) Post-qualification admissions	81.9	77.7	93.7	80.0	54.3	80.4	94.5	98.4
Students	31,406	1,224	4,614	25,567	7,852	7,851	7,851	7,852

Students applying only to mathematics programs. Each row reports means under that design over 200 simulated markets. Utility is in kilometers of travel; the placement rate is the percent of students in the column placed at a program; unplaced students have utility 0. Under 2018 shares, takers hold an unconditional offer and are placed on it, decliners hold one and are not placed on it, and the remaining students hold none; each student keeps that group in every design. Q1–Q4 are quartiles of achieved age-18 attainment (Q4 highest). Welfare is match utility plus the unconditional-offer premium $z_i'\lambda_i$ for students placed on an unconditional offer; "Excluding the premium" reports match utility alone. Under post-qualification admissions there are no unconditional offers, so welfare equals match utility.

In our policy simulations, unconditional offers benefit students who take them and low-achieving students through the unconditional-offer premium, but reduce average welfare because of their negative effects on other students. Students who take unconditional offers gain 19.0% (201.3km), while those who decline one lose 0.6% (9.9km) and those who never receive one lose 1.2% (15.7km). Most of the benefits to takers come from the unconditional-offer premium: without it (b′), takers only gain 8.5% (89.8km).

However, when splitting by achieved attainment, only the lowest quartile gains; the second, third, and top quartiles lose 1.1% (12.4km), 0.9% (13.7km), and 0.6% (11.6km). This is consistent with a private information mechanism, in which low-achieving students are more likely to choose to take unconditional offers than high-achieving students because they expect to miss any grade conditions on conditional offers. In aggregate, unconditional offers reduce welfare by 0.5% (6.4km).

Post-qualification admissions, in contrast, raise welfare for every subgroup relative to a ban, mostly through better matches rather than more placements. The overall placement rate is essentially unchanged (82.0% under the ban, 81.9% under post-qualification admissions). On net, students' welfare increases by an average of 14.2% (187.0km). The top quartile's placement rate does not change, but these students attend programs they prefer.[29] The middle two quartiles gain in both placement and match quality: the placement rates of the second and third quartiles rise by 0.6pp and 1.9pp, respectively, and their average utilities rise by 17.8% (205.8km) and 15.8% (254.4km). The lowest quartile gains slightly in welfare, 1.5% (9.5km), but loses places: its placement rate falls by 3.7pp, so the gain comes from the students who remain placed attending programs they prefer.

[29] Students may not have even applied to these programs at baseline, as they had not yet received their Stage-3 taste shock. Under post-qualification admissions, students' behavior can reflect these preferences.

Table 10: Decomposition of welfare gains from post-qualification admissions

Panel A. Welfare, by design

	No new information	S	P	S and P
Two stages	1,319	1,356 (+2.8%)	1,364 (+3.4%)	1,440 (+9.2%)
One stage (T)	1,041 (–21.1%)	1,402 (+6.3%)	1,105 (–16.3%)	1,506 (+14.2%)

Panel B. Contribution of each change, averaged over orderings

S	P	T	Total
+219 (+16.6%)	+74 (+5.6%)	–106 (–8.1%)	+187 (+14.2%)

Mean welfare of all students applying only to mathematics programs under each combination of three changes to the ban, with the percent change relative to the ban in parentheses. The top-left cell is the ban (a); the bottom-right cell, which combines S, P, and T, is post-qualification admissions (c). S, P, and T are student information, program information, and timing, as described in the text. Panel B reports each change's Shapley value: its marginal effect averaged over the six orders in which the three changes can be added, so the three values sum to the total up to rounding. Table A.8 reports the decomposition by student group. Means over 200 simulated markets. Utility is in kilometers of travel; unplaced students have utility 0.

Post-qualification admissions change three features of the ban at once, and through additional simulations, we can determine the marginal value of each feature. First, under post-qualification admissions, students know their age-18 results r_i and their Stage-3 taste shocks ε_{is} when they apply (S). Second, programs make offers on achieved rather than predicted scores (P). Third, students apply once, at a single date, and are matched by deferred acceptance (T).[30] To measure what each change contributes, we simulate all eight combinations of the three. Panel A of Table 10 reports the total welfare and percent change relative to a ban for each combination of features. The second row adds the single-stage feature (T), while the columns add students' information (S), programs' information (P), or both, so the ban sits at the top left and post-qualification admissions (c) at the bottom right; Appendix F.8 defines each design in more detail.

Each change on its own has little or negative impact on students. Relative to the ban, students' information (S) raises welfare by 2.8% (36.6km), and programs' information (P) raises it by 3.4% (44.5km). A single stage on its own (T), in which programs rank applicants on predicted scores and students rank programs before they learn their taste shocks, lowers welfare by 21.1% (278.7km).

However, the changes for each feature are complementary: each raises welfare more (or decreases it less) when another is already in place. For example, even though student information and program information raise welfare relative to a ban by 2.8% and 3.4% on their own, when combined, welfare is raised by 9.2% (120.8km). Because of these complementarities, a change's contribution depends on the order in which the changes are added, so we calculate the *Shapley value* of each change, measuring its average contribution to welfare across all six orders in which the three changes (S, P, T) can be added. Averaged across orders, students' information (S) in-

[30] The single-letter abbreviations stand for Student information, Program information, and Timing.

creases welfare by 16.6% (219.0km) and programs' information (P) increases it by 5.6% (74.3km), while the single stage (T) subtracts 8.1% (106.3km); together they sum to the total gain of 14.2% (187.0km). Student information is thus the primary driver of the gain from post-qualification admissions. We additionally note that the single-stage admission process increases welfare if and only if it is also paired with student information.

This decomposition implies that the majority of the welfare gain from post-qualification admissions comes from better information. Student welfare can be increased by 9.2% with changes to student and program information sets, which is about two-thirds of the total gain that a full move to post-qualification admissions creates (+14.2%). Implementing the two information channels can take the form of two distinct policy levers. First, students and programs must learn more about student ability; this could occur through an earlier release of age-18 exam results, or through accurate teacher-predicted grades on age-18 exams. Second, students need to apply with better information about their preferences over programs, information which could be conveyed through guidance counselors or university fairs. These two levers are complementary, and are most effective when implemented together. The change to a single-stage mechanism can further increase student welfare by 5.0pp (66.2km). However, this change is more disruptive to the application system; moreover, the mechanism change is only welfare-improving if the two information features are implemented first. Different application systems can improve student welfare, but doing so requires students and programs to have more information about student ability.

In the appendix, we report additional results on the effects of post-qualification admissions on student welfare and program enrollment. Table A.8 splits welfare gains by academic quartile and unconditional-offer status, with benefits largely accruing to the top three quartiles. Table A.9 reports welfare by sex and Low-SES status; changes are similar across these groups, and post-qualification admissions raise welfare by 13.9% to 14.8% in each. Tables A.10 and A.11 report program enrollment and the achievement of entering students by program selectivity. Post-qualification admissions barely change the total enrollment for each selectivity quartile, but do change who enrolls: the mean achievement of entering students rises by 0.280SD at second-quartile programs and falls by 0.055SD at the most selective quartile.

6 Conclusion

The design of admissions systems matters. In the context of UK university admissions, we find that unconditional offers have no incentive effects and positive match effects for targeted students. However, through a model to evaluate the match effects for non-targeted students, we find that unconditional offers decrease total student welfare and transfer utility from high-achieving students to low-achieving students. Post-qualification admissions can improve student welfare by 14.2%, with benefits accruing to higher-achieving students. Most of this gain comes from students knowing their results and preferences when they apply.

This paper suggests that debates around the UK university application process should be reopened. The UK Department for Education considered moving to post-qualification admissions in 2022, but ultimately decided against it over many concerns, particularly the compressed timeline that universities would need to admit students. This paper provide empirical benefits of a system-

atic change, and shows that it can provide large benefits to students. Our decomposition also suggests that much of the gain may be available without the compressed timeline: in the model, giving students and programs better information within the current two-stage system captures about two-thirds of it.

Our results suggest multiple avenues for future research. In this paper, we take universities' behavior as given; what competitive forces lead to the growth of unconditional offers? Descriptively, programs are more likely to use unconditional offers when their competitors did so in the prior year. The dynamics of competition between universities are important to study, and development and estimation of dynamic best-response models which incorporate private information can shed light on these issues.

Finally, private information about ability is necessary for the strong selection into take-up that is present in our setting. A recent literature suggests that parents often have inaccurate beliefs about children's ability (Dizon-Ross, 2019; Wang, Cheng and Smyth, 2022) — so where does students' private information come from? Perhaps there is a wedge between parent and child perceptions of ability, or perhaps the structure of secondary education in the UK has allowed students to gain stronger signals of their own performance. Identifying reasons behind and ways to close this gap could help improve the efficiency of familial education investments.

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A Supplementary Information

Table A.1: Characteristics of programs that adopt unconditional offers

Panel A. Program types

	(1) Top tercile	(2) Middle tercile	(3) Bottom tercile	(4) STEM	(5) Arts	(6) Enrollment
Uses unconditional offers	–0.389***	0.082***	0.307***	–0.112***	0.035***	–1.927**
	(0.012)	(0.013)	(0.013)	(0.013)	(0.011)	(0.975)
Constant	0.477***	0.271***	0.252***	0.398***	0.174***	32.19***
	(0.010)	(0.009)	(0.009)	(0.010)	(0.008)	(0.745)

Panel B. Baseline program selectivity, enrolling class characteristics

	(7) Offer rate	(8) Rate ranked first	(9) Age-16 English (z)	(10) Age-16 math (z)	(11) Teacher-predicted age-18 (z)	(12) Achieved age-18 (z)
Uses unconditional offers	0.092***	–0.065***	–0.355***	–0.457***	–0.538***	–0.530***
	(0.006)	(0.004)	(0.013)	(0.015)	(0.016)	(0.016)
Constant	0.758***	0.297***	0.298***	0.413***	0.038***	0.173***
	(0.005)	(0.003)	(0.011)	(0.012)	(0.014)	(0.013)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column is a separate regression of the named program characteristic, measured in the baseline year, on an indicator for whether more than 5% of a program's offers are unconditional. Observations are programs; the sample is programs averaging at least 20 offers per year over 2010–2012 that did not use unconditional offers in 2013 or earlier. $N = 4{,}990$, except in columns (9)–(12), where it is 4,976. Standard errors in parentheses.

Table A.2: Characteristics of unconditional offer recipients and takers, by timing of exposure

	Receiving an unconditional offer			Take-up, among offered		
	(1) No offer	(2) Received offer	(3) Difference	(4) Decliner	(5) Taker	(6) Difference
Panel A. First-year exposure						
Age-16 English (z)	−0.007	0.086	0.093***	0.129	0.076	−0.053***
	(0.003)	(0.006)	(0.007)	(0.013)	(0.006)	(0.015)
Age-16 math (z)	0.022	0.062	0.040***	0.114	0.050	−0.064***
	(0.003)	(0.007)	(0.008)	(0.015)	(0.007)	(0.017)
Teacher-predicted age-18 (z)	−0.299	0.101	0.400***	0.251	0.068	−0.183***
	(0.003)	(0.006)	(0.007)	(0.014)	(0.007)	(0.015)
Achieved age-18 (z)	−0.282	−0.037	0.245***	0.191	−0.089	−0.280***
	(0.003)	(0.007)	(0.008)	(0.014)	(0.007)	(0.016)
Panel B. Later-year exposure						
Age-16 English (z)	−0.083	−0.113	−0.030***	−0.091	−0.119	−0.028***
	(0.002)	(0.003)	(0.004)	(0.007)	(0.004)	(0.006)
Age-16 math (z)	−0.106	−0.206	−0.100***	−0.175	−0.213	−0.039***
	(0.003)	(0.003)	(0.005)	(0.008)	(0.004)	(0.007)
Teacher-predicted age-18 (z)	−0.442	−0.227	0.215***	−0.151	−0.243	−0.092***
	(0.003)	(0.003)	(0.005)	(0.008)	(0.004)	(0.007)
Achieved age-18 (z)	−0.395	−0.331	0.064***	−0.154	−0.368	−0.214***
	(0.003)	(0.003)	(0.005)	(0.008)	(0.005)	(0.007)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Columns are defined as in Table 2. Panel A repeats the first-exposure figures; Panel B reports the same quantities for students in a year after the first in which any program in their application set makes unconditional offers.

Table A.3: Balance of student characteristics at first exposure to unconditional offers

	(1) Age-16 English (z)	(2) Age-16 math (z)	(3) Independent school	(4) Female	(5) Low-SES	(6) Teacher-predicted age-18 (z)
First year exposed	0.001	0.007	0.041	−0.001	−0.002	0.022***
	(0.007)	(0.007)	(0.026)	(0.005)	(0.004)	(0.008)
Later years exposed	−0.007	0.014	−0.044	−0.004	0.011	0.017
	(0.014)	(0.016)	(0.055)	(0.008)	(0.010)	(0.017)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column re-estimates Equation (5) with a student characteristic measured before application as the outcome, and without the academic and demographic controls. Indicators for missing age-16 and teacher-predicted scores are estimated alongside and not reported. Observations are students; the sample is years 2010–2019, main-scheme applicants whose application sets contain no program that used unconditional offers in 2013 or earlier and whose programs each average at least 20 offers per year over 2010–2012. $N = 370{,}973$, except in column (5), where it is 368,907. Standard errors are clustered at the application set level.

Figure A.1: Time series of exposure to unconditional offers. (a) Share of students in each exposure group over time; (b) share of students with an unconditional offer, split by exposure group.

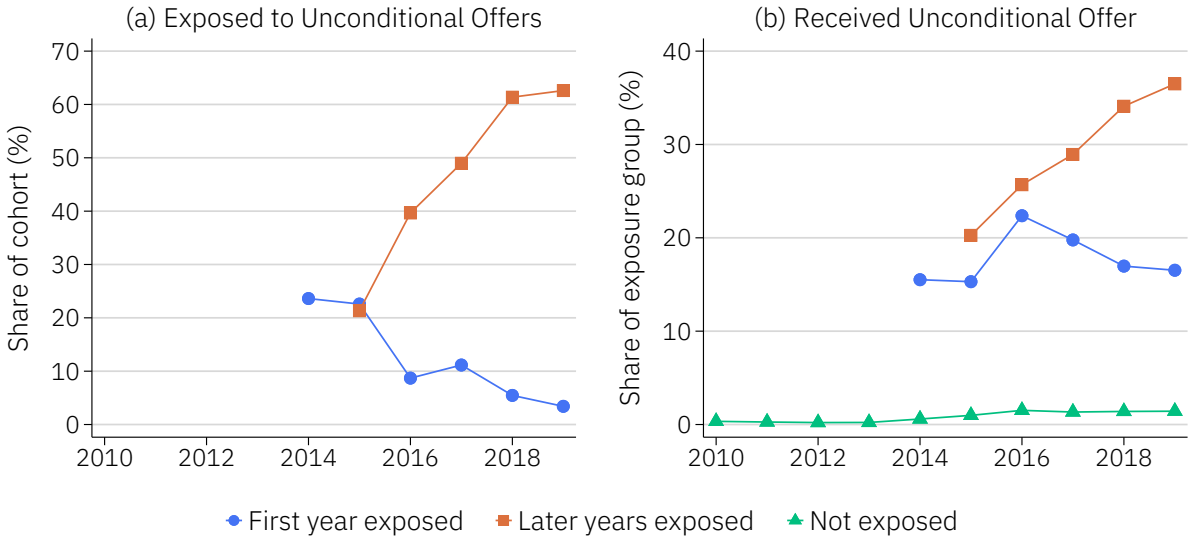


Table A.4: Always-taker robustness of the decomposition

Assumed always-taker effect	0 SD	−0.05 SD	−0.10 SD	−0.15 SD
ATT	0.030	0.026	0.022	0.018
	(0.045)	(0.045)	(0.045)	(0.045)
PI	−0.164	−0.160	−0.156	−0.152
	(0.046)	(0.046)	(0.046)	(0.046)
$\hat{\theta}$	−0.224	−0.193	−0.162	−0.131
	(0.332)	(0.332)	(0.332)	(0.332)
θ cap	0.324	0.355	0.386	0.417

Relaxes the assumption in Section 3.1 that there are no always-takers, and recomputes bounds on the share of the conditional gap which is composed of moral hazard. Table 5, column (2) puts this share at $\hat{p}_A = 0.0129$ (SE 0.00201), against a complier share of $\hat{\pi} = 0.143$; write $\xi_C = \hat{\pi}/(\hat{\pi} + \hat{p}_A) = 0.917$ and $\xi_A = \hat{p}_A/(\hat{\pi} + \hat{p}_A) = 0.083$ for the resulting weights on compliers and always-takers among the treated. Each column assumes always-takers have the stated moral-hazard effect (in place of the complier LATE of Table 6) and recomputes ATT $= \xi_A(\text{assumed effect}) + \xi_C(\hat{\rho}/\hat{\pi})$, PI $= \hat{\delta}_X -$ ATT, and $\theta =$ ATT$/\hat{\delta}_X$. Rather than treat the assumed effect as known with certainty, we give it a standard error scaled for the always-takers' smaller population. Standard errors on ATT, PI, and θ propagate this alongside the analytic clustered standard errors of $\hat{\pi}$, $\hat{\rho}$, and $\hat{\delta}_X$, under an assumption of independence between all four, extending the approach of column (1) of Table A.5. θ cap is the one-sided 95% upper bound on θ, obtained the same way as in Table A.5 by inverting $w(\theta_0)$ on the resulting blended estimate. $\hat{\pi}$, $\hat{\rho}$, $\hat{\delta}_X$, $\hat{w}$, and its t-statistic are unaffected by the assumed always-taker effect and are omitted; see Table 6 or Table A.5's Main column for their values. Sample includes 1,700,457 students for the gap, 366,288 for the exposure effects, and 97,730 application-set clusters.

Table A.5: Robustness of the decomposition

	Main	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\pi}$	0.143	0.143	0.177	0.141	0.146	0.144	0.144	0.149	0.145
$\hat{\rho}$	0.005	0.005	0.001	0.004	0.005	0.004	0.003	0.002	0.004
$\hat{\delta}_X$	−0.134	−0.134	−0.122	−0.128	−0.133	−0.138	−0.139	−0.134	−0.136
$\hat{w}$	0.024	0.024	0.022	0.022	0.025	0.024	0.023	0.022	0.024
	(0.007)	(0.007)	(0.008)	(0.009)	(0.007)	(0.007)	(0.007)	(0.008)	(0.007)
t	3.43	3.50	2.83	2.29	3.70	3.52	3.27	2.70	3.43
ATT	0.033	0.033	0.003	0.026	0.036	0.026	0.019	0.012	0.026
PI	−0.167	−0.167	−0.125	−0.154	−0.169	−0.165	−0.159	−0.146	−0.162
$\hat{\theta}$	−0.244	−0.244	−0.023	−0.205	−0.271	−0.191	−0.140	−0.088	−0.194
θ cap	0.346	0.327	0.561	0.650	0.281	0.352	0.420	0.563	0.365
Clusters	97,730	97,730	97,004	59,709	103,379	100,998	93,898	5,007	81,892

$\hat{\pi}$ is the first stage, $\hat{\rho}$ the reduced form and $\hat{\delta}_X$ the taker–decliner achievement gap; $\hat{w} = \hat{\rho} - \hat{\pi}\hat{\delta}_X$, with its standard error in parentheses and t its t-statistic; ATT $= \hat{\rho}/\hat{\pi}$ is the moral-hazard effect and PI $= \hat{\delta}_X - \hat{\rho}/\hat{\pi}$ private information; $\hat{\theta}$ is the moral-hazard share of the gap and θ cap its one-sided 95% upper bound, obtained by inverting $w(\theta_0)$ in every column. *Main* is the specification of Table 6: a 5% threshold for a program counting as using unconditional offers, programs averaging at least 20 offers per year over 2010–2012, clustering on the application set, and standard errors from a 200-replication cluster bootstrap. The following columns report a variety of robustness checks, changing elements of our primary specification. (1) replaces the bootstrap with standard errors propagated linearly from the three clustered inputs under an assumption of independence. (2) raises the threshold for using unconditional offers from 5% to 10%. (3) restricts to 2010–2015. (4), (5), and (6) set the baseline offer requirement to a mean of at least 0, 10, and 30 offers per year rather than 20. (7) clusters on the least-selective university in the application set rather than on the set itself, which is the only column that speaks to correlation across sets that share a university. (8) drops observations after a set's first year of exposure, and is equivalent to ATT(g, g) from Callaway and Sant'Anna (2021). Columns (2)–(8) all assume independence, as in (1); the gap in standard errors between (1) and Main shows the implication of this assumption.

Table A.6: Treatment effects across the distribution of achieved attainment

Quantile	(1) Smoothing $(-.01, .01)$	(2) Smoothing $(-.05, .05)$	(3) Smoothing $(-.1, .1)$
0.1	–0.005	–0.005	–0.023
	(0.019)	(0.018)	(0.020)
0.2	–0.008	–0.026*	–0.029**
	(0.015)	(0.015)	(0.014)
0.3	–0.008	–0.003	–0.029**
	(0.014)	(0.014)	(0.013)
0.4	0.003	–0.006	–0.016
	(0.015)	(0.014)	(0.014)
0.5	0.014	0.018	0.003
	(0.014)	(0.014)	(0.014)
0.6	0.007	0.017	0.020
	(0.016)	(0.016)	(0.016)
0.7	0.005	0.015	0.026
	(0.016)	(0.017)	(0.016)
0.8	0.041**	0.029*	0.055***
	(0.018)	(0.017)	(0.018)
0.9	0.039**	0.027	0.018
	(0.018)	(0.018)	(0.019)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column reports the coefficient on first exposure from a quantile regression of standardized achieved age-18 attainment on the same controls and fixed effects as Table 6's $\hat{\rho}$ (year fixed effects, application-set × post-first-exposure fixed effects, academic and demographic controls), estimated separately at each decile 0.1–0.9, with standard errors in parentheses. Columns differ in the smoothing bandwidth of the smoothed quantile regression, given in each column header. $N = 368{,}874$.

Table A.7: Convergence diagnostics for the structural estimates

	ESS	$\hat{R}$	Geweke
Preference stage — 50,000 draws			
Constant	107	1.046	−2.13
Can live at home	2,331	1.002	1.64
× Low-SES	2,042	1.002	0.97
σ^2_η	384	1.006	−0.57
Same region	368	1.002	0.18
σ^2_ν	118	1.007	0.53
Belief acceptance		0.630	
Converged		5 of 6	
Signal stage — 250,000 draws			
λ_i at mean ability	52,510	1.000	−0.17
Gradient, per SD	49,711	1.000	0.26
Female	54,743	1.000	0.66
σ^2_ε	5,781	1.000	−0.19
σ^2_ψ	73	1.026	4.11
Belief acceptance		0.938	
Converged		4 of 5	

Diagnostics for Table 8. A parameter counts as converged when split-$\hat{R} \leq 1.01$ and the effective sample size is at least 20; the sampler separately flags $|\text{Geweke}| > 2$. In the preference stage only the constant fails, and it is weakly identified by construction. In the signal stage only σ^2_ψ fails, and its posterior standard deviation in Table 8 is likely understated.

Table A.8: Decomposition of welfare gains from post-qualification admissions, by student group

		By uncond. offer under 2018 shares			By achieved attainment			
Design	All	Taker	Decliner	No uncond.	Q1	Q2	Q3	Q4
Panel A. Welfare (km)								
(a) Ban on unconditional offers	1,319	1,059	1,676	1,268	645	1,156	1,612	1,865
(S) Students know r_i, ε_{is} at application	1,356	1,194	1,745	1,294	653	1,166	1,659	1,946
(P) Programs offer on achieved scores	1,364	1,098	1,683	1,319	591	1,227	1,693	1,945
(T) One stage, students rank before ε_{is}	1,041	942	1,329	994	399	710	1,360	1,694
(SP) S and P	1,440	1,163	1,788	1,391	590	1,277	1,798	2,095
(ST) S and T	1,402	1,278	1,745	1,347	592	1,061	1,829	2,126
(PT) P and T	1,105	856	1,397	1,064	410	933	1,388	1,688
(c) Post-qualification admissions (SPT)	1,506	1,241	1,845	1,458	654	1,362	1,866	2,143
Panel B. Contribution of each change, averaged over orderings (km)								
S: students' information	+219	+240	+259	+211	+116	+213	+271	+276
P: programs' information	+74	−19	+54	+82	−6	+180	+67	+56
T: one stage	−106	−40	−144	−103	−101	−187	−84	−54
Total: (c) minus (a)	+187	+182	+169	+190	+10	+206	+254	+278
Students	31,406	1,224	4,614	25,567	7,852	7,851	7,851	7,852

Panel A reports mean welfare under each design. S: students know their age-18 results r_i and Stage-3 taste shocks ε_{is} when they apply. P: programs make Stage-2 offers on the admissions index with achieved scores instead of predicted scores. T: students apply once, at a single date, and are matched by deferred acceptance; students rank programs before they learn ε_{is}, and programs rank applicants on predicted scores. Designs without T keep the two-stage timing and the Stage-4 grade condition; design (c), which combines S, P, and T, is post-qualification admissions. Panel B reports each change's Shapley value: its marginal effect averaged over the six orders in which the three changes can be added, so the three rows sum to the total up to rounding. No design in this table has unconditional offers, so welfare equals match utility. Means over 200 simulated markets. Utility is in kilometers of travel; unplaced students have utility 0. Under 2018 shares, takers hold an unconditional offer and are placed on it, decliners hold one and are not placed on it, and the remaining students hold none; each student keeps that group in every design. Q1–Q4 are quartiles of achieved age-18 attainment (Q4 highest).

Table A.9: Student welfare by sex and socioeconomic status

		Sex		SES	
Design	All	Female	Male	Low-SES	Other
Panel A. Welfare (km)					
(a) Ban on unconditional offers	1,319	1,308	1,325	1,250	1,339
(b) Unconditional shares of 2018	1,313	1,305	1,317	1,245	1,332
(b′) Excluding the premium	1,309	1,297	1,315	1,238	1,328
(c) Post-qualification admissions	1,506	1,490	1,515	1,434	1,526
Panel B. Placement rate (%)					
(a) Ban on unconditional offers	82.0	81.6	82.3	79.6	82.7
(b) Unconditional shares of 2018	82.1	81.7	82.2	79.8	82.7
(c) Post-qualification admissions	81.9	81.4	82.2	79.3	82.6
Students	31,406	10,746	20,660	6,778	24,628

Low-SES: student lives in an IMD quintile 1–2 area. Each row reports means under that design over 200 simulated markets. Utility is in kilometers of travel; unplaced students have utility 0. Welfare is match utility plus the unconditional-offer premium; "Excluding the premium" reports match utility alone. Under post-qualification admissions welfare equals match utility.

Table A.10: Where students enroll, by program

		By uncond. offers in 2018		By program selectivity			
Design	All	Adopters	Non-adopters	Q1	Q2	Q3	Q4
Panel A. Students enrolled							
(a) Ban on unconditional offers	25,766	3,212	22,554	2,933	3,723	7,274	11,836
(b) Unconditional shares of 2018	25,772	3,213	22,559	2,934	3,724	7,276	11,837
(c) Post-qualification admissions	25,727	3,208	22,519	2,928	3,718	7,263	11,817
Panel B. Mean achieved attainment of incoming students (SD)							
(a) Ban on unconditional offers	0.945	0.572	0.998	0.051	0.312	1.021	1.317
(b) Unconditional shares of 2018	0.941	0.499	1.004	0.026	0.320	1.014	1.316
(c) Post-qualification admissions	0.965	0.665	1.007	0.046	0.592	1.043	1.261
Programs	59	15	44	15	15	14	15

Adopters are programs with a positive unconditional-offer share in 2018. Q1–Q4 are quartiles of programs' mean incoming attainment in 2010–2013 (Q4 most selective). Programs with no simulated enrollment under the ban are excluded. Each row reports means under that design over 200 simulated markets.

Table A.11: Enrollment by program selectivity and student attainment

	Students by achieved attainment			
	Q1	Q2	Q3	Q4
(a) Ban on unconditional offers				
Programs Q1	1,734	813	282	104
Programs Q2	1,670	1,285	536	231
Programs Q3	816	2,053	2,302	2,103
Programs Q4	333	2,118	4,156	5,230
(b) Unconditional shares of 2018				
Programs Q1	1,765	795	274	100
Programs Q2	1,649	1,298	544	233
Programs Q3	843	2,052	2,290	2,091
Programs Q4	342	2,101	4,156	5,238
(c) Post-qualification admissions				
Programs Q1	1,848	631	259	190
Programs Q2	1,185	1,240	741	552
Programs Q3	697	2,091	2,387	2,088
Programs Q4	531	2,354	4,036	4,896

Rows are program selectivity quartiles (2010–2013 mean incoming attainment, Q4 most selective); columns are student achieved-attainment quartiles (Q4 highest). Programs with no simulated enrollment under the ban are excluded. Cells are mean students enrolled over 200 simulated markets.

B Data and Variable Construction

In this section, we describe in detail the definition and construction of variables.

Table B.1: Variable definitions

Variable	Definition
Unconditional offer	Indicator equal to 1 if an offer is marked as unconditional using UCAS flags, defined at the offer level.
Major	A specific JACS3.0 principal subject code; the Joint Academic Coding System (v3.0) was the standard subject-classification scheme in UK higher education prior to 2019.
Program	A particular combination of university and major. Programs may consist of multiple courses (the true unit of application); we aggregate offers and admissions across all courses in a given program.
Application set	The set of programs applied to by a given student, defined at the student level.
Uses unconditional	Indicator equal to 1 if more than 5% of a program's offers in a given year are unconditional offers, defined at the year-program level.
Exposed to unconditional offers	Indicator equal to 1 if any program in a given application set uses unconditional offers, defined at the application set level.
First exposure	Indicator for whether an application set is exposed to unconditional offers for the first time, defined at the year-application set level.
Later exposure	Indicator for whether an application set is exposed to unconditional offers in a period after first exposure, defined at the year-application set level.
Take-up	Indicator for take-up of an unconditional offer, which occurs either through ranking a binding unconditional offer first, or ranking a standard unconditional offer first or second. Referred to as "Take-up" or "Uncond. take-up" in the exhibits. Defined at the student level.
Age-18 scores (z)	A student's aggregate achievement on their age-18 examinations (which may include both A-levels and BTECs). If a student took more than three such exams, we consider the three exams that were most commonly taken by other students in their primary major of application. Grades are converted to a numerical value using "tariff points": A* = 56, A = 48, B = 40, C = 32, D = 24, E = 16, U = 0. BTEC grades are converted using the same UCAS tariff scale, applied to whichever qualification size a student holds (e.g., for a BTEC National Extended Diploma, sized to three A-levels: D*D*D* = 168, D*D*D = 160, D*DD = 152, DDD = 144, DDM = 128, DMM = 112, MMM = 96, MMP = 80, MPP = 64, PPP = 48; for a BTEC National Diploma, sized to two A-levels: D*D* = 112, DD = 96, DM = 80, MM = 64, MP = 48, PP = 32). We normalize scores to mean 0, standard deviation 1, within groups of students taking the same number of A-levels and the same number of BTEC exams.
Teacher-predicted age-18 scores (z)	A student's aggregate achievement, as predicted by their teachers, on their age-18 examinations. A small minority of students have a completed A-level exam at the time of their applications; for these students, we replace the relevant predicted grade with their achieved grade. Variable construction proceeds identically to Age-18 scores (z).

Continued on next page

Table B.1 – continued from previous page

Variable	Definition
Teacher-predicted age-18 scores (categorical)	A student's aggregate achievement, as predicted by their teachers, on their age-18 examinations. Identical to Teacher-predicted age-18 scores (z), but rather than standardizing the scores, we create an indicator variable for each unique combination of (score 1, test type 1, score 2, test type 2, score 3, test type 3).
Age-16 test scores (z)	A student's achievement on their age-16 examinations (GCSEs). We convert percentiles on the exam to a z-score using the inverse normal CDF. Additionally includes a separate indicator for missing values.
Socioeconomic quintile	Quintile (1–5, 1 most deprived) of the Index of Multiple Deprivation (IMD) associated with a student's postcode.
Low-SES	Indicator for whether a student's postcode belongs to the bottom two quintiles of IMD.
Female	Indicator in UCAS data of whether a given student is female.
Independent school	Indicator for whether a student attends an independent school.
Expected achievement (z)	Fitted values from a regression of Age-18 scores (z) on Teacher-predicted age-18 scores (categorical), Age-16 test scores (including indicator for missing values), Low-SES, and Female using data from 2010–2013. Absent moral hazard, this is the best prediction available of student achievement given their prior observable characteristics.
Distance (d_{is})	Straight-line distance in kilometers between a student's postcode and the program's institution.
Commuting range	Indicator equal to 1 if $d_{is} < 40\text{km}$, i.e., close_{is} in Section 4.1.4.
Region match (m_{is})	Indicator for whether a student's home region and a program's region are the same, using the 12 International Territorial Level 1 (ITL1) regions defined by the Office for National Statistics.
Matriculation	Indicator for whether a student enrolls at a university in a given year (e.g., at age 18, 19, ...).
Years enrolled	Cumulative number of years a student is observed enrolled in HESA by a given age (e.g., by age 24), used in Section 3.2.
Degree completion / classification	Indicator for completing a degree by a given age, and the classification received (first-class honors, 2:1, 2:2, third-class honors), from HESA.

C A Model of Unconditional Offers

Takers score 0.139SD below decliners with the same academic attainment, demographics, and school (Table 4, column 3). Two forces could produce this gap: students who know they would place higher may decline their offers (private information), or takers may prepare less for the examination (moral hazard).

We build a stylized model of the unconditional-offer market in which both forces are present. It shows that unconditional offers can be made, and taken, alongside either force; that either force alone produces a gap between takers and decliners of the kind we observe, so the gap does not tell

them apart; and that although adopters anticipate the gap, the programs above the highest adopter end up with weaker intakes than with conditional offers alone.

A unit mass of students is matched to three programs $k \in \{a, b, c\}$, each with capacity $\frac{1}{4}$; a student whom no program admits takes an outside option, d. A student's ability is $A = v_1 + v_2$, with v_1 and v_2 independent and uniform on $[0, 1]$: v_1 is public (age-16 scores, teacher-predicted grades), while v_2 is unobserved by all parties until the age-18 examination. With probability p, independent of ability, a student privately observes v_2 after receiving her offers and before ranking them; otherwise she observes nothing. A student who takes an unconditional offer prepares less for the examination: her realized ability is $A - s$, with $s \geq 0$. Because her place is secured, the loss falls on the program that admitted her and on no one's placement.[31] Finally, a share q of unconditional-offer holders decline for reasons unrelated to ability, such as a preferred offer elsewhere, and rank their conditional offers instead.[32] Programs are aware of p and s, but not which students received a signal. Programs maximize the total realized ability of their intake.[33] A student who enrolls at k receives utility u_k, and one who takes the outside option u_d, with $u_a > u_b > u_c > u_d$, and evaluates lotteries with a strictly increasing, strictly concave U.

Timing follows the UK admissions cycle. Students apply to all programs before v_2 is realized. Programs can make two types of offers: conditional offers, which depend on the revelation of v_2, and unconditional offers, which guarantee admission without regard to v_2. Students can either rank their conditional offers or commit to one unconditional offer and forfeit all other offers.

An equilibrium is a profile in which, at the prevailing admission thresholds, no program can raise its payoff by changing its set of unconditional offers once the other programs withdraw the offers that the change leaves unprofitable.[34] Each student's take-up decision maximizes her expected utility, and conditional admission clears the market. A program's strategy is the set of students it gives unconditional offers. In equilibrium this set is an interval of observable ability, the program's *unconditional-offer range*. The remaining conditional admission thresholds are set by market clearing. To simplify the analysis, we assume that students break any ties in favor of unconditional offers, and that programs make unconditional offers only when they are strictly profitable. We characterize the equilibrium of this market in Proposition C.1. Four equilibria, one for each combination of the two forces, are shown in Figure C.1. In each, one or both of the two lower-ranked programs make unconditional offers to a range of observably strong students, and the remainder of the class is admitted conditionally.

> *Proposition C.1 (Equilibrium unconditional-offer ranges).* With $u = (3, 2.5, 2, 1)$, $U = \ln$, and $q = 0.2$, as in Figure C.1, in every equilibrium:

[31] We take no stand on the cost of shirking to the student herself, which matters to the extent that attainment affects later outcomes (Section 3.2); in the model, students value only their placement.

[32] The share q multiplies the value of every unconditional offer and every taker mass by $1 - q$; it changes no sign, floor, or ceiling at the margin, but at positive scale it narrows b's unconditional-offer range and can widen c's (Appendix D). We set $q = 0.2$ in Figure C.1, so that every panel has decliners to compare takers with.

[33] This is equivalent to maximizing average quality and enforcing capacity $\frac{1}{4}$. Total attainment, counting the students who take the outside option, is fixed when $s = 0$ and falls by s times the mass of takers when $s > 0$.

[34] This is the anticipatory equilibrium of Wilson (1977), introduced for insurance markets with private information, where an equilibrium in the usual sense can fail to exist (Rothschild and Stiglitz, 1976). A program values each offer by its effect on its intake at the prevailing admission thresholds and does not account for how its own offers move those admission thresholds (Appendix D.1).

(i) Program a never makes unconditional offers. If b or c makes unconditional offers, it offers to an interval of observable ability, $[\underline{x}_k(p,s), \bar{x}_k)$; it breaks even at the floor, and uninformed students stop accepting at the ceiling. When both offer, the two intervals are disjoint, with c's below b's.

(ii) Unconditional-offer ranges are widest at $p = s = 0$. Program k makes unconditional offers if and only if $s < \hat{s}_k(p)$, where the exit threshold $\hat{s}_k$ is strictly decreasing in p, positive at $p = 0$ whenever k offers at all, and reaches zero at some $p < 1$.

Proof. See Appendix D.6. □

An equilibrium exists (Proposition D.8), and it is unique at every parameter point we compute.[35] Which program's exit threshold lies inside the other's depends on exact utility values (Appendix D.6); under the setup of Figure C.1, b's lies inside c's, so as either force grows both programs offer, then only c, then neither. Beyond both exit thresholds the assignment with conditional offers alone is an equilibrium, and the only one at every point we compute.

Unconditional offers grant insurance to students uncertain about their placement; this can benefit a program when it attracts students who would otherwise have enrolled above it, though unconditional offers do not always benefit adopting programs relative to an economy with only conditional offers. In every panel of Figure C.1, c gains relative to the assignment with conditional offers alone, and a and b lose. Program c's range holds students who would otherwise have placed at b; b's unconditional offers attract students from a, but this gain is offset by the loss of students to c. Though b would prefer that unconditional offers be banned, it uses unconditional offers anyway because staying out would be worse: in panel (a), b's mean intake ability is 1.138 with conditional offers alone, 1.112 when it offers alongside c, and 1.088 if b stays out and c best responds. Private information and moral hazard both reduce the value of these offers to programs, but in different ways. Private information leads students with high v_2 to decline, reducing the expected value of an unconditional offer to the program. Moral hazard acts as a "tax" on unconditional offers, lowering the value of every unconditional offer by s times its take-up rate. Once the two forces together leave the offer at a program's ceiling unprofitable, that program stops offering altogether.

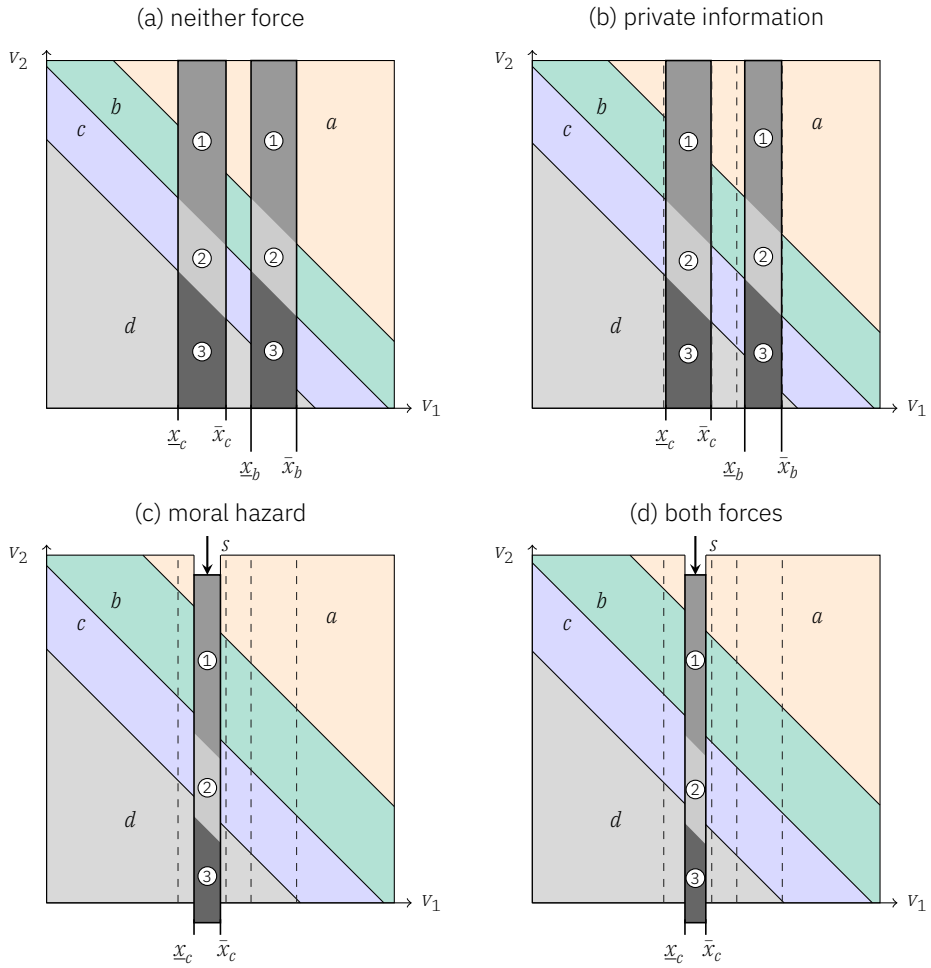
Both forces generate a gap between takers and decliners. Under private information, declining is itself the signal: the informed decliners are the students who know they would place above the program, so the gap is the distance between the strong students who leave and the average student who stays. This creates a "winner's curse" from the program's perspective: an informed student who takes an unconditional offer knows she is weak, while the uninformed take regardless. Under moral hazard, takers and decliners are the same students until the examination, and the gap is what the examination reveals. Proposition D.10 gives the gap in closed form, as the sum of the tax s and a private-information term that, when some holders decline for other reasons ($q > 0$), is nonzero if and only if $p > 0$; the same gap arises from a continuum of (p, s) pairs, and so the gap itself is not sufficient to distinguish the two. The panels of Figure C.1 show this: $(p, s) = (0.104, 0)$

[35] An analytical proof is shown in the online appendix.

in (b) and $(0, 0.0567)$ in (c) produce the same gap of -0.139SD between takers and decliners, the conditional gap of Table 4 (column 3). Together, in panel (d), they produce -0.278SD.

In both cases, programs respond by shrinking their unconditional-offer ranges, but the two forces have different welfare consequences for programs and students, and thus different policy implications. Private information decreases the benefits of unconditional offers but leaves total attainment, counting the students who take the outside option, constant: it redistributes. Unconditional offers also change who attends a program. Takers who would otherwise have taken the outside option, students with strong observables who do poorly on the examination, take the places of students with weaker observables whose scores clear c's admission threshold with conditional offers alone but not the higher threshold the takers create. In panel (a) of Figure C.1, takers displace 3.4% of students, the mean ability of students at the three programs falls by 0.007, and that of students who take the outside option rises by 0.021 (Appendix D.3). Ex-post student welfare (as measured after final matching) is constant under either force, since capacities fix the placements, and so is total ex-ante student welfare (as measured during the ranking phase): every student is placed or takes the outside option, so students' expected utilities sum to $\frac{1}{4}\sum_k U(u_k)$ in every equilibrium. Unconditional offers therefore only redistribute among students. An informed student takes an offer only when it beats her placement, so under private information informed students do better, and uninformed students worse, than with conditional offers alone; under moral hazard alone every student is uninformed, and their total is unchanged. Moral hazard, by contrast, lowers total attainment, by s times the mass of takers. Beyond placements, the effect of moral hazard on student welfare is ambiguous and depends on the sign of returns to age-18 exams, which the model does not price; returns encompass many components, such as human capital, non-pecuniary costs of studying for tests, and direct labor market effects. Under either force, a ban on unconditional offers raises the programs' total intake ability, because with conditional offers alone the programs enroll the students with the highest ability, and it gives the displaced students back their places. Under private information, it also redistributes among students, from informed to uninformed, and among programs, toward those that lose from the offers, a and b; under moral hazard, it also recovers the attainment lost to shirking, with ambiguous impacts on students.

Figure C.1: Equilibrium unconditional-offer ranges and take-up in the (v_1, v_2) plane, by force.



Notes: Axes are observable ability v_1 and the unobserved component v_2; ability $A = v_1 + v_2$ is constant along diagonals. Colored bands are the conditional-admission assignment (the outside option d, then c, b, a, from the bottom left), separated by the solid diagonals, the equilibrium admission thresholds $\underline{\alpha}_k$. The vertical gray strips are the unconditional-offer ranges of c (left) and b (right), from floor $\underline{x}_k$ to ceiling $\bar{x}_k$; the dashed verticals in (b)–(d) mark the ranges of (a), for comparison. Within each strip, ① is the gain cell, takers who would otherwise have placed above k, each displacing a worse conditional admit; ② the neutral cell, takers who would have enrolled at k anyway; ③ the loss cell, takers who would have placed below k, each displacing a better conditional admit. Informed students, absent in (a) and (c), decline in ① and accept in ② and ③. The share q of holders who decline regardless, for reasons unrelated to ability, is spread uniformly over every cell and is not drawn, for ease of viewing. Under moral hazard, (c) and (d), takers score s less than they would have, so each strip is drawn shifted down by s, at the takers' realized v_2. Panels use $u = (3, 2.5, 2, 1)$, $U = \ln$, $q = 0.2$, $p = 0.104$, and $s = 0.0567$. In (b) and (c), this results in a taker–decliner achievement gap of -0.139SD (Table 4, column 3). Program b makes no unconditional offers in (c) and (d). The four equilibria are detailed in Appendix D.3.

D Appendix for the Theoretical Model

D.1 Timing

$t = 0$	apply	Students submit applications to all three programs.
$t = 1$	offer	Programs make offers: conditional (admitting if the realized A meets the program's admission threshold) or unconditional (made before the examination, and so conditioned on v_1 alone).
$t = 2$	rank	Students privately learn v_2 with probability p, then rank up to three offers. Students can either rank all conditional offers or rank a single unconditional offer and forfeit all remaining offers. A share q of unconditional-offer holders, drawn independently of ability and information, rank their conditional offers regardless.
$t = 3$	accept	Students who ranked conditional offers take a test, revealing $A = v_1 + v_2$. Students are accepted to the best-ranked program for which they meet the conditions. Students who took an unconditional offer score $A - s$, and keep their place.

Equilibrium. Students best respond by Lemma D.2. Each program takes the admission thresholds as given and values an offer at target x by $V_k(x)$, the change in its payoff from a marginal offer there at the prevailing admission thresholds (Lemma D.3).[36] A program's offers are an equilibrium response if no change to them raises its payoff once the other programs withdraw the offers that the change leaves unprofitable (Wilson, 1977), and programs make only strictly profitable offers (Appendix C). At a target where no other program makes unconditional offers, no offer is withdrawn, so k makes offers there exactly where $V_k(x) > 0$. At a target where program j makes unconditional offers, an offer from a program ranked below j has no takers, since every student who would accept it prefers j's. An offer from a program k ranked above j takes all of j's takers there, since every student who would accept j's offer prefers k's; j's offer then has no takers and is withdrawn, and k's offer is worth $V_k(x)$. Each target therefore receives unconditional offers from at most one program: the highest-ranked program k with $V_k(x) > 0$, if there is one.[37] Because unconditional-offer ranges cover a nonnegligible share of capacity, takers shift the admission thresholds, so an equilibrium is a joint fixed point of unconditional-offer ranges and market clearing.[38] Proposition C.1 characterizes the unconditional-offer sets, Appendix D.3 computes the fixed point, and Proposition D.8 shows that it exists; at every parameter point we compute, it is unique.

[36] A program that also accounted for the effect of its own offers on the admission thresholds would behave differently. In our computations for c, its extra takers raise its own admission threshold, which makes keeping conditional offers riskier for uninformed students and so raises its ceiling. Such a program makes unconditional offers below its break-even target and, with $p > 0$, loss-making offers above its ceiling, whose informed takers raise its admission threshold further.

[37] Without the anticipated withdrawal, k would value an uninformed holder of j's offer at $A - s - \underline{\alpha}_k$ rather than $\varphi_k(A) - s$, since she would otherwise enroll at j. Such offers can pay at the top of a lower adopter's range: under the setup of Figure C.1, for b inside c's range in panels (a) and (b) and for a inside b's range in panel (a), so those panels would not be equilibria. But j's offer then has no takers and is withdrawn, after which k's offer loses. This cycle is the nonexistence problem of insurance markets with private information (Rothschild and Stiglitz, 1976).

[38] Conditional admission is not strategic: admission thresholds clear the market. Strategic conditional admission would not help either. Committing to reject an available student only passes her down the ranking and fills the seat from below, since the clearing admission thresholds above do not depend on whom a program rejects beneath them.

D.2 Lemmas

We first prove three lemmas about a single unconditional offer. Appendix D.3 extends them to unconditional-offer ranges of positive mass, Appendix D.4 proves existence, and Appendix D.6 proves Propositions C.1 and D.10.

Define the *target* of an unconditional offer as the observable ability $x = v_1$ of the student who receives it. A program's *ceiling* $\bar{x}_k$ is the highest target at which uninformed students accept its offer (Lemma D.2), and its *floor* $\underline{x}_k$ is the lowest target at which the offer breaks even (Lemma D.4); both depend on p and s, which we suppress. Program k's *exit thresholds* are $\hat{p}_k$, the value of p at which its unconditional-offer range vanishes when $s = 0$, and $\hat{s}_k(p)$, the value of s at which it vanishes given p. Primed objects (admission thresholds $\underline{\alpha}'_k$, ceilings $\bar{x}'_k$, floors $\underline{x}'_k(p)$, exit thresholds $\hat{p}'_k$ and $\hat{s}'_k(p)$) are evaluated at the conditional-only benchmark of Lemma D.1, where Lemmas D.2 and D.3 analyze a single offer; we write $\underline{x}'_k$ for $\underline{x}'_k(0)$. Unprimed objects are equilibrium values.

Our lemmas hold beyond the baseline economy (equal capacities, hence quartile admission thresholds, and $u = (3, 2.5, 2, 1)$ with $U = \ln$), which we use only for numerical evaluations. The proofs rely on three conditions, flagged where used and all satisfied at the baseline economy.

(C1) At the prevailing admission thresholds, every band is narrower than the support of v_2, so that students who do not observe v_2 have uncertainty about their program assignment.

(C2) At the prevailing admission thresholds and every target x between k's floor and ceiling, $\underline{\alpha}_k - x$ and $\bar{\alpha}_k - x$ lie in $(0, 1)$, so that an offeree at target x may place below, at, or above k depending on her v_2.

(C3) Unconditional offers are viable for some k: $\bar{x}'_k > \underline{x}'_k$, so that some target is both profitable for k and accepted by uninformed students.

(C1) for the bands of b and c, and (C2) wherever a program's value is positive below its ceiling, hold at the benchmark and in every equilibrium (Lemma D.6). Preferences enter only through the ceilings $\bar{x}'_k$; the floors $\underline{x}'_k(p)$ are pure geometry. Two kinds of claims depend on the baseline: the exit order in Proposition C.1, which reverses when u_b is close enough to u_a (Appendix D.6), and the results at positive scale that we establish by computation (Appendix D.3). The two frictions enter in two places only. Moral hazard s is a per-taker cost in Lemma D.3 and nowhere else: a taker's place is secured, so her lower score moves no admission threshold. The random-decline share q multiplies every taker mass, in Lemma D.3 and in Equation (19), by $1 - q$, and nothing else: a random decliner is an ordinary conditional applicant. Every sign below, and every floor, ceiling, and exit threshold of the marginal analysis, is therefore invariant to q. At positive scale q changes the unconditional-offer ranges, though not always downward: it narrows b's ($\hat{p}_b$ falls from 0.575 to 0.557 at $q = 0.2$) but can widen c's, which at $p = s = 0$ is $[0.378, 0.516)$ with $q = 0.2$ and $[0.397, 0.524)$ with $q = 0$. The tables report $q = 0$ unless stated.

> *Lemma D.1 (Benchmark assignment).* With conditional offers only, if each student enrolls at the highest-ranked program that admits her, the assignment that clears the market is unique and assortative: program k enrolls the students with $A \in [\underline{\alpha}'_k, \bar{\alpha}'_k)$, and students with $A < \underline{\alpha}'_c$ take the outside option. Here the admission thresholds $(\underline{\alpha}'_a, \underline{\alpha}'_b, \underline{\alpha}'_c) = (2 - \sqrt{\tfrac{1}{2}},\ 1,\ \sqrt{\tfrac{1}{2}})$ are the quartiles of the distribution of A,

and $\bar{\alpha}'_k$ is the admission threshold of the program ranked directly above k: $\bar{\alpha}'_c = \underline{\alpha}'_b$, $\bar{\alpha}'_b = \underline{\alpha}'_a$, and $\bar{\alpha}'_a = 2$, the top of the support. The assignment is the same for every p.

Proof. The ability A has distribution $F(\alpha) = \frac{\alpha^2}{2}$ for $\alpha \leq 1$ and $1 - \frac{(2-\alpha)^2}{2}$ for $\alpha \geq 1$, strictly increasing on $[0, 2]$. Students share the ranking $u_a > u_b > u_c > u_d$, so in any market-clearing outcome admission thresholds are ordered and each intake is the interval between consecutive admission thresholds. Capacities of $\frac{1}{4}$ then force $1 - F(\underline{\alpha}'_a) = \frac{1}{4}$, $1 - F(\underline{\alpha}'_b) = \frac{1}{2}$, and $1 - F(\underline{\alpha}'_c) = \frac{3}{4}$; strict monotonicity of F makes the solution unique. The argument uses only the common ranking and the continuity and strict monotonicity of F; with general capacities, the admission thresholds are the corresponding quantiles. Nothing is chosen before the examination, so the assignment is the same for every p. □

Write $\mu(A)$ for the program assigned to ability A under Lemma D.1, with $\mu(A) = d$ for the outside option. Program k's band is $[\underline{\alpha}'_k, \bar{\alpha}'_k)$: its own admission threshold is the lower edge, and the admission threshold of the program directly above is the upper edge, so an overline on α is the underline of the next program up. A taker who would have placed below k costs k its worst student, of ability $\underline{\alpha}'_k$. A taker who would have placed above k costs k its *best* student: she leaves a vacancy above, which is refilled from the top of k's band, so k keeps her at the price of a student of ability $\bar{\alpha}'_k$. So the *net contribution* of a student of ability A who enrolls at k by taking an unconditional offer is

$$\varphi_k(A) \;\equiv\; \max\{A - \bar{\alpha}'_k,\, 0\} - \max\{\underline{\alpha}'_k - A,\, 0\},$$

her distance outside k's band, signed: positive if she would have placed above k, zero if at k, negative if below. At other admission thresholds, φ_k is defined in the same way with those admission thresholds.

Lemma D.2 (Acceptance). Consider a single unconditional offer from $k \in \{b, c\}$ to a student with $v_1 = x$. One offer has measure zero, so admission thresholds stay at the benchmark values of Lemma D.1; primes mark objects evaluated at those admission thresholds.

(i) An uninformed holder accepts if and only if $x \leq \bar{x}'_k$. The *ceiling* $\bar{x}'_k$ is the unique solution in $(0, \underline{\alpha}'_k)$ of $W_k(x) = U(u_k)$, where $W_k(x) \equiv E\left[U\left(u_{\mu(A)}\right) \mid v_1 = x\right]$ is the *lottery utility* of keeping the conditional offers: continuous, piecewise linear, and strictly increasing in x.[39]

(ii) For every increasing U, an informed holder accepts if and only if she would have placed at or below k, that is, $A < \bar{\alpha}'_k$. Up to the measure-zero boundary $A = \bar{\alpha}'_k$, this is $\varphi_k(A) \leq 0$.

(iii) At any prevailing admission thresholds that satisfy (C1), including those induced by a positive mass of offer holders, an uninformed holder accepts if and only if $W_k(x) \leq U(u_k)$, so the targets she accepts form an interval that

[39] The ceiling carries a prime because the lottery depends on the admission thresholds through μ; at the equilibrium admission thresholds of Appendix D.3 it is a different number, $\bar{x}_k$.

ends at the ceiling, and an informed holder if and only if $A < \bar{\alpha}_k$, with W_k, μ, φ_k, and the ceiling evaluated at those admission thresholds.

Proof. Conditional on $v_1 = x$, $A \sim U[x, x+1]$, so the lottery utility is

$$W_k(x) \;=\; \int_0^1 U\big(u_{\mu(x+t)}\big)\, dt,$$

continuous and piecewise linear, with derivative $U\big(u_{\mu(x+1)}\big) - U\big(u_{\mu(x)}\big)$ wherever x and $x+1$ are interior to bands. Raising the target shifts probability from the worst reachable band to the best, so the derivative is positive as long as $\mu(x+1)$ is ranked strictly above $\mu(x)$, which (C1) guarantees. At $x = 0$ the support $[0,1]$ of A meets only d's and c's bands, so $W_c(0) = \underline{\alpha}'_c\, U(u_d) + (1 - \underline{\alpha}'_c)\, U(u_c) < U(u_c)$;[40] at $x = \underline{\alpha}'_c$ the support $[\underline{\alpha}'_c, \underline{\alpha}'_c + 1]$ lies weakly above $\underline{\alpha}'_c$ and strictly above $\underline{\alpha}'_b$ with positive probability, so $W_c(\underline{\alpha}'_c) > U(u_c)$. Thus, W_c crosses $U(u_c)$ exactly once on $(0, \underline{\alpha}'_c)$, at $\bar{x}'_c$; the indifferent student at the ceiling accepts by convention. The same argument with u_b in place of u_c on $(0, \underline{\alpha}'_b)$ gives $\bar{x}'_b$: $W_b(0) = W_c(0) < U(u_c) < U(u_b)$, and at $x = \underline{\alpha}'_b$ the support lies weakly above $\underline{\alpha}'_b$ and strictly above $\underline{\alpha}'_a$ with positive probability. For (ii), an informed holder knows A and compares the certain u_k to the certain $u_{\mu(A)}$: she accepts if and only if $\mu(A)$ is k or worse, that is, $A < \bar{\alpha}'_k$, for any increasing U, and at $\mu(A) = k$ she accepts by convention. Since $\varphi_k(A) \le 0$ exactly when $A \le \bar{\alpha}'_k$, this is the last claim of (ii). For (iii), each student is atomistic and takes the prevailing admission thresholds as given; she compares $W_k(x)$ to $U(u_k)$ as in (i), W_k is strictly increasing by (C1), and the argument for (ii) uses only the order of the admission thresholds. □

At the baseline, $\bar{x}'_c = 0.495$ and $\bar{x}'_b = 0.698$.

For an offer at target x, the offeree's v_2 falls into one of three cells: the *loss cell* $v_2 < \underline{\alpha}'_k - x$, of length $\ell_k(x) \equiv \underline{\alpha}'_k - x$, where she would have placed below k; the *neutral cell* $\underline{\alpha}'_k - x \le v_2 < \bar{\alpha}'_k - x$, where she would have placed at k; and the *gain cell* $v_2 \ge \bar{\alpha}'_k - x$, of length $g_k(x) \equiv x + 1 - \bar{\alpha}'_k$, where she would have placed above k. At other admission thresholds, the cells are defined in the same way with those thresholds. Under (C2) all three cells are nonempty.

Lemma D.3 (Value of a marginal unconditional offer). Let program k make unconditional offers to every student with $v_1 \in [x, x+\varepsilon)$, a strip on which no other program makes unconditional offers, with students following Lemma D.2. The prevailing admission thresholds may be the benchmark admission thresholds of Lemma D.1 or those induced by any unconditional offers, including k's own (Appendix D.3); φ_k and the cells are evaluated at them. Holding fixed the acceptance decisions of the holders of all other unconditional offers, the change in k's payoff is $\varepsilon\, V_k(x) + O(\varepsilon^2)$ at

[40] In general, $W_k(0) < U(u_k)$ is the condition that some uninformed student accepts, which is part of viability (C3).

every target $x \neq \bar{x}_k$, where

$$V_k(x) \;=\; (1-q)\Big[(1-p)\,\mathbf{1}\{x<\bar{x}_k\}\int_0^1 \varphi_k(x+v_2)\,dv_2 \\ +\; p\int_0^1 \min\{\varphi_k(x+v_2),\,0\}\,dv_2 \;-\; s\,\tau_k(x)\Big],$$

and $\tau_k(x) = (1-p)\,\mathbf{1}\{x<\bar{x}_k\} + p\int_0^1 \mathbf{1}\{\varphi_k(x+v_2)\le 0\}\,dv_2$ is take-up at x among holders not subject to the random decline.

Proof. Capacities bind and clearing is assortative, so when a taker of ability A enters k's intake exactly one student leaves it, and she is always at an admission threshold. If the taker would have placed above k, the vacancy she leaves is filled from the top down, each program taking the best student of the program below, until the program directly above k takes k's best student, of ability $\bar{\alpha}_k$. If she would have placed below k, k sheds its worst student, of ability $\underline{\alpha}_k$, and the surplus passes down in the same way to the program, or the outside option, that she vacated. If she would have placed at k, she replaces herself. In each case her net contribution to k is $\varphi_k(A)$, and no other program's intake enters k's payoff. The taker then scores $A-s$ rather than A; since her place is secured, the shortfall changes no assignment and is a cost of s to k alone. Holders outside the strip keep their decisions by hypothesis. The takers have mass $O(\varepsilon)$, so admission thresholds move by $O(\varepsilon)$ and each displaced student has ability equal to the admission threshold up to $O(\varepsilon)$; offerees in the strip whose acceptance flips because admission thresholds moved lie within $O(\varepsilon)$ of the boundary $\varphi_k = 0$, or of the ceiling, which $x \neq \bar{x}_k$ keeps outside the strip for small ε; either set has measure $O(\varepsilon^2)$. By Lemma D.2, for $x \neq \bar{x}_k$ the uninformed accept if and only if $x < \bar{x}_k$ and are uniform in v_2, while the informed accept if and only if $\varphi_k(A) \le 0$; the share q who decline regardless are uniform in v_2 and independent of information, so they scale every term by $1-q$ and enter the conditional pool as ordinary applicants. Summing $\varphi_k(A)-s$ over the takers in the strip $[x, x+\varepsilon)$ therefore changes k's payoff by $\varepsilon\,V_k(x) + O(\varepsilon^2)$. □

In Lemma D.3, an uninformed taker contributes her full net contribution, an informed taker, who accepts exactly when $\varphi_k \le 0$, contributes only its negative part, and every taker costs s. The lemma gives the value of a target in closed form. With the cell lengths ℓ_k and g_k both in $(0,1)$ (C2), $\int_0^1 \varphi_k = \frac{1}{2}\,(g_k^2 - \ell_k^2)$, $\int_0^1 \min\{\varphi_k, 0\} = -\frac{1}{2}\ell_k^2$, and the informed accept on the loss and neutral cells, of combined length $1-g_k$, so $\tau_k = 1 - p\,g_k$. Below the ceiling $\bar{x}_k$, where uninformed students accept,

$$\frac{V_k(x)}{1-q} \;=\; \frac{1}{2}\,[(1-p)\,g_k(x)^2 - \ell_k(x)^2] - s\,(1-p\,g_k(x)) \;=\; G_k(x) - p\,\frac{g_k(x)^2}{2} - s\,(1-p\,g_k(x))\,, \quad (18)$$

where $G_k(x) \equiv \frac{1}{2}\,[g_k(x)^2 - \ell_k(x)^2]$ is the value with neither force: private information removes a fraction p of the gain cell and nothing else, and moral hazard charges s per taker on the take-up rate

$1-p\,g_k$. Above the ceiling only informed students accept, so $V_k = -(1-q)\,p\left[\frac{1}{2}\ell_k^2 + s(1-g_k)\right] \leq 0$. Raising the target by dx lengthens the gain cell and shortens the loss cell by dx each, so below the ceiling V_k is strictly increasing in x, with derivative $(1-q)\left[(1-p)\,g_k + \ell_k + s\,p\right] > 0$; it is strictly decreasing in s, and strictly decreasing in p wherever $g_k > 2s$, since an informed gain-cell student who declines costs k her average net contribution $g_k/2$ and saves it the s she would have forgone. We write $V_k(\bar{x}_k)$ for Equation (18) evaluated at the ceiling, the value of the last target that uninformed students accept.

Remark (a). For a, $\bar{\alpha}_a = 2$ is the top of the support, so the gain cell is empty and every taker has $\varphi_a \leq 0$. Hence every term of V_a in Lemma D.3 is nonpositive, so $V_a \leq 0$ at every target and every p and s, and a makes no unconditional offers. This holds also at targets where another program makes unconditional offers, since a program makes offers only where its own value is positive (Appendix D.1). These arguments hold at any admission thresholds.

D.3 Positive scale

Lemmas D.2 and D.3 value a single unconditional offer at fixed admission thresholds, which is enough to sign every claim about a marginal offer in Propositions C.1 and D.10 (Appendix D.6), because an unconditional-offer set of mass ε moves every admission threshold by $O(\varepsilon)$. In equilibrium, an unconditional-offer range of positive mass moves the admission thresholds it is evaluated against. This section computes that equilibrium; Lemma D.4 reduces each adopter's strategy to one number, and Equation (19) then gives the admission thresholds.

> *Lemma D.4 (Interval unconditional-offer sets).* Fix any admission thresholds satisfying (C1) and (C2), and any p and s. For $k \in \{b, c\}$, the value $V_k(x)$ of Lemma D.3 is nondecreasing in x below the ceiling $\bar{x}_k$, strictly so when $p < 1$, and nonpositive above it. At these admission thresholds, k makes no unconditional offers if $V_k(\bar{x}_k) \leq 0$. Otherwise its unconditional-offer set is the interval $[\underline{x}_k, \bar{x}_k)$ on which $V_k > 0$, up to its endpoints and less any targets at which a higher-ranked program makes unconditional offers, and its floor breaks even, $V_k(\underline{x}_k) = 0$, unless $V_k > 0$ at every target below the ceiling, in which case $\underline{x}_k = 0$.
>
> *Proof.* Below the ceiling, raising the target raises $\varphi_k(x + v_2)$ weakly for every v_2, and strictly whenever $x + v_2$ lies outside k's band, which by (C1) has positive probability; it also weakly lowers informed take-up. Hence V_k is nondecreasing, and strictly increasing when $p < 1$; under (C2), Equation (18) gives $V_k' = (1-q)\left[(1-p)\,g_k + \ell_k + s\,p\right] > 0$. Above it only informed students accept; by Lemma D.2(ii) each has $\varphi_k \leq 0$, and each costs s, so $V_k \leq 0$. In equilibrium, k makes offers exactly at the targets where $V_k > 0$ and no higher-ranked program makes offers (Appendix D.1). If $V_k(\bar{x}_k) \leq 0$, monotonicity gives $V_k \leq 0$ at every target, so k makes no offers. By monotonicity the targets with $V_k > 0$ form an interval that ends at the ceiling, and since V_k is continuous below the ceiling, its floor solves $V_k(\underline{x}_k) = 0$ unless $V_k > 0$ throughout. So k's unconditional-offer set is $[\underline{x}_k, \bar{x}_k)$, up to its endpoints and less any targets at which a higher-ranked program makes offers. □

Market clearing pins down the admission thresholds. Let $K \subseteq \{b, c\}$ be the set of adopters, each $k \in K$ offering on the strip $[\underline{x}_k, \bar{x}_k)$ of length L_k and midpoint m_k; every offeree in the strip not subject to the random decline takes if uninformed, and takes if informed exactly when $A < \bar{\alpha}_k$ (Lemma D.2); the share q who decline regardless rank their conditional offers. Write $\bar{F}(\alpha) \equiv 1 - F(\alpha)$ for the mass of students with $A \geq \alpha$ and κ_j for capacities. Students with $A \geq \underline{\alpha}_j$ either take an unconditional offer or enroll conditionally at j or above, so for each $j \in \{a, b, c\}$, where $i < j$ means that i is ranked below j,

$$\bar{F}(\underline{\alpha}_j) \;=\; \sum_{i \geq j} \kappa_i \;+\; (1-q)(1-p) \sum_{i \in K,\, i<j} L_i \left(1 - \underline{\alpha}_j + m_i\right) \;-\; (1-q) \sum_{i \in K,\, i \geq j} L_i \left(\underline{\alpha}_j - m_i\right). \tag{19}$$

Equation (19) is a seat count, and s does not appear in it: a taker's lower score moves no admission threshold, because her place is already hers. The seats at j and above, $\sum_{i \geq j} \kappa_i$, are filled either conditionally, by students with $A \geq \underline{\alpha}_j$ who took no unconditional offer, or unconditionally, by the takers of adopters at or above j. The mass $\bar{F}(\underline{\alpha}_j)$ therefore equals the seats, plus the takers with $A \geq \underline{\alpha}_j$ at adopters below j (counted in $\bar{F}$, but filling no seat at j or above), minus the takers with $A < \underline{\alpha}_j$ at adopters at or above j (filling a seat, but not counted in $\bar{F}$). In words, the admission threshold of j falls by the takers that adopters below j capture from above it (uninformed only, since informed takers of i have $A < \bar{\alpha}_i \leq \underline{\alpha}_j$), and rises by the takers that adopters at or above j admit from below it (informed and uninformed alike, since $A < \underline{\alpha}_j \leq \bar{\alpha}_i$).

Both terms take these closed forms because, for a taker at target t, the probability that her ability is at least $\underline{\alpha}_j$ is $t+1-\underline{\alpha}_j$ whenever $\underline{\alpha}_j$ lies strictly inside her range of abilities, $[t, t+1]$: for the adopter's own admission thresholds this is (C2), and for the others it requires $\underline{x}_c > \underline{\alpha}_a - 1$ and $\bar{x}_b < \underline{\alpha}_c$. Both conditions hold at every solution below; the two strips are also disjoint there, so no student holds two unconditional offers.

An equilibrium with adopter set K is a solution of Equation (19) for $j = a, b, c$ together with, for each $k \in K$, the acceptance condition $W_k(\bar{x}_k) = U(u_k)$ and the floor condition $V_k(\underline{x}_k) = 0$, both evaluated at the equilibrium admission thresholds: $3 + 2|K|$ equations in as many unknowns. In addition, no target may be profitable for a non-adopter $j \in \{b, c\} \setminus K$: $V_j(\bar{x}_j) \leq 0$ at the equilibrium admission thresholds (Lemma D.4).

The acceptance condition is explicit whenever the ceiling lies in $(\underline{\alpha}_a - 1, \underline{\alpha}_c)$. For targets in this interval, the lottery utility, which is the same for b and c, is $W_c(x) = U(u_d)(\underline{\alpha}_c - x) + U(u_c)(\underline{\alpha}_b - \underline{\alpha}_c) + U(u_b)(\underline{\alpha}_a - \underline{\alpha}_b) + U(u_a)(x + 1 - \underline{\alpha}_a)$, so $W_c(\bar{x}_c) = U(u_c)$ and $W_c(\bar{x}_b) = U(u_b)$ give

$$\bar{x}_c \;=\; \omega_c\, \underline{\alpha}_c + \omega_b(\underline{\alpha}_b - 1) + \omega_a(\underline{\alpha}_a - 1), \qquad \bar{x}_b \;=\; \bar{x}_c + \omega_b, \tag{20}$$

where $\omega_k \equiv [U(u_k) - U(u_{k'})]/[U(u_a) - U(u_d)]$ is the share of the utility gap between a and d that lies between k and the option k' ranked directly below it, so the weights are positive and sum to one; at the baseline, $(\omega_a, \omega_b, \omega_c) = (0.166, 0.203, 0.631)$, and we write $\omega_{ab} \equiv \omega_a + \omega_b$.

Given the strip endpoints, Equation (19) is piecewise quadratic in $\underline{\alpha}_j$; the acceptance condition makes the ceiling affine in the admission thresholds (Equation (20)); and by Equation (18) the floor condition $V_k(\underline{x}_k) = 0$ is, at $s = 0$, $\sqrt{1-p}\, g_k = \ell_k$, the gain cell equal to $1/\sqrt{1-p}$ times the loss cell, which is linear in the floor: $\underline{x}_k = [\underline{\alpha}_k - \sqrt{1-p}\,(1 - \bar{\alpha}_k)]/(1 + \sqrt{1-p})$. With $s > 0$ the floor condition

is the quadratic $(1-p)\,g_k^2-\ell_k^2=2s\,(1-p\,g_k)$, whose left side is increasing in the floor and right side nonincreasing, so it has one admissible root, increasing in s. The equilibrium is therefore a fixed point in the $2|K|$ endpoints. Solving, at $q=0$, along p with $s=0$ and along s with $p=0$:[41]

p	s	$\underline{\alpha}_a$	$\underline{\alpha}_b$	$\underline{\alpha}_c$	$\underline{x}_b$	$\bar{x}_b$	$\underline{x}_c$	$\bar{x}_c$	M_b	M_c
0	0	1.136	0.997	0.796	0.566	0.727	0.397	0.524	64%	51%
0.2	0	1.212	0.987	0.770	0.621	0.722	0.400	0.519	36%	43%
0.35	0	1.250	0.986	0.754	0.657	0.717	0.411	0.514	20%	34%
0.5	0	1.276	0.986	0.739	0.692	0.712	0.427	0.509	6%	25%
0.6	0	1.286	0.988	0.730	—	—	0.442	0.505	—	18%
0.7	0	1.290	0.994	0.720	—	—	0.463	0.501	—	10%
0.8	0	1.293	0.999	0.709	—	—	0.490	0.496	—	1%
0	0.02	1.185	0.981	0.770	0.608	0.716	0.401	0.513	43%	45%
0	0.04	1.231	0.969	0.750	0.654	0.708	0.411	0.505	22%	38%
0	0.05	1.252	0.965	0.742	0.678	0.706	0.418	0.503	11%	34%
0	0.07	1.277	0.971	0.728	—	—	0.442	0.499	—	23%
0	0.09	1.287	0.990	0.713	—	—	0.476	0.496	—	8%

The table shows 12 of the 356 parameter points at which we compute the equilibrium.[42] For each candidate set of adopters we find every solution of Equation (19), with the acceptance and floor conditions, by scanning each floor on a grid of step 0.01, refined near the ceiling,[43] and keep the solutions at which each non-adopter's offer at its ceiling has nonpositive value; exactly one remains at every point. In every equilibrium found, b makes unconditional offers only if c does. Along each line in p and in s, the length of each range falls as either force grows, and $\hat{s}_b(p)$ falls in p and exceeds $\hat{s}'_b(p)$ at every p on a grid of step 0.01 below $\hat{p}_b$: from 0.060 at $p=0$ to 0.003 at $p=0.55$ with $q=0$, and from 0.056 to 0.001 with $q=0.2$.

The signs of the two sums in Equation (19) order the admission thresholds against the benchmark. No adopter lies below c, so only the negative sum enters c's condition and $\underline{\alpha}_c>\underline{\alpha}'_c$; no adopter lies at or above a, so only the positive sum enters a's and $\underline{\alpha}_a<\underline{\alpha}'_a$. For b, only the positive sum enters when c is the sole adopter, so $\underline{\alpha}_b<\underline{\alpha}'_b$; when b also adopts, its own takers below $\underline{\alpha}_b$ enter with the opposite sign, and the computed fixed point settles the comparison, $\underline{\alpha}_b<\underline{\alpha}'_b$ in every row of the table. Beyond $\hat{p}_c$ the offer at each program's benchmark ceiling has nonpositive value, so making no offers is an equilibrium (Lemma D.4), and at every such point we compute it is the only one.

[41] As $L_k\to 0$ the admission thresholds return to the benchmark, so c's unconditional-offer range vanishes exactly where the benchmark floor reaches the benchmark ceiling, $\hat{p}_c=0.816$ (Appendix D.6). Adopter k fills $M_k=(1-p)\,L_k+p\int_{\underline{x}_k}^{\bar{x}_k}(\bar{\alpha}_k-t)\,dt$ of its seats with unconditional admits and the rest conditionally at $\underline{\alpha}_k$; the last columns report M_k as a share of capacity.

[42] With $q=0$, the points are every $p\in\{0,0.01,\dots,0.99\}$ at $s=0$, every $s\in\{0.0025,0.005,\dots,0.1225\}$ at $p=0$, and every $p\in\{0.1,0.2,\dots,0.8\}$ with every $s\in\{0.01,0.02,\dots,0.1\}$; with $q=0.2$, every $p\in\{0,0.05,\dots,0.95\}$ at $s=0$, every $s\in\{0.005,0.01,\dots,0.12\}$ at $p=0$, every $p\in\{0.1,0.2,\dots,0.8\}$ with every $s\in\{0.01,0.02,\dots,0.1\}$, and the three parameter pairs of Figure C.1.

[43] With both programs adopting, the scan covers c's floor on $[0,0.56]$ and b's on $[0.45,0.80]$ and skips overlapping ranges, which Lemma D.9 rules out. At every point a solver started from five other points finds no further solution, and at ten points a search from 60 random starting admission thresholds, using the general forms of the clearing conditions, finds exactly one equilibrium.

The box collects the four equilibria of Figure C.1: the unconditional-offer ranges, and everything market clearing implies from them. All four use $q = 0.2$, so that every panel has decliners; $p = 0.104$ and $s = 0.139/\sqrt{6} \approx 0.0567$ are chosen so that each force alone reproduces the conditional taker–decliner achievement gap of -0.139SD (Table 4, column 3), with one standard deviation of A equal to $1/\sqrt{6}$. The box's gaps and shortfalls average the comparisons of Proposition D.10 over the targets in c's range.

The equilibria of Figure C.1.

1. The equilibrium is defined by unconditional-offer ranges. Program a makes no unconditional offers; every uninformed offer holder accepts, an informed one accepts only if she knows she would place at or below the adopter, and a share q decline regardless.

	(a) neither $p = 0,\ s = 0$	(b) private information $p = 0.104,\ s = 0$	(c) moral hazard $p = 0,\ s = 0.0567$	(d) both $p = 0.104,\ s = 0.0567$
c offers unconditionally to $v_1 \in$	$[0.378,\ 0.516)$	$[0.384,\ 0.514)$	$[0.424,\ 0.500)$	$[0.439,\ 0.499)$
b offers unconditionally to $v_1 \in$	$[0.588,\ 0.719)$	$[0.611,\ 0.717)$	—	—

2. Through market clearing, we derive conditional admission thresholds $\underline{\alpha}_k$; these yield take-up among offer holders, the taker–decliner achievement gap, and takers' shortfall relative to the observable projection in c's range (Proposition D.10; b's in (b) are 0.763, -0.139SD, 0.033SD); the share of each adopter's intake arriving unconditionally; mean realized intake ability relative to the conditional-only benchmark, $(1.529, 1.138, 0.862)$ for a, b, c and 0.471 for the students who take the outside option; and the share of students who attend a program with unconditional offers but not with conditional offers alone (an equal share does the reverse).

		(a) neither	(b) private information	(c) moral hazard	(d) both
Admission threshold $\underline{\alpha}_a$		1.192	1.218	1.277	1.281
$\underline{\alpha}_b$		0.983	0.983	0.970	0.978
$\underline{\alpha}_c$		0.773	0.763	0.730	0.724
Take-up in c's range		0.800	0.761	0.800	0.759
Taker–decliner achievement gap (SD)		0	−0.139	−0.139	−0.278
Takers' shortfall vs. projection (SD)		0	0.033	0.139	0.173
Unconditional share of intake, b / c		42% / 44%	32% / 40%	— / 24%	— / 18%
Mean intake ability vs. benchmark,	a	−0.045	−0.032	−0.004	−0.003
	b	−0.026	−0.025	−0.024	−0.017
	c	+0.050	+0.040	+0.006	+0.004
	outside option	+0.021	+0.018	+0.008	+0.006
Total attainment vs. benchmark		0	0	−0.003	−0.003
Students who switch into the programs		3.4%	3.1%	1.5%	1.2%

Every adopter's intake also exceeds what it would be if it alone stopped offering. Relative to the benchmark, c and the outside option gain while a and b lose in every panel; b offers in (a) and (b) and still loses, because c's offers capture the top of its band; it withdraws in (c) and (d) because $s = 0.0567$ exceeds its exit threshold, which at $q = 0.2$ is $\hat{s}_b(0) = 0.056$ and $\hat{s}_b(0.104) = 0.047$. Total attainment is fixed at $s = 0$ and falls by s times the mass of takers otherwise. The programs' total falls even at $s = 0$. The students who switch in are takers with $A < \underline{\alpha}'_c$ (in panel (a), mean A of 0.590 and mean v_1 of 0.472), and those who switch out take no unconditional offer and have A in $[\underline{\alpha}'_c, \underline{\alpha}_c)$ (mean A of 0.741, mean v_1 of 0.298). The students with $A \geq \underline{\alpha}'_c$ are the set of mass $\frac{3}{4}$ with the highest total ability, so any such switch lowers the programs' total.

With $q = 0$, program b's unconditional-offer range does not vanish at $\hat{p}'_b = 0.443$, where its benchmark floor reaches its benchmark ceiling (Appendix D.6) and where it still supplies 11% of

b's intake, but at $\hat{p}_b = 0.575$, the root of $V_b(\bar{x}_b) = 0$ at the lone-adopter equilibrium admission thresholds: b exits into a market where c still makes unconditional offers, which lower both $\underline{\alpha}_a$ and, by more, $\underline{\alpha}_b$, so that at b's ceiling the gain cell is larger and the loss cell smaller than at the benchmark. For c the two exits coincide, $\hat{p}_c = \hat{p}'_c$, because no unconditional-offer range remains at c's exit, so the last offer faces the benchmark admission thresholds. The exits in s behave the same way. At $p = 0$, c's range vanishes at $\hat{s}_c = \hat{s}'_c = 0.100$, the value of its benchmark ceiling target per taker, $G_c(\bar{x}'_c)$; b's vanishes at $\hat{s}_b = 0.060$ at scale against $\hat{s}'_b = 0.036$ at the margin, again because c's remaining offers enlarge b's gain cell and shrink its loss cell. The exit threshold $\hat{s}_k(p)$ falls in p: at the margin it is $\hat{s}'_k(p) = [(1-p)\,g_k^2 - \ell_k^2] \,/\, [2(1-p\,g_k)]$ at the ceiling, and at scale $(\hat{s}_b, \hat{s}_c) = (0.051, 0.092)$ at $p = 0.104$ and $(0.042, 0.084)$ at $p = 0.2$.

D.4 Existence

Appendix D.3 computes the equilibrium at chosen values of p, s, and q, and finds exactly one at each. We show that an equilibrium exists at every value by a fixed-point argument on the admission thresholds. The argument uses an auxiliary rule: a program's takers may not exceed its capacity, so a program whose takers would otherwise exceed $\frac{1}{4}$ offers on the targets nearest its ceiling, where its offers are worth most, until its takers fill its capacity. The rule keeps each program's takers at most $\frac{1}{4}$, so that the seat counts in the proof of Proposition D.8 can be met, and Lemma D.7 shows that it never binds. Let D be the set of admission thresholds with $1/\sqrt{2} \le \underline{\alpha}_c \le \underline{\alpha}_b \le \underline{\alpha}_a \le 2 - 1/\sqrt{2}$, a compact convex set on which the bands of b and c have width at most $2 - \sqrt{2} < 1$.

Lemma D.5 (Floor cells). Fix $p \in [0,1]$, $s \ge 0$, a program $k \in \{b, c\}$, and a width $w_k \in [0, 1)$ of its band. The floor condition $(1-p)g^2 - \ell^2 = 2s(1-pg)$, with $\ell + g = 1 - w_k$, has at most one root with $\ell, g > 0$. If it has one, write $\ell = \underline{\ell}(w_k)$. Then:

(i) $\ell \le \sqrt{1-p}\,g$, so $\ell \le \frac{1-w_k}{2} \le g$; and $s < g$.

(ii) $\underline{\ell}$ is continuously differentiable, with slope in $(-1, 0)$, and $\underline{\ell}(w_k) \to 0$ as w_k approaches a width at which the root vanishes.

Proof. The difference of the two sides, $(1-p)g^2 - (1 - w_k - g)^2 - 2s(1-pg)$, has derivative $2[(1-p)g + \ell + sp] > 0$ in g, so it has at most one root. (i) Since $2s(1-pg) \ge 0$, $\ell^2 \le (1-p)g^2$. If $s \ge g$, then $(1-p)g^2 > 2g(1-pg)$, so $g(1+p) > 2$, which is impossible since $g < 1$. (ii) Since the difference rises in g, the implicit function theorem makes $\underline{\ell}$ continuously differentiable, and differentiating the floor condition gives $-\underline{\ell}' = [(1-p)g + sp]/[(1-p)g + sp + \ell]$. This lies in $(0, 1)$ because $\ell > 0$ and $(1-p)g > 0$: at $p = 1$ the floor condition reads $-\ell^2 = 2s(1-g)$ and has no root. The difference is continuous and negative at $g = 0$, so the root can leave $(0, 1 - w_k)$ only through $g = 1 - w_k$, that is, with $\underline{\ell}(w_k) \to 0$. □

Lemma D.6 (Where equilibria lie). (i) With or without the rule, every equilibrium has admission thresholds in D.

(ii) At admission thresholds in D, (C1) holds for the bands of b and c; for $k \in \{b, c\}$, target 0 lies below k's ceiling and $V_k(0) < 0$; and (C2) holds at every target below k's ceiling at which $V_k > 0$.

Proof. (i) The common ranking orders the admission thresholds, as in the proof of Lemma D.1, and a makes no unconditional offers (the remark after Equation (18)). The mass of students with $A \geq \underline{\alpha}_c$ is $\frac{3}{4}$ less the takers with $A < \underline{\alpha}_c$, so $\underline{\alpha}_c \geq 1/\sqrt{2}$, and the mass with $A \geq \underline{\alpha}_a$ is $\frac{1}{4}$ plus the takers with $A \geq \underline{\alpha}_a$, so $\underline{\alpha}_a \leq 2 - 1/\sqrt{2}$. Hence the admission thresholds lie in D.

(ii) The bands of b and c have width at most $2 - \sqrt{2} < 1$, which is (C1) for them. At target 0, $A \geq \underline{\alpha}_c$ with probability at most $1 - 1/\sqrt{2}$, so, as $U(u_d) = \ln 1 = 0$, the lottery utility is at most $(1 - 1/\sqrt{2})\, U(u_a)$, which is below $U(u_c) = \omega_c U(u_a)$; target 0 therefore lies below both ceilings. There k's gain cell, at most $1 - 1/\sqrt{2}$, is shorter than its loss cell, at least $1/\sqrt{2}$, so $\int_0^1 \varphi_k(v_2)\, dv_2 < 0$ and $\int_0^1 \min\{\varphi_k(v_2), 0\}\, dv_2 < 0$, and $V_k(0) < 0$ by Lemma D.3. At target $\underline{\alpha}_k$ an uninformed holder places at k or above, and above with positive probability because k's band is narrower than one, so she declines and k's ceiling lies below $\underline{\alpha}_k$. At a target below the ceiling, the loss cell is therefore positive and the gain cell shorter than one, and $V_k > 0$ requires a nonempty gain cell, since otherwise every taker has $\varphi_k \leq 0$, and so a loss cell shorter than one. This is (C2). □

At admission thresholds in D, write $w_b \equiv \underline{\alpha}_a - \underline{\alpha}_b$ and $w_c \equiv \underline{\alpha}_b - \underline{\alpha}_c$ for the widths of the two bands. By Lemma D.4 and Lemma D.6(ii), the targets with $V_k > 0$ form an interval $(\underline{x}_k, \bar{x}_k)$, possibly empty. When it is not empty, its floor is positive and solves $V_k(\underline{x}_k) = 0$, which by Equation (18) is the floor condition of Lemma D.5; so k makes unconditional offers only if the floor condition has a root, and the loss cell at its floor is then $\underline{\ell}(w_k)$. Write $(\underline{\ell}_k, \underline{g}_k)$ and $(\bar{\ell}_k, \bar{g}_k)$ for the loss and gain cells at the floor and at the ceiling, $L_k = \underline{\ell}_k - \bar{\ell}_k$ for the length of the interval (zero when it is empty), and n_c for the mass of c's band.

Lemma D.7 (The rule is slack). In every equilibrium, with or without the rule, $L_b < \frac{1}{4}$ and $L_c < 0.23$. So each program's takers have mass below $\frac{1}{4}$, the rule does not bind, and the equilibria with and without the rule are the same.

Proof. Range lengths. By Lemma D.5(i), $L_k \leq \frac{1 - w_k}{2} - \bar{\ell}_k$. Since $U(u_d) = 0$, the lottery utility at target x is $U(u_a)[\omega_c \Pr(A \geq \underline{\alpha}_c) + \omega_b \Pr(A \geq \underline{\alpha}_b) + \omega_a \Pr(A \geq \underline{\alpha}_a)]$. At target 1 each of these probabilities is at least $1/\sqrt{2}$ on D, so the lottery utility there is at least $U(u_a)/\sqrt{2}$, which exceeds $U(u_c)$ since $\omega_c < 1/\sqrt{2}$, and $W_c(\bar{x}_c) = U(u_c)$. At c's ceiling the probabilities are $1 - \bar{\ell}_c$, $\max\{1 - \bar{\ell}_c - w_c, 0\}$, and $\max\{1 - \bar{\ell}_c - w_c - w_b, 0\}$, so $\omega_c \bar{\ell}_c = \omega_b \max\{1 - \bar{\ell}_c - w_c, 0\} + \omega_a \max\{1 - \bar{\ell}_c - w_c - w_b, 0\}$. The right side falls in $\bar{\ell}_c$ and is at least its value without the maxima, so $\bar{\ell}_c \geq \omega_{ab}(1 - w_c) - \omega_a w_b$. At b's ceiling the probabilities are $\min\{1 - \bar{\ell}_b + w_c, 1\}$, $1 - \bar{\ell}_b$, and $\max\{1 - \bar{\ell}_b - w_b, 0\}$, and $W_b(\bar{x}_b) \leq U(u_b)$, with equality unless $\bar{x}_b = 1$, so $\omega_a \max\{1 - \bar{\ell}_b - w_b, 0\} \leq \omega_b \bar{\ell}_b + \omega_c \max\{\bar{\ell}_b - w_c, 0\}$. The left side is at least $\omega_a(1 - \bar{\ell}_b - w_b)$. If $\bar{\ell}_b < w_c$, the right side is $\omega_b \bar{\ell}_b$, which gives $\bar{\ell}_b \geq \frac{\omega_a}{\omega_{ab}}(1 - w_b)$; otherwise it is $(1 - \omega_a)\bar{\ell}_b - \omega_c w_c$, which gives $\bar{\ell}_b \geq \omega_a(1 - w_b) + \omega_c w_c$. So $\bar{\ell}_b$ is at least the smaller of the two bounds; and if

$w_c \le \frac{\omega_a}{\omega_{ab}}(1-w_b)$, the first case is impossible, so $\bar{\ell}_b \ge w_c$, that is, $\bar{x}_b \le \underline{\alpha}_c$. Hence

$$\begin{aligned} L_c &\le \hat{L}_c, & \hat{L}_c &\equiv \left(\tfrac{1}{2}-\omega_{ab}\right)(1-w_c)+\omega_a w_b, \\ L_b &\le \max\left\{\hat{L}_b,\ \left(\tfrac{1}{2}-\tfrac{\omega_a}{\omega_{ab}}\right)(1-w_b)\right\}, & \hat{L}_b &\equiv \left(\tfrac{1}{2}-\omega_a\right)(1-w_b)-\omega_c w_c. \end{aligned} \tag{21}$$

By Lemma D.6(i), $w_b \le 2-\sqrt{2}$, so $L_c \le \frac{1}{2}-\omega_{ab}+\omega_a(2-\sqrt{2}) < 0.23$, and $\frac{1}{2}-\frac{\omega_a}{\omega_{ab}} < 0.06$. It remains to show $\hat{L}_b < \frac{1}{4}$, that is, $(\frac{1}{2}-\omega_a)w_b+\omega_c w_c > \frac{1}{4}-\omega_a$.

Band mass. Program c enrolls $\frac{1}{4}$: the students in its band, less those who take b's offer, plus its own takers from outside its band. An offeree at a target t in c's range has A uniform on $[t, t+1]$, which contains c's band because $t < \underline{\alpha}_c$ and $t+1-\underline{\alpha}_b \ge \underline{g}_c > 0$, so she lies outside the band with probability $1-w_c$; and c's offers lie in $(\underline{x}_c, \bar{x}_c)$. Hence $n_c \ge \frac{1}{4}-(1-q)(1-w_c)L_c$. The density of A is at most one, so $n_c \le w_c$, and therefore $w_c - \frac{1}{4} + (1-w_c)\hat{L}_c \ge 0$. On the triangle $(\frac{1}{2}-\omega_a)w_b+\omega_c w_c \le \frac{1}{4}-\omega_a$, the left side rises in both widths, so it is largest on the edge where the inequality binds; there it is a convex quadratic in w_b, equal to -0.018 at $w_b = 0$ and -0.077 at $w_b = (\frac{1}{4}-\omega_a)/(\frac{1}{2}-\omega_a)$. So no equilibrium has widths in the triangle, and $L_b < \frac{1}{4}$.

A program's takers have mass at most the length of the targets at which it offers, which lie in $(\underline{x}_k, \bar{x}_k)$, so each program's takers have mass below $\frac{1}{4}$ and the rule does not bind. A program's best response with the rule then coincides with its best response without it, so the two models have the same equilibria. □

Proposition D.8 (Existence). In the baseline economy, for every $p \in [0,1]$, $s \ge 0$, and $q \in [0,1)$, an equilibrium exists.

Proof. We find an equilibrium with the rule; by Lemma D.7, it is an equilibrium.

Offers. Fix admission thresholds in D. The lottery utility is continuous in the target and the thresholds and nondecreasing in the target, and it is strictly increasing wherever it equals $U(u_b)$ or $U(u_c)$. There, x and $x+1$ lie in different bands: if $[x, x+1]$ lay within one band, that band would be d's or a's, since the bands of b and c are narrower than one, and the lottery utility would be $U(u_d)$ or $U(u_a)$. So each ceiling, the target at which the lottery utility reaches $U(u_k)$, or 1 if it does not, is positive (Lemma D.6(ii)) and continuous in the thresholds. Below the ceiling, V_k is continuous in the target and the thresholds, and strictly increasing in the target when $p < 1$ (Lemma D.4); since $V_k(0) < 0$ (Lemma D.6(ii)), the floor $\underline{x}_k$ is continuous in the thresholds and reaches the ceiling as the interval $(\underline{x}_k, \bar{x}_k)$ vanishes, and when $p = 1$ the interval is empty. Program a makes no offers (the remark after Equation (18)). In equilibrium with the rule (Appendix D.1), b offers on the part of $(\underline{x}_b, \bar{x}_b)$ nearest its ceiling that its capacity allows. Since $U(u_b) > U(u_c)$ and the lottery utility is nondecreasing, b's ceiling is at least c's, so b's offers cover the top of c's interval, if any of it, and c offers on the part of $(\underline{x}_c, \bar{x}_c)$ below b's offers, nearest its upper end, that its capacity allows. The mass of takers on a range of targets is continuous in its endpoints and in the thresholds, and strictly increasing in its length, so the ranges and the masses M_b and M_c of the two programs' takers are continuous in the thresholds.

Clearing. Given the thresholds and these offers, let $R(\alpha)$ be the mass of students with $A \geq \alpha$ who take no unconditional offer; it is continuous in α and the thresholds jointly, and nonincreasing in α. The seats at j and above are filled by the takers at those programs and by the students who take no unconditional offer and whose ability is at least j's admission threshold, so admission thresholds $\underline{\alpha}'$ clear the market for these takers exactly when

$$R(\underline{\alpha}'_a) = \tfrac{1}{4}, \qquad R(\underline{\alpha}'_b) = \tfrac{1}{2} - M_b, \qquad R(\underline{\alpha}'_c) = \tfrac{3}{4} - M_b - M_c. \tag{22}$$

Since $M_b, M_c \leq \frac{1}{4}$ by the rule, the right sides rise from $\frac{1}{4}$ to less than $R(0) = 1 - M_b - M_c$, so the solutions of each equation form a nonempty closed interval, and the three intervals are ordered. Consider the correspondence that maps the thresholds to the set of ordered solutions $\underline{\alpha}'$ of Equation (22). This set is nonempty, convex, and closed, the correspondence has a closed graph because both sides of each equation are continuous in the thresholds, and by the argument for Lemma D.6(i) it maps D into itself.

Fixed point. By Kakutani's theorem (Kakutani, 1941), the correspondence has a fixed point in D. There the programs' offers are equilibrium responses with the rule, students best respond (Lemma D.2), and the admission thresholds clear the market, so the fixed point is an equilibrium with the rule, and by Lemma D.7 it is an equilibrium. □

The scan in Appendix D.3 and the proof of Proposition C.1 use the following lemma.

Lemma D.9 (Disjoint ranges). In every equilibrium in which both b and c make unconditional offers, $\bar{x}_c \leq \underline{x}_b$: c's range ends at its ceiling, below b's floor.

Proof. Suppose, for contradiction, that the ranges overlap: $\underline{x}_b < \bar{x}_c$, that is, $\underline{\ell}_b > w_c + \bar{\ell}_c$. By Lemma D.5(i) and the bound on $\bar{\ell}_c$ in the proof of Lemma D.7, this requires $(\frac{1}{2} - \omega_a)w_b + \omega_c w_c < \frac{1}{2} - \omega_{ab}$. On that triangle $w_c < \frac{\omega_a}{\omega_{ab}}(1 - w_b)$, so $\bar{x}_b \leq \underline{\alpha}_c$. Also b's ceiling is below 1: by Lemma D.5(i) each offeree has $A < \underline{\alpha}_c$ with probability at most $\frac{1}{2}$, so by Lemma D.7 the takers with $A < \underline{\alpha}_c$ have mass at most $\frac{1}{2}(L_b + L_c) < 0.24$; then $1 - F(\underline{\alpha}_c) > 0.51$, so $\underline{\alpha}_c < 1$, and $W_b(1) \geq U(u_a)(\omega_c + \omega_{ab}/\sqrt{2}) > (1 - \omega_a)\,U(u_a) = U(u_b)$. The lottery utility rises at most at rate $U(u_a) - U(u_d)$, so $\bar{x}_b - \bar{x}_c \geq \omega_b$, and at target $\underline{\alpha}_b - 1$ it is at most $U(u_c)$, so $\bar{x}_c \geq \underline{\alpha}_b - 1$. Hence b makes offers at every target in $[\bar{x}_c, \bar{x}_b)$ (Appendix D.1 and Lemma D.7), an offeree there has A in c's band with probability w_c, and b's takers include $(1 - q)\,w_c\,\omega_b$ in c's band. As in the proof of Lemma D.7, $w_c \geq n_c \geq \frac{1}{4} - (1 - q)[(1 - w_c)L_c - \omega_b w_c] \geq \frac{1}{4} - \max\{(1 - w_c)\hat{L}_c - \omega_b w_c,\ 0\}$. On the triangle the difference of the two outer sides rises in both widths and is convex along the edge $(\frac{1}{2} - \omega_a)w_b + \omega_c w_c = \frac{1}{2} - \omega_{ab}$, where it is largest at $w_b = 0$, at -0.002. So the outer inequality fails, a contradiction: the ranges do not overlap, and c's range ends at its ceiling. □

D.5 The taker–decliner achievement gap

In Proposition C.1, programs choose their unconditional-offer ranges knowing p and s. Even so, the students who take an offer differ from what their observables predict, and from those who decline it. Fix an adopter k and a target x in its range, and write $g_k(x)$ for the length of the gain cell, the share of offerees at x who would have placed above k.

Proposition D.10 (The taker–decliner achievement gap). In any equilibrium of Proposition C.1, at every target x in adopter k's unconditional-offer range:

(i) Take-up is $(1-q)\,(1-p\,g_k(x))$: private information lowers it, moral hazard does not.

(ii) Takers fall short of the ability their observables project by

$$\underbrace{\frac{p\,g_k(1-g_k)}{2(1-p\,g_k)}}_{\text{private information}} + \underbrace{s}_{\text{moral hazard}},$$

where the first term is positive if and only if $p > 0$. Either force alone produces a shortfall, and, at given admission thresholds, the same shortfall arises from a continuum of (p, s) pairs.

(iii) If $p > 0$ or $q > 0$, so that some holders decline, takers score below decliners by

$$\Delta_k(x) \;=\; \underbrace{\frac{p\,g_k(1-g_k)}{2}\left[\frac{1}{1-p\,g_k} + \frac{1-q}{q+(1-q)\,p\,g_k}\right]}_{\text{private information}} + \underbrace{s}_{\text{moral hazard}}.$$

If $q > 0$, then at given admission thresholds the same gap arises from a continuum of (p, s) pairs.

(iv) Each adopter's floor breaks even in realized ability, so its range already accounts for the shortfall. A program above the highest adopter enrolls a weaker intake than in the benchmark whenever a positive mass of uninformed takers has ability in its benchmark band; under the setup of Figure C.1 this holds for every program above the highest adopter, and, by computation, b's intake is also weaker when it adopts. Total attainment falls by s times the mass of takers.

Under private information the decliners include the informed students who would have placed above k; under moral hazard alone they are the q who decline for other reasons, and the gap is s. At the calibration of Figure C.1, $p = 0.104$ alone and $s = 0.0567$ alone each produce a gap of -0.139SD, and together -0.278SD. Private information redistributes: at $p = 0.104$, c gains 0.040 in mean intake ability (0.050 with neither force), a loses 0.032, and b loses 0.025 although it adopts, because c's offers capture the top of b's band; the students who take the outside option gain 0.018. Moral hazard also lowers the total: at $s = 0.0567$, c's gain from adopting shrinks to

0.006, b withdraws and still loses 0.024, and total attainment falls by s times the mass of takers (the box in Appendix D.3).

D.6 Proofs of the propositions

Proof of Proposition C.1. Step 1: who adopts. Program a makes no offers by the remark after Equation (18). For $k \in \{b, c\}$, Equation (18) gives $V_k \geq 0$ if and only if

$$(1-p)\, g_k(x)^2 - \ell_k(x)^2 \;\geq\; 2s\,(1 - p\, g_k(x))\,,$$

which at $s = 0$ is $\sqrt{1-p}\, g_k(x) \geq \ell_k(x)$, the gain cell at least $1/\sqrt{1-p}$ times the loss cell. We take $s = 0$ for the closed forms and return to s in Step 2. Since V_k is strictly increasing in x below the ceiling and nonpositive above it, the strictly profitable targets form the interval $\left(\underline{x}'_k(p), \bar{x}'_k\right)$ with the floor

$$\underline{x}'_k(p) \;=\; \frac{\underline{\alpha}'_k - \sqrt{1-p}\;(1-\bar{\alpha}'_k)}{1+\sqrt{1-p}},$$

the break-even target, excluded by convention. At $p = 0$ the floor is $\underline{x}'_k = \frac{1}{2}(\underline{\alpha}'_k + \bar{\alpha}'_k) - \frac{1}{2}$, the target whose two cells are equal; at the baseline, $\underline{x}'_c = 0.354 < \bar{x}'_c = 0.495$ and $\underline{x}'_b = 0.646 < \bar{x}'_b = 0.698$, so both middle programs adopt (C3).[44] The two ranges are disjoint, c's below b's: $\bar{x}'_c < \underline{x}'_b$ at the benchmark, and $\bar{x}_c < \underline{x}_b$ at every computed equilibrium (Appendix D.3). These are the unconditional-offer ranges of the marginal analysis; the equilibrium ranges differ, because they shift the admission thresholds (b's is wider and c's narrower at $p = s = 0$), and are reported in Appendix D.3.

Step 2: shrinking and exit. The floor is strictly increasing in s and in p. It solves $h(x) \equiv (1-p)\, g_k(x)^2 - \ell_k(x)^2 - 2s\,(1 - p\, g_k(x)) = 0$, where h is increasing in x and decreasing in s, and $\partial h/\partial p = -g_k\,(g_k - 2s)$, which is negative at the root because there $g_k - 2s = \left[g_k(1-g_k) + \ell_k^2\right]/(1 - p\, g_k) > 0$. The ceiling depends on neither force, so the range shrinks from below as either force grows. At $s = 0$ it is nonempty exactly while the ceiling target still pays, $\sqrt{1-p}\, g_k(\bar{x}'_k) > \ell_k(\bar{x}'_k)$, that is,

$$p \;<\; \hat{p}'_k \;=\; 1 - \left[\frac{\ell_k(\bar{x}'_k)}{g_k(\bar{x}'_k)}\right]^2.$$

At the baseline, $\hat{p}'_b = 0.443$ and $\hat{p}'_c = 0.816$, so the ladder is $\{b, c\} \to \{c\} \to \emptyset$. Program c's exit is exact at any scale, $\hat{p}_c = \hat{p}'_c$, because b exits first at the baseline, so no unconditional-offer range remains at c's exit and the marginal offer faces the benchmark admission thresholds; b's is not, since b exits into a market where c still makes unconditional offers, and Appendix D.3 computes $\hat{p}_b = 0.575$. In s, the ceil-

[44] (C3) is a joint condition on preferences and geometry: a low enough u_k pushes $\bar{x}'_k$ below the floor and removes k from the adopter set, and for some utility vectors so does weak enough risk aversion, though not for $u = (3, 2.5, 2, 1)$, where the ceilings stay above the floors even for a risk-neutral student. It fails for a under every preference, since a's gain cell is empty and no target is profitable for a (the remark after Equation (18)).

ing target pays exactly while $s < \hat{s}'_k(p) = [(1-p)\,g_k^2 - \ell_k^2] \,/\, [2(1-p\,g_k)]$ evaluated at $\bar{x}'_k$, an exit threshold that is positive exactly for $p < \hat{p}'_k$ and strictly decreasing in p, since

$$\frac{d\hat{s}'_k}{dp} \;=\; -\frac{g_k\,[g_k(1-g_k)+\ell_k^2]}{2(1-p\,g_k)^2} \;<\; 0;$$

at $p = 0$ it is $G_k(\bar{x}'_k)$, 0.100 for c and 0.036 for b. As with p, c's exit threshold is exact at scale and b's is not: Appendix D.3 computes $\hat{s}_b = 0.060$ at $p = 0$.

Step 3: no offers beyond the exit, at any scale. At $p = 1$ every holder is informed, so by Lemma D.3, at any admission thresholds and every target, $V_k = (1-q)\left[\int_0^1 \min\{\varphi_k(x+v_2), 0\}\,dv_2 - s\,\tau_k(x)\right] \le 0$. No target is strictly profitable, so no program adopts, and the outcome is Lemma D.1.[45] For $p \ge \hat{p}_c$, or $s \ge \hat{s}_k(p)$ for each k, the offer at each program's benchmark ceiling has nonpositive value (for b because $\hat{p}'_b < \hat{p}_c$ and, at every p we compute, $\hat{s}'_b(p) < \hat{s}_b(p)$), so by Lemma D.4 the benchmark is an equilibrium, and at every such point we compute it is the only one (Appendix D.3).

Positive scale. Steps 1 and 2 establish the shape of the ranges at the margin, for the exit thresholds $\hat{p}'_k$ and $\hat{s}'_k$. At positive scale, the ranges in (i) are intervals whose floors break even and which are disjoint, by Lemmas D.4, D.6, D.7 and D.9, and in (ii) the facts that the ranges are longest at $p = s = 0$, that b offers if and only if $s < \hat{s}_b(p)$, that c offers if and only if $s < \hat{s}_c(p)$ (which needs the computed fact that b never offers alone), that no program offers beyond the exit thresholds (Step 3), and that $\hat{s}_b$ falls in p, rest on the computation in Appendix D.3, which covers $q = 0$ and the setup of Figure C.1, $q = 0.2$. □

Remark (the order of exit). The order of exit is the same under both forces, and one number decides it. With equal capacities the bands have a common width $\bar{w} = \underline{\alpha}'_a - \underline{\alpha}'_b = \underline{\alpha}'_b - \underline{\alpha}'_c$, so at any target $\ell_k + g_k = 1 - \bar{w}$ for $k \in \{b, c\}$, and both exits are decreasing functions of the loss cell at the ceiling alone: $\hat{p}'_k = 1 - [\ell_k/(1-\bar{w}-\ell_k)]^2$, and $\hat{s}'_k(p)$ has numerator $(1-p)(1-\bar{w}-\ell_k)^2 - \ell_k^2$, falling in ℓ_k, over denominator $2\,[1-p(1-\bar{w}-\ell_k)]$, rising in it. Hence b exits first, in p and in s at every p, if and only if $\ell_b(\bar{x}'_b) > \ell_c(\bar{x}'_c)$: its ceiling lies deeper below its own admission threshold, because the insurance it sells, against falling to c, is worth less to an uninformed student than the insurance c sells, against falling to d. At the baseline $0.302 > 0.212$; the inequality, and with it both orders at once, reverses at $u_b = 2.63$. The nesting is a consequence of equal capacities, not of the two forces. The s-intercept $G_k(\bar{x}'_k) = \frac{1}{2}(g_k^2 - \ell_k^2)$ scales with the size of the cells, while the p-intercept $1-(\ell_k/g_k)^2$ depends only on their ratio; with a common ℓ_k+g_k the two move together. With unequal capacities $\ell_k + g_k$ differs across programs, a program can have both cells longer than the other's, and the exit thresholds can cross.[46]

Proof of Proposition D.10. Fix a target x in k's range, so $x < \bar{x}_k$, with cells evalu-

[45] If v_2 were public, an unconditional offer could condition on A and would be a conditional offer, so Lemma D.1 would again apply.

[46] At capacities $(0.4, 0.2, 0.2, 0.2)$ and $u_b = 2.6$, c exits first in s ($\hat{s}_c(0) = 0.089 < \hat{s}_b(0) = 0.100$) but b exits first in p (the exit thresholds cross near $p = 0.6$), marginally and at scale alike; such crossings occur only within a narrow band of u_b around the point where the ordering flips.

ated at the equilibrium admission thresholds (Lemma D.2(iii)). Of the holders at x, a share q decline regardless; of the rest, the uninformed accept, and the informed accept exactly on the loss and neutral cells, $v_2 < 1 - g_k$, by Lemma D.2(ii). Take-up is therefore $(1-q)\left[(1-p) + p(1-g_k)\right] = (1-q)(1-p\,g_k)$, which is (i); s does not enter because a taker's score is realized after she accepts.

For (ii), v_2 is uniform on $[0,1]$ at every target, so the observable projection of a holder's v_2 is $\frac{1}{2}$. Takers have mass $(1-q)(1-p\,g_k)$ and mean

$$v_2 \;=\; \frac{(1-p)\,\frac{1}{2} + p\,\frac{(1-g_k)^2}{2}}{1-p\,g_k} \;=\; \frac{1}{2} - \frac{p\,g_k(1-g_k)}{2(1-p\,g_k)},$$

and score s less than that. The private-information part is positive exactly when $p > 0$, since $g_k \in (0,1)$ by (C2); at given admission thresholds, so that g_k is fixed, the shortfall is continuous and strictly increasing in both p and s, so each of its level sets in the (p,s) plane is a curve.

For (iii), decliners have mass $q + (1-q)\,p\,g_k$: the q who decline regardless, with mean v_2 of $\frac{1}{2}$, and the informed on the gain cell, with mean $1 - g_k/2$; their mean

$$v_2 \;=\; \frac{\frac{q}{2} + (1-q)\,p\,g_k\left(1 - \frac{g_k}{2}\right)}{q + (1-q)\,p\,g_k} \;=\; \frac{1}{2} + \frac{(1-q)\,p\,g_k(1-g_k)}{2\left(q + (1-q)\,p\,g_k\right)}.$$

Subtracting the takers' realized mean gives $\Delta_k(x)$. At $p = 0$ the decliners are the q alone and the gap is s; at $q = 0$ the private-information term is $(1-g_k)/\left[2(1-p\,g_k)\right]$, which does not vanish as $p \to 0$, because the only decliners are then the informed on the gain cell however few they are. For $q > 0$ and at given admission thresholds, the gap Δ_k is continuous and strictly increasing in both p and s, so the same gap arises along a curve of (p,s) pairs.

For (iv), the floor condition $V_k(\underline{x}_k) = 0$ of Lemma D.4 is stated in realized values, so the equilibrium range already nets out both the informed decliners and the s. Let j be a program above the highest adopter. Only the captured-from-above sum enters j's condition in Equation (19), so $\underline{\alpha}_j < \underline{\alpha}'_j$ (Appendix D.3), and the same holds for every program above j, so $\bar{\alpha}_j \le \bar{\alpha}'_j$. Its intake is therefore a set of mass $\frac{1}{4}$ inside $\{A < \bar{\alpha}'_j\}$ that excludes, by hypothesis, a positive mass of students in its benchmark band $[\underline{\alpha}'_j, \bar{\alpha}'_j)$: uninformed takers of adopters below j. Under the setup of Figure C.1 such takers exist for every such j, because at every equilibrium we compute, every target in an adopter's range exceeds $\underline{\alpha}'_a - 1 = 1 - 1/\sqrt{2}$, so an uninformed taker's ability, uniform on $[x, x+1]$, reaches every benchmark band above the adopter's. The benchmark intake $[\underline{\alpha}'_j, \bar{\alpha}'_j)$ is the unique mass-$\frac{1}{4}$ subset of $\{A < \bar{\alpha}'_j\}$ with the highest mean, so j's mean realized ability is below it. For b as an adopter the comparison is numerical: its mean intake ability is below the benchmark in panels (a) and (b) of Figure C.1 and at every p in the table of Appendix D.3. Finally, every student enrolls somewhere and capacities sum to one, so the sum of realized intake abilities is the total ability of the population less s times the mass of takers. □

E Optimality of the Application Portfolio under Correlated Admissions

The Stage-1 estimator of Section 4.3 requires the observed portfolio to satisfy the one-shot-swap (OSS) inequalities of Larroucau and Rios (2020). As noted in Section 4.3.2, under nested admission (Assumption 2) these inequalities are necessary for optimality but not sufficient; the exact conditions are the dynamic-programming inequalities of Ali and Shorrer (2025). This appendix writes both sets of inequalities in one notation, characterizes the gap between them, and uses it to justify our use of the OSS inequalities in estimation.

E.1 Setup

Fix a student and drop the subscript i. Following Implication 1, program s has Stage-1 utility $u_s \equiv U_{is}$ and admission probability $q_s \equiv q_{is}$, the joint probability that s makes her an offer and admits her (Section 4.3.1). As in Implication 1, admission events are nested: a student admitted to a program is also admitted to every program with a higher q_s.

Nested events with these probabilities are generated by a single score. Let S be uniform on $[0,1]$, and let the student be admitted to s if and only if $S < q_s$, so a more selective program has a lower q_s. Because one score governs every admission, rejection is informative. After rejection by program s', the score is uniform on $[q_{s'}, 1]$, so for any less-selective program s ($q_s > q_{s'}$) the admission probability given rejection by s' is

$$\frac{q_s - q_{s'}}{1 - q_{s'}}.$$

After rejection by every program in a set $\mathcal{R}$, the score is uniform on $[\max_{s\in\mathcal{R}} q_s,\ 1]$, which depends only on the least-selective program that rejected the student.

For a portfolio $A = \{s_1, \dots, s_m\}$ with $q_{s_1} \le \cdots \le q_{s_m}$, and a student who knows only that $S \ge t$, the expected value of the best admission is

$$V(A;\, t) \;=\; \mathbb{E}\Big[\max_{s\in A,\ S<q_s} u_s \;\Big|\; S \ge t\Big] \;=\; \frac{1}{1-t}\sum_{l=1}^{m}\big[\max\{q_{s_l}, t\} - \max\{q_{s_{l-1}}, t\}\big]\max_{j\ge l} u_{s_j}, \tag{23}$$

with $q_{s_0} \equiv 0$. At $t = 0$ this is the portfolio value of Implication 1. Let $A^* = \{s_1^*, \dots, s_K^*\}$ be the observed portfolio, indexed so that $q_{s_1^*} < \cdots < q_{s_K^*}$; by Implication 2, $u_{s_1^*} > \cdots > u_{s_K^*}$, so the inner maximum in Equation (23) is $u_{s_l^*}$. Let $\tilde{V}(n;\, t) = \max_{|A|=n} V(A;\, t)$ be the value of the best n-program portfolio for a student who knows that $S \ge t$, and let $A^*_{m+1:K} = \{s^*_{m+1}, \dots, s^*_K\}$ be the observed programs less selective than s^*_m.

E.2 The two sets of inequalities

E.2.1 Larroucau–Rios one-shot swaps (LR)

When admissions are independent across programs, the Marginal Improvement Algorithm of Chade and Smith (2006) solves the portfolio problem, and Larroucau and Rios (2020) show that optimality of A^* can be characterized by a polynomial-sized set of one-shot-swap inequalities: for each

$s \in A^*$ and each $\ell \notin A^*$,

$$V(A^*;\,0) \;\geq\; V((A^* \setminus \{s\}) \cup \{\ell\};\,0)\,. \tag{LR}$$

These $K(|\mathcal{S}| - K)$ inequalities compare portfolios that differ in a single program, each valued at the prior. They are the inequalities of Equation (16), which our estimator imposes with V evaluated under nested admission, as here.

E.2.2 Ali–Shorrer dynamic programming (AS)

Under nested admission, the optimal portfolio instead satisfies the dynamic-programming recursion of Ali and Shorrer (2025): at each level $m \in \{1, \dots, K\}$, the choice s_m^* must be optimal given rejection by all more selective programs already chosen. Let $t_m \equiv q_{s_{m-1}^*}$, with $q_{s_0^*} \equiv 0$. Then, for each level m and each alternative j with $q_j > t_m$,

$$\begin{aligned} &\frac{q_{s_m^*} - t_m}{1 - t_m}\, u_{s_m^*} + \frac{1 - q_{s_m^*}}{1 - t_m}\, \tilde{V}\big(K - m;\, q_{s_m^*}\big) \\ &\qquad\geq\; \frac{q_j - t_m}{1 - t_m}\, u_j + \frac{1 - q_j}{1 - t_m}\, \tilde{V}\big(K - m;\, q_j\big)\,. \end{aligned} \tag{AS}$$

The left-hand side is the value of choosing s_m^* at level m: it admits the student with probability $(q_{s_m^*} - t_m)/(1 - t_m)$, and upon rejection she continues *optimally* knowing that $S \geq q_{s_m^*}$. The right-hand side is the same value for choosing j in place of s_m^*.

E.3 The gap

The LR inequalities hold the observed tail $A^*_{m+1:K}$ fixed, while the AS inequalities reoptimize the continuation after rejection. To compare them, condition on rejection by $s_1^*, \dots, s_{m-1}^*$, which both portfolios share, and write the swap of s_m^* for j with the tail held fixed:

$$\begin{aligned} &\frac{q_{s_m^*} - t_m}{1 - t_m}\, u_{s_m^*} + \frac{1 - q_{s_m^*}}{1 - t_m}\, V\big(A^*_{m+1:K};\, q_{s_m^*}\big) \\ &\qquad\geq\; \frac{q_j - t_m}{1 - t_m}\, u_j + \frac{1 - q_j}{1 - t_m}\, V\big(A^*_{m+1:K};\, q_j\big)\,. \end{aligned} \tag{LR′}$$

When $t_m < q_j \leq q_{s_{m+1}^*}$ and $u_{s_{m+1}^*} \leq u_j \leq u_{s_{m-1}^*}$, (LR′) is the one-shot swap of s_m^* for j in Equation (LR), divided by $1 - t_m$; the upper bound on u_j holds whenever A^* satisfies Equation (LR), since otherwise j would dominate s_{m-1}^*. Comparing Equation (LR′) with Equation (AS), the right-hand side of (AS) exceeds that of (LR′) by

$$\Delta_m(j) \;\equiv\; \frac{1 - q_j}{1 - t_m}\Big[\tilde{V}\big(K - m;\, q_j\big) - V\big(A^*_{m+1:K};\, q_j\big)\Big] \;\geq\; 0, \tag{24}$$

since $\tilde{V}(K - m;\, q_j)$ is the value of the *optimal* $(K - m)$-program portfolio given rejection by j, which weakly exceeds the value of the fixed tail. The left-hand sides coincide when A^* satisfies (AS) at every level: by backward induction the observed tail is then the optimal continuation, $\tilde{V}(K - m;\, q_{s_m^*}) = V(A^*_{m+1:K};\, q_{s_m^*})$. Hence (AS) implies (LR′), and the slack in (LR′) exceeds the slack in (AS)

by exactly $\Delta_m(j)$: the OSS inequalities check only whether a single swap improves on the observed portfolio holding the rest fixed, whereas the exact condition lets the continuation be reoptimized after the swap.

The gap arises only because rejection is informative: when admissions are independent, rejection carries no information, and the OSS inequalities characterize the optimum exactly (Larroucau and Rios, 2020). Under nested admission, $\Delta_m(j)$ is scaled by $(1-q_j)/(1-t_m)$, the probability that program j rejects the student given rejection by s^*_{m-1}. Because the bracket in Equation (24) is at most $\max_s u_s$, $\Delta_m(j) \leq \frac{1-q_j}{1-t_m} \max_s u_s$, which is small for safety programs with high admission probability, whatever the reoptimization gain.

Rejection by a program with admission probability t moves the chance of admission to any other program by at most t: for $q > t$ the change is $t(1-q)/(1-t) \leq t$, and for $q \leq t$ the chance falls from q to zero. Because students condition on rejection by their most selective programs, where $\hat{q}_{is}$ is lowest, the posteriors in Equation (AS) stay close to the prior, and the OSS inequalities closely approximate the exact conditions. Equivalently, one may interpret the estimator as identifying preferences from the requirement that the observed portfolio be a *local* optimum of the portfolio problem (for which the OSS inequalities are necessary) rather than a global one, with Equation (24) bounding the difference between the two.

F Structural Model Estimation Details

This appendix records how the demand-side estimator of Section 4.3 is implemented: the construction of the revealed-preference cone, the Gibbs sampler, the treatment of estimated admission beliefs, and the warm start. We write $\mathcal{L}_i(p)$ for the matrix of student i's revealed-preference inequalities (the constraints $\mathcal{L}_i$ of Section 4.3.2) evaluated at the offer probabilities p, so the augmented utilities satisfy the cone $\mathcal{L}_i(p)\, U_i > 0$.

F.1 Constraint construction

For each student the observed portfolio implies three blocks of rows (Section 4.3.2). The ranking rows order utilities inversely to the admission probabilities $\hat{q}_{is}$ (Implication 2). The positivity rows require $U_{is} > 0$ for $s \in A_i$. For a student with five programs, the one-shot-swap rows of Equation (16) compare A_i with every portfolio that replaces one chosen program by one non-chosen program, so the comparison set is the complement of the portfolio rather than the set of all portfolios; each swap row involves the chosen programs and exactly one non-chosen alternative, which has a single negative coefficient. For a student who applies to fewer than five programs, the addition rows instead compare A_i with every portfolio that adds one non-chosen program ℓ; under nested admission each reduces to $U_{i\ell} \leq U_{is}$, where s is the chosen program just less selective than ℓ, or to $U_{i\ell} \leq 0$ if ℓ is less selective than every chosen program. The swap and addition rows use the total acceptance probabilities $\hat{q}_{is} = \hat{p}_{is}[\hat{g}_{is} + (1-\hat{g}_{is})\,\hat{a}^o_{is}]$ of Section 4.3.1, built from the Firth-penalized offer and conversion probits, so both the offer and the conversion probability enter the portfolio stage, as Implication 1 requires. For the ranking stage we also form, for each observed pair of ranked choices, the probability that the second choice converts given that the first does not, by truncating the common μ_i above the first choice's threshold in Equation (8).

F.2 Gibbs sampler

We draw from the joint posterior of $(\delta, \bar{\delta}, \sigma_\delta^2, \bar{\beta}^c, \Gamma^c, \sigma_\eta^2, \sigma_\nu^2, \{\eta_i^c\}, \{U_i\})$ by a blocked sampler. Each sweep alternates:

(a) **Utilities** $U_i \mid \delta, \beta_i^c, \sigma_\nu^2$, truncated to $\mathcal{L}_i(\hat{p})\, U_i > 0$: conditional on the chosen utilities the alternatives are independent and drawn as exact one-dimensional truncated normals, while the (at most five) chosen utilities are drawn jointly by hit-and-run within the cone. The program fixed effects δ and the coefficient $\beta_i^c = \bar{\beta}^c + \Gamma^c \,\text{Low-SES}_i + \eta_i^c$ enter only through this draw's mean.

(b) **Program fixed effects** $\delta \mid U, \beta^c, \sigma_\nu^2$: a conjugate Normal-means update. The residual $U_{is} + d_{is} - \beta_i^c \,\text{close}_{is} - \beta^m m_{is}$ is a noisy observation of δ_s with precision σ_ν^{-2}, so δ_s is drawn from a Normal with posterior precision $n_s \sigma_\nu^{-2} + \sigma_{\bar{\delta}}^{-2}$, where n_s counts the latent utilities in which s appears. The hyperparameters of the prior $\delta_s \sim N(\bar{\delta}, \sigma_\delta^2)$ are themselves drawn (Normal and inverse-Gamma), shrinking thinly applied-to programs toward the population mean rather than letting them take extreme values fit to a handful of students.

(c) **Preference parameters**: η_i^c (Normal), $(\bar{\beta}^c, \Gamma^c)$ in one joint Normal draw, σ_ν^2 (inverse-Gamma), and σ_η^2 under the half-t prior described below. The nesting of Equation (13) is imposed by masking: η^c and σ_η^2 are sampled only in the specification carrying an i.i.d. component, and Γ^c only where the close coefficient loads on Low-SES status.

The cone is homogeneous in U, so the scale is pinned by fixing the linear distance coefficient at $\beta^d = -1$ and estimating σ_ν^2 freely, which denominates utilities in kilometers of travel. For the heterogeneity scales we use a weakly informative half-t prior and update the $(\eta_i^c, \sigma_\eta^2)$ block by an Ancillarity-Sufficiency Interweaving Strategy (ASIS). The same interweave is applied to the $(\delta, \sigma_\delta^2)$ block to improve mixing for both popular and unpopular programs.

F.3 Estimated beliefs

On real data $\mathcal{L}_i(\hat{q})$ is built from the estimated joint acceptance probabilities $\hat{q}_{is}$ of Section 4.3.1, which enter the row *coefficients*, so sampling error that reorders two nearby programs misspecifies rows. As in Section 4.3.2 we treat q as latent and resample it jointly with U, keeping every utility draw over an exact cone. We augment the sampler with an independence-Metropolis step: propose q', recomputed from the same joint formula, by drawing the offer-probit parameters (γ_s, c_s) from their asymptotic distribution, holding the conversion-probit parameters $(\gamma_s^{\text{admit}}, \alpha_s, c_s^R)$ fixed at their point estimates, then accept if and only if the observed portfolio is still optimal under the proposal — with this proposal the Metropolis ratio reduces to that feasibility indicator — rebuilding the rows for accepted students and otherwise keeping the incumbent. The test is reorder-aware, and initializing at $\hat{q}$ nests the fixed-belief sampler. Proposals are drawn separately for each program–cohort cell; cells with perfect separation draw from a Beta$(r+1, n-r+1)$ distribution, where r of the cell's n applicants received offers, and cells without a fitted probit keep a belief of 0.3.

F.4 Warm start

The free-scale sampler is initialized from a pooled preliminary run in which all students share one close-to-home coefficient. The results of this run provide a population mean near the true parameter, per-student deviations of each student's pooled utilities on their characteristics, and a feasible utility vector to begin the estimation procedure. The fixed effects are started at the program-level means of the pooled utilities, shrunk toward the overall mean so that unpopular programs start near $\bar{\delta}$ and popular ones start near their own mean. The pooled run is estimated on a free scale and handed off by a single global rescaling onto $\beta^d = -1$, exact by scale-invariance of the cone.

F.5 Ranking stage

The ranking stage uses the same device. Each candidate ranking's expected value is linear in the program latents, so the requirement that the realized pair maximize expected utility gives linear winner inequalities $\mathcal{W}_i U_i \geq 0$, which we sample with the same hit-and-run kernel. Conditional on the latents, we draw the taste-shock variance σ_ε^2 (inverse-Gamma) and the coefficient λ_i on the unconditional-offer premium (a multivariate-normal regression on z_i of the utility gap that carries the premium, as in Equation (9)). This stage also carries the private signal: ψ_i is augmented per student, and σ_ψ is drawn by slice sampling on $(0, \sigma_r)$ under a half-t prior, given $\{\psi_i\}$ and the realized $\{r_i\}$, with the target set to zero unless every observed ranking remains feasible. Because ψ_i enters the ranking inequalities through $\Phi(\cdot)$ rather than linearly, its draw is a Metropolis step within the block. Stage-1 uncertainty is propagated by drawing each β_i^c and the fixed effects δ from their Stage-1 posterior. This procedure was validated on simulated data.

F.6 Validity of the sequential estimator

Within each block, the estimator is a data-augmentation Gibbs sampler of the kind McCulloch and Rossi (1994) introduced for discrete choice. Where the full conditional is not conjugate (the belief resampling at Stage 1 and the signal draw at Stage 3), we use Metropolis steps. The resulting Metropolis-within-Gibbs kernel leaves the target posterior invariant and is ergodic under the conditions of Tierney (1994). We report posterior means and standard deviations after burn-in. We assess convergence using split-$\hat{R}$, effective sample sizes, and Geweke diagnostics, reported in Table A.7.

We estimate the three blocks in order: the supply side (Stages 2 and 4), then Stage 1, then Stage 3. Each block conditions on the estimates of the blocks before it and carries forward part of their uncertainty. The Stage-1 sampler resamples the offer-probit component of $\hat{q}_{is}$ from its asymptotic distribution and holds the conversion-probit component at its point estimate (Appendix F.3); because a student's belief draw is accepted only if her portfolio stays feasible, her applications restrict the beliefs used in Stage 1, but they do not change the supply-side estimates used anywhere else. The Stage-3 sampler draws β_i^c and δ from their Stage-1 posterior (Appendix F.5), and the rankings do not update the Stage-1 parameters.

This cut rests on two features of the model. First, students hold true beliefs about admission (Section 4.1.2), so the supply-side estimates are also estimates of students' beliefs. Second, Stage-1 beliefs depend on the supply side only through the scalar q_{is}. Its marginal components

$\hat{p}_{is}$ and $\hat{a}_{is}$ are correctly specified whatever the correlation across programs, because the common shock μ_i is integrated out of each marginal (Section 4.3.1). Combining them into $\hat{q}_{is}$ through the conditional conversion probability $\hat{a}^o_{is}$ additionally uses the bivariate-normal structure of μ_i.

The blocks follow estimation order, not the game's timing. The conversion probit is fit on first-ranked choices, so it uses a Stage-3 decision, and its fitted values enter Stage-1 beliefs. This creates no circularity, because $\hat{q}_{is}$ is evaluated for every applicant rather than at the student's own realized ranking. The cost is that the conversion probit uses only first-ranked choices, not the full at-risk sample of Section 4.3.1.

F.7 Run lengths and priors

The Stage-1 block is a Gibbs sampler over $(\delta, \bar{\delta}, \sigma^2_\delta, \bar{\beta}^c, \Gamma^c, \sigma^2_\eta, \sigma^2_\nu, \{\eta^c_i\})$ and the utilities $\{U_i\}$, run for 50,000 iterations with 5,000 discarded as burn-in. Priors are Normal-inverse-Gamma for $(\delta, \sigma^2_\delta)$, Normal for $(\bar{\beta}^c, \Gamma^c)$ with a half-t prior on σ_η, and inverse-Gamma for σ^2_ν. The Stage-3 block is run for 250,000 iterations with 25,000 discarded.

F.8 Counterfactuals

We detail the application system designs used in Section 5.

For each program–cohort we compute the *expected capacity* $\hat{N}_{st}$ that program s expects to fill in cohort t, given the offers it made at Stage 2 in the estimated baseline and its own beliefs about yield and cutoff clearing. Summing over the applicants who hold an offer from s, this is the probability that an applicant ranks s first and clears its condition, plus the probability that she ranks s second, misses her first-choice condition, and clears s's. The second term is computed as a joint rather than as a product of marginals: missing one condition and clearing another are both driven by the same realization of r_i, so the two events are negatively dependent and multiplying the marginals overstates the rebound. We hold $\hat{N}_{st}$ fixed across designs.

For each design, we compute a fixed point of the best-response map in student and program strategies. We iterate until three conditions hold: the weighted average gap between each program's expected intake and $\hat{N}_{st}$ is below a tolerance, aggregate intake is within tolerance of total capacity, and every program's intake is within one sampling standard deviation of $\hat{N}_{st}$. We cap the loop at 1,000 iterations and keep the best iterate among those that meet these conditions. One market under design P of Table 10 reaches the cap without meeting the first condition: its weighted gap is 1.2%, against a tolerance of 1%, while the other two conditions hold. Students choose portfolios at Stage 1 and rankings at Stage 3 to maximize expected utility, as in Section 4.1.4 and Section 4.1.2. Their Stage-1 admission beliefs use the offer cutoffs and unconditional-offer shares that programs play in the design, computed on a preliminary pool in which students apply on tastes alone rather than on the pool that actually applies. Programs choose scalar cutoffs, taking the applicant pool as given, to maximize the expected sum of enrolling students' quality subject to expected enrollment not exceeding $\hat{N}_{st}$, where a student's quality to program s is measured by s's own estimated weights on her characteristics.

Two features of the program side matter. First, the relevant index differs by stage, because the program's information does. At Stage 2 the program has not yet observed r_i (which is why it

makes conditional offers), so it maximizes expected quality given its Stage-2 information, and the offer cutoff $\bar{\pi}_s$ is a threshold on π_{is} in Equation (11). At Stage 4, r_i is realized, and the conversion cutoff $\bar{\pi}_s^R$ is a threshold on the admissions index in Equation (8). Second, following Kapor (2025), a program that maximizes expected enrolled quality subject to capacity uses a threshold rule on its index, so its choice reduces to a scalar cutoff.

Except where noted, the data-generating process in each design is identical to the setup described in Section 4. Program fixed effects $\{\delta_s\}$ are recalibrated in the simulated baseline to match observed application counts and then held fixed across designs; all other preference parameters are held at their estimated values, and the choice set consists of the same set of programs as used in parameter estimation. Some offer holders decline all their offers, or rank no second choice, for reasons outside the model; in each design we add such random declines so that the decline and no-second-choice rates, by number of offers held, match the data.

In simulation, the offer score of Equation (11) is implemented as $\pi_{is} = \kappa_s \tilde{z}_{is} + \mu_i$, where $\tilde{z}_{is}$ is the index $z_i^{\text{offer}\prime}\gamma_s$ centered within each program–cohort and κ_s is a per-program scale fit to how sharply offers separate offer holders from other applicants in the data. The offer threshold varies by program–cohort, $\bar{\pi}_{st}$, so an offer from s is the event $\mu_i \geq L_{is} \equiv \bar{\pi}_{st} - \kappa_s \tilde{z}_{is}$; students update on these thresholds at Stage 3.

1. **Ban on unconditional offers.** In Stage 2 we set $\bar{\pi}_s^U = \infty$ for all program–cohorts, so that every offer is conditional and every first choice is tested against $\bar{\pi}_s^R$. Programs re-optimize $\bar{\pi}_s$ and $\bar{\pi}_s^R$.

2. **Unconditional behavior consistent with 2018.** In Stage 2, $\bar{\pi}_s^U$ is set program–cohort by program–cohort so that the share of s's offers that are unconditional equals its 2018 share; what is held fixed across cohorts is the share rather than the threshold, since the applicant pool differs by cohort. Programs re-optimize $\bar{\pi}_s$ and $\bar{\pi}_s^R$ subject to that constraint.

3. **Post-qualification admissions.** There is a single application stage, taken after age-18 results are released, so that r_i is common knowledge. Three things follow. The signal ψ_i is moot, since students and programs observe r_i alike. The conditional–unconditional distinction disappears, since there is no future performance left to condition on: every place a program offers is unconditional, and the single cutoff $\bar{\pi}_s^R$ on the admissions index of Equation (8) is the admission decision. Because application and ranking now coincide, a student knows her full idiosyncratic taste $v_{is} + \varepsilon_{is}$ when she chooses, rather than only v_{is}. Programs rank the full pool of applicants by that index and admit to capacity; students rank programs; the outcome is the stable matching, which we compute as the student-proposing deferred acceptance outcome. The only uncertainty a student faces at the point of choosing is her own μ_i, which programs observe and she does not.

4. **Decomposition designs.** Post-qualification admissions combine three changes to the ban, and Table 10 simulates every combination of them.

    - *S, student information.* Each student knows her age-18 result r_i and her Stage-3 taste shocks ε_{is} when she applies, so the signal ψ_i is redundant.

- *P, program information.* Programs make Stage-2 offers on the admissions index of Equation (8), which uses achieved scores, instead of the offer score π_{is}.
- *T, single stage.* Students apply once, at a single date, and are matched by deferred acceptance with no limit on the number of programs they list. Students therefore rank programs on their tastes alone, and their own results do not affect their lists. Without S, students rank programs before they learn ε_{is}; without P, programs rank applicants on predicted scores.

Designs without T keep the two-stage timing and the Stage-4 grade condition. Post-qualification admissions (c) combine S, P, and T. Each change's Shapley value is its marginal effect averaged over the six orders in which the three changes can be added.